

Trio[®]**VIBRATING GRIZZLY FEEDER****TF-Series Vibrating Grizzly Feeder**

Installation, Operation and Maintenance Manual

WEIR**Minerals**www.global.weir

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VIBRATING GRIZZLY FEEDER

TF-Series Vibrating Grizzly Feeder

Installation, Operation and Maintenance Manual



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Project Information

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TF-Series Vibrating Grizzly Feeder

Installation, Operation and Maintenance Manual



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1 Introduction

The Installation, Operation, and Maintenance Manual (IOM) included in the shipment must be read thoroughly before installing or operating the Vibrating Grizzly Feeder (VGF). All instructions regarding maintenance must be retained for reference.

Weir Trio TF- series feeders are two-shaft feeders, they generate their power through two timed shafts with gears. It serves the basic purpose of removing fine material upstream of the crusher as well as spreading the material, so it is distributed evenly to the crusher at the desired rate.

1.1 Scope of the Manual

This manual describes the installation, operation, and maintenance of Weir Trio Vibrating Grizzly Feeders. The manual must be used as a reference in conjunction with the Weir Trio product-specific training. This manual is written for all types of the Weir Trio Vibrating Grizzly Feeders. The feeders series as below [Table 1-1](#).

Model Number	Power Requirement kW (hp)
TF4010	11 (15)
TF4012	15 (20)
TF4014	15 (20)
TF4016	18.5 (25)
TF4616	18.5 (25)
TF4620	22 (30)
TF5217	30 (40)
TF5220	30 (40)
TF5224	37 (50)
TF5820	45 (60)
TF6420	45 (60)
TF6424	45 (60)
TF7220	55 (75)
TF7224	75 (100)

Table 1-1: Vibrating Grizzly Feeders Models

1.2 Intended Audience

This manual is intended for customers and the personnel responsible for installation/maintenance appointed by the customer.

1.3 Disclaimer

Please contact your local Weir Trio representative for any assistance.

- This manual provides general information about the product. The product purchased may contain equipment variations to this manual.
- Operating conditions specified in the manual may vary to the operating conditions of the product installed on-site.
- Illustrations shown in the manual are for reference only and may not match the specific product.

	 WARNING
	PERSONNEL INJURY The feeder must be installed, operated, and maintained only by personnel trained and have sufficient knowledge about the hazards that may occur during feeder operation.

1.4 Overview of the IOM Manual

Table 1-2 describes the organisation of the manual.

Chapter	Description
Introduction	Contains an introduction about the feeders and the scope of the manual.
Safety	Lists the safety guidelines that must be adhered to while installing, operating and maintaining the feeders.
Technical Data	Contains the technical aspects of the feeders.
Description	Describes the working principle and different components of the feeders.
Transport and Storage	Provides information on safety guidelines and procedures to follow while delivering the feeders to a customer location, bringing the feeders to a workshop for maintenance, and storing the feeders.
Installation	Lists the procedures to install the feeders.
Commissioning and Operation	Contains the information on feeders pre-start-up procedures, feeders start-up, problems that may occur during the feeders start-up and shut-down procedures.
Assembly	Contains safety guidelines and procedures to be followed while performing feeders assembly.
Disassembly	Contains safety guidelines and procedures to be followed while performing feeders disassembly.
Maintenance	Lists running and overhaul maintenance procedures.
Lubrication	Contains details about the lubrication requirements, specifications, lubrication checks and service intervals.
Decommissioning and Disposal	Provides guidelines regarding feeders out of service, and disposing or selling it.
Troubleshooting	Provides information regarding various problems that may occur during feeders operation and respective solutions.
Special Tools	Lists special tools that are required for assembly and disassembly.
Appendix	Contains additional information such as torque specification, V-belt drive terminology and reference documents for customers.

Table 1-2: Overview of the Manual

1.5 Contact Information

If you need assistance, you can email us today at trio@mail.weir or contact your local Weir Trio® representative by visiting: <https://www.global.weir/map/>.

1.6 Glossary of Abbreviation

Abbreviation	Description
ATEX	Equipment for Potentially Explosive Atmospheres
COP	Code of Practice
COR	Chain of Responsibility
EMC	Electromagnetic Compatibility
GA	General Arrangement
IOM	Installation, Operation and Maintenance
ISO	International Organisation for Standardisation
LOTO	Lock Out and Tag Out
MSHA	Mine Safety and Health Administration
OSHA	Occupational Safety and Health Administration
PED	Pressure Equipment Directive
PPE	Personal Protective Equipment
RPM	Revolutions Per Minute
SDS	Safety Data Sheets
QTY	Quantity
VI	Visual Inspection

Table 1-3: Abbreviations

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2 Safety

The instructions in this chapter should be followed for safe and efficient operation.

2.1 Safety Symbols

The safety symbols used in this manual are indicated in [Table 2-1](#).

Signal	Warning Level	Description
 DANGER	DANGER	Indicates a hazard with a high level of risk which, if not avoided, could result in death or catastrophic equipment damage.
 WARNING	WARNING	Indicates a hazard with a medium level of risk which, if not avoided, could result in serious or moderate injury, or serious equipment damage.
 CAUTION	CAUTION	Indicates a hazard with a low level of risk which, if not avoided, could result in moderate or minor injury, or moderate equipment damage.
 NOTICE	NOTICE	Indicates important information for the user.

Table 2-1: Safety Symbols

2.2 Safety Sign Maintenance

Keep the operator's safety in mind at all times.

- Replace any missing or damaged safety signs.
- Use mild soap and water to clean safety signs.
- Do not use solvent-based cleaners, as they may damage the safety sign material.

2.3 Safety Signs and Icons on the Equipment

The safety regulations follow a set of common-sense rules. Personnel who work with or near heavy equipment must be aware of these rules.

For creating such awareness to the personnel, safety signs are fixed on the equipment. These symbols remind personnel about common hazards and safety procedures. The language of the icons is simple and direct (refer to [Table 2-2](#)).

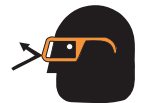
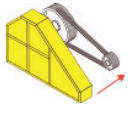
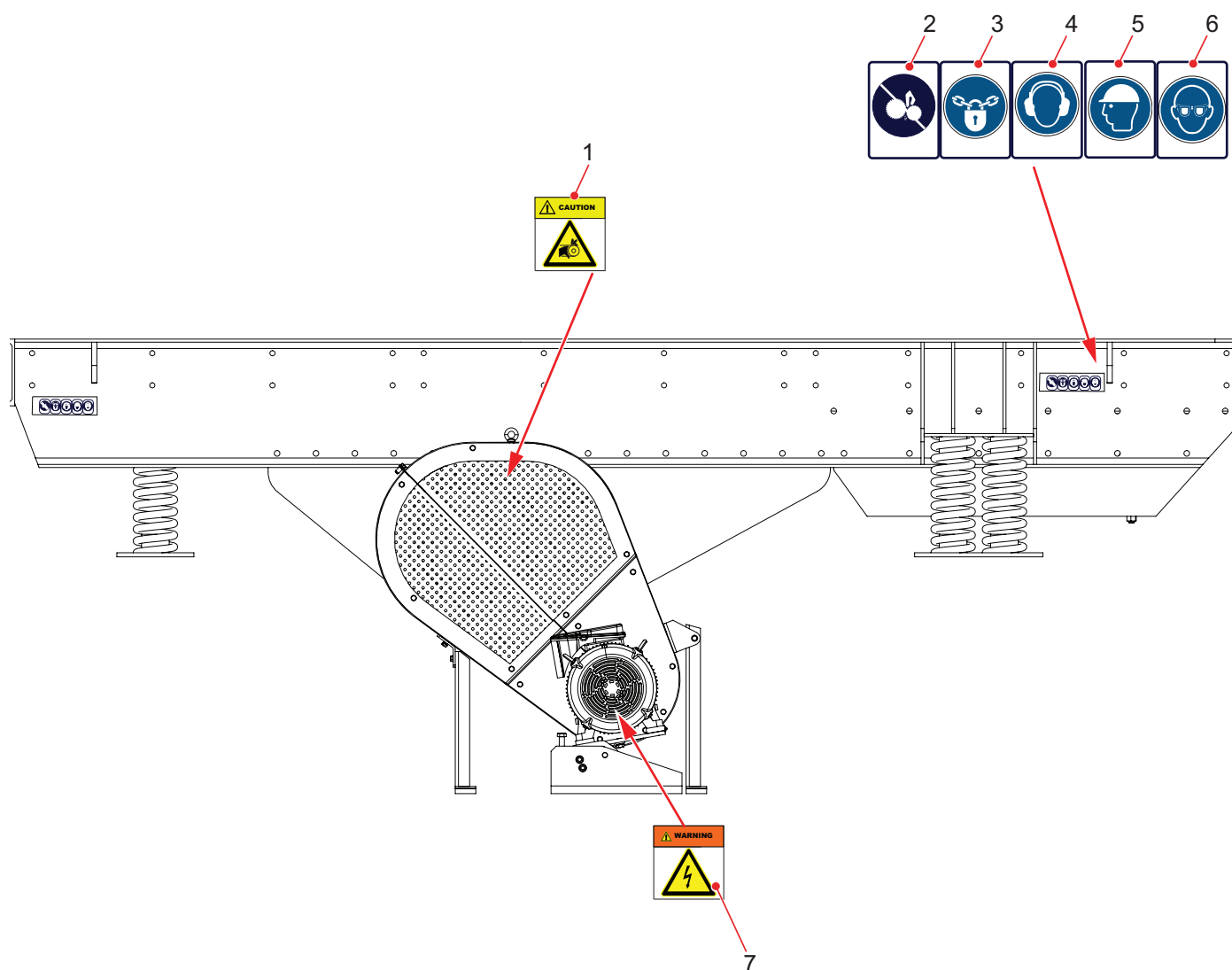
Symbols		Description
		Read the manual.
		Safety alert symbol.
		Lock out power before servicing machine.
		Wear hearing protection when you are near this equipment.
		Wear eye protection when you are near this equipment.
		Wear respiratory protection when you are near this equipment.
		Wear head protection when you are near this equipment.
		Entanglement hazard. Death or serious injury can result by getting caught in belting.
		Make sure that all guards are in place.
		Risk of electric shock.
		Risk of Crushing.

Table 2-2: Safety Pictogram



1	Entanglement hazard (Belt Drive)	5	Wear Head Protection
2	Crushing of Hands	6	Wear Eye Protection
3	Lock Out	7	Electricity Hazard
4	Wear Ear Protection		

Figure 2-1: Safety Signs on Grizzly Feeders

2.4 Personnel Safety

This equipment is designed with the safety of all personnel in mind. Never attempt to change, modify, eliminate or bypass any of the safety devices installed at the factory. Guards, covers and shields installed around moving parts at the factory are meant to prevent accidental injury to operators and other personnel. Do not remove them.

	 DANGER
	RISK OF PERSONNEL INJURY OR DEATH SOMETIMES, AT A CUSTOMER'S REQUEST, EQUIPMENT IS SHIPPED WITHOUT CERTAIN FEATURES SUCH AS A FEEDER SHEAVE. IN SUCH CASES IT IS THE CUSTOMER'S RESPONSIBILITY TO GUARD THE MACHINE PROPERLY BEFORE OPERATION STARTS. AN UNGUARDED MACHINE CAN LEAD TO PERSONAL INJURY OR DEATH.

	 WARNING
	RISK OF INJURY DUE TO SHARP EDGES • Before starting any work, identify and examine any parts that could be hazardous due to sharp edges. All metal components must be considered to have sharp edges. • Worn out components can have sharp or jagged edges. Handle worn parts carefully, to prevent damage to slings or personnel injury. • Assess the components before moving or lifting and use mechanical devices wherever possible.

The recommended safety practices are as follows:

	 WARNING
	GLOVES MUST BE WORN Appropriate gloves must be worn when handling parts with sharp edges. For example, puncture-resistant or cut-resistant gloves. If in doubt, please ask your Safety Representative for advice.

2.4.1 Management Responsibilities

- Make sure that everyone working on or near this equipment is familiar with safety precautions. Assign a supervisor at the job site responsible for job safety.
- Give crew members specific safety responsibilities and instruct them to report any unsafe conditions to their supervisor.

2.4.2 Operator Responsibilities

	 DANGER
	RISK DUE TO ALCOHOLIC BEVERAGES AND MEDICATION THE PERSONNEL WORKING ON OR NEAR THE EQUIPMENT HAVE TO BE ALERT. DO NOT USE ALCOHOLIC BEVERAGES BEFORE COMING TO WORK OR WHILE ON THE JOB. BE AWARE OF MEDICINES, TRANQUILLIZERS OR OTHER DRUGS WHICH CAN MAKE YOU SLEEPY OR LESS ALERT.

- Read all danger, warning, caution and notice signs.
- Always lock out and tag out involved energy sources before performing maintenance or adjustments on this equipment. Make it impossible for anyone to start this machine while others work on it or in it.

- Never remove any guard, cover or shield when this equipment is in motion.
- Replace guards, covers and shields after completing the relevant tasks.
- Block parts as necessary to prevent their sudden movement while people are working on the machine.
- Never attempt to clear away jammed feed material, discharge material or other stoppages while the machine is running. Stop the equipment, lock out and tag out before touching this machine with tools or hands.
- Wear proper personal protective equipment, including eye protection, hard hat and safety shoes, whenever you are near this machine while it is running.
- Wear eye protection when handling fuel, oil, cleaning fluid or brake fluid. They can damage the eyes.
- Wear respirator whenever required.
- Wear safety glasses when drilling, grinding or hammering metal.
- Keep the pockets free of objects as it can fall into the machinery.
- Wear gloves while changing cables, to protect your hands.
- Wear safety glasses and protective clothing while using high-pressure air.
- Wear goggles and protective clothing when handling molten metals like zinc, babbitt, lead, etc.
- Wear ear protective devices when required or reduce exposure time as required.
- Dress appropriately in every way. Never wear loose clothes, long hair, coattails, jewellery, pockets full of tools or any other item that could get caught in moving parts.
- Know where your fellow workers are. Always look around and inside this machine before starting it. Make sure nobody is in the way of moving parts or working on the machine.
- Lift with your legs, not with your back. Keep the weight close to your body. If the load is more than 40 lbs., get someone to help you.
- Never engage in horseplay when near this machine, or any other.
- Report any defective machinery or equipment and unsafe conditions or activity to your supervisor immediately.
- Don't limit safety practices to the few rules listed here. Think safety and act safely at all times.
- Most of all, know your equipment. Understand the machinery, the conditions under which it operates, and what it is capable of doing.

2.4.3 Work Area Safety

- Keep the work area as neat and clean as practical, free of debris and tools. Oily and wet floors, steps, and handrails are slippery. In winter, watch out for ice and snow.
- Keep all product safety signs clean, clear, and current.
- Make sure all electrical equipment is properly grounded. Wet spots near electrical current are especially dangerous.
- Store hazardous materials in restricted access areas and mark them clearly. Local, regional regulations require special labelling of certain materials.
- Do not let any material build up on or around the feeder.
- Know where fire extinguishers and other fire-suppression equipment are located. Learn how to use them effectively.
- Be alert and wary around any hydraulic pressurised system. High-pressure oils and are very dangerous.
- Take appropriate precautions when torch cutting or welding. Avoid breathing the fumes when performing these types of procedures as it may cause a health hazard. Perform such procedures in an outdoor, well-ventilated area with either a separate clean air supply provided to the mechanic or with the local exhaust of fumes. Please refer to OSHA or MSHA standards as appropriate.
- Never do hot work on feeders without the approval of weir minerals.
- Rubber lining materials are highly flammable and hot work with cause fires if heat is applied to rubber directly or through heat conduction. Remove rubber if at risk and contact weir to re-apply rubber lining.

2.4.4 Equipment and Tools

	 DANGER
	GENUINE PARTS • GENUINE PARTS AND ACCESSORIES ARE DESIGNED, TESTED AND INCORPORATED INTO THE PRODUCTS TO HELP ENSURE THEY MAINTAIN CONTINUED PRODUCT QUALITY AND PERFORMANCE IN USE. • AS WEIR TRIO DOES NOT TEST THE PARTS AND ACCESSORIES SOURCED FROM OTHER VENDORS, THE INSTALLATION OF SUCH PARTS AND ACCESSORIES ADVERSELY AFFECTS THE PERFORMANCE AND SAFETY FEATURES OF THE PRODUCTS. • THE FAILURE TO PROPERLY SELECT, INSTALL OR USE AUTHORISED PARTS AND ACCESSORIES IS CONSIDERED MISUSE. DAMAGE OR FAILURE CAUSED BY MISUSE IS NOT COVERED BY OUR WARRANTY. • IN ADDITION, ANY MODIFICATION OF WEIR TRIO PRODUCTS OR REMOVAL OF ORIGINAL COMPONENTS MAY IMPAIR THE SAFETY OF THESE PRODUCTS IN THEIR USE.

- Clean tools that are properly labelled and stored are safer tools. Keep your tools in good order.
- Use proper tools. Handle tools and heavy parts sensibly.
- Keep drive belts and sheaves in good condition. Frayed belts or cracked sheaves are not only dangerous, but they can also cost you downtime.
- Always use mechanical assistance to lift heavy loads. Never overload a hoist, crane, jack or other lifting devices. Check to lift tackle regularly; replace it at the first sign of stretch, fraying, or other wear.
- Keep your equipment clean, free of dirt and grease, so that loose, cracked, or broken parts are more easily spotted. Replace defective parts as soon as they are discovered.
- To prevent slipping, wipe hand lever and knobs clean of oil or grease.

2.4.5 Training and Knowledge

	 WARNING
	RISK OF INJURY Do not operate this equipment until you have been trained in its operation or maintenance. This equipment may only be operated or maintained by trained personnel who have demonstrated their ability to do so safely.

Proper training and equipment knowledge is essential to the safe operation of this machine. Carefully read the entire manual before attempting to operate. Keep this manual for future reference.

2.5 Plant Safety

	 DANGER LOCK OUT, TAG OUT & TRY OUT THE WORK AREA • THE ELECTRICAL, HYDRAULIC, PNEUMATIC OR MECHANICAL POWER SOURCE OF THE FEEDING EQUIPMENT MUST BE LOCKED OUT, TAGGED OUT AND TRY OUT WHENEVER ANY MAINTENANCE ACTIVITY IS CARRIED OUT ON IT. • ACCIDENTAL START-UP OF THE EQUIPMENT CAN RESULT IN FATAL ACCIDENT.
	 DANGER INSTALL DRIVE GUARDS • FEEDER SHEAVES ARE DESIGNED FOR MAXIMUM RIM SPEEDS. IF THESE SPEEDS ARE EXCEEDED, SHEAVE MAY EXPLODE AND CAN CAUSE SEVERE PERSONAL INJURY OR EVEN DEATH. • FEEDERS ARE DRIVEN BY VIBRATOR. • TAKE SPECIAL CARE TO SEE THAT ALL PARTS OF FEEDER BODY ARE CLEAR OF MOVING PARTS AND PINCH POINTS BEFORE STARTING OR OPERATING MACHINERY. • MAKE SURE THAT ALL ROTATING SHEAVES, PULLEYS AND BELTS ARE PROPERLY GUARDED TO PREVENT BEING CAUGHT BY THEM. • BE SURE THAT THE FEED AND DISCHARGE SYSTEMS FOR MATERIAL PROCESSING EQUIPMENT ARE DESIGNED AND MAINTAINED SO THAT PROCESSED MATERIALS (ROCKS, GRAVEL, COAL, ETC.) ARE RETAINED WITHIN THE SYSTEM. • NEVER SIT OR STAND ON ANY COMPONENT OF THE MACHINERY WHILE IT IS IN OPERATION TO PREVENT INJURIES ARISING FROM VIBRATION. • VERIFY THAT THE SUPPORTING STRUCTURE(S) PROVIDED BY THE COMPANY FOR THE MACHINERY IS SUFFICIENT TO SUPPORT BOTH THE TOTAL WEIGHT OF THE EQUIPMENT AND THE DYNAMIC LOAD AND ALSO STRESSES CREATED BY THE EQUIPMENT.

**DANGER****FEEDER PLANT NOISE EMISSION**

- FOR THE MOTOR ONLY SOUND LEVELS, REFER TO THE RELEVANT PROPRIETARY ELECTRICAL MOTOR MANUAL.
- FOR AN IN-SERVICE FEEDER, INCLUDING ITS DRIVE, THE ACTUAL NOISE LEVELS GENERATED ARE DEPENDENT ON A NUMBER OF VARIABLE FACTORS, SUCH AS DRIVE TYPES, SLURRY TYPES, SURROUNDING EQUIPMENT, WHETHER IT IS INSTALLED INDOOR / OUTDOORS, TYPE OF FOUNDATION OR ANY COMBINATION OF THE ABOVE AND LOCAL ENVIRONMENTAL FACTORS. THESE FACTORS ARE BEYOND THE CONTROL OF WEIR TRIO.
- SOUND EMISSION LEVELS SHOULD BE ASSESSED IN SERVICE BY THE USER REFER TO DIRECTIVE 2003/10/EC FOR MINIMUM.
- HEALTH & SAFETY EXPOSURE REQUIREMENTS OF EUROPE SUITABLE PRECAUTIONS SUCH AS THE USE OF PERSONAL PROTECTIVE EQUIPMENT (PPE) SHOULD BE TAKEN TO PROTECT OPERATORS FROM EXPOSURE TO NOISE IN ACCORDANCE WITH LOCAL HEALTH, SAFETY AND ENVIRONMENTAL REGULATIONS.
- FEEDING EQUIPMENT AND AUXILIARY EQUIPMENT FOUND IN AND AROUND THE FEEDING EQUIPMENT SUCH AS CHUTES, TRANSFER STATIONS, CONE CRUSHER ARE NOISY THAN THE FEEDERS.
- WEAR EAR PROTECTIVE DEVICES IF THE NOISE LEVEL IS HIGHER THAN THE LIMITS.
- IF EAR PROTECTIVE DEVICES ARE UNDESIRABLE, REDUCE THE AMOUNT OF NOISE EXPOSURE THAT AN OPERATOR OR MAINTENANCE PERSONNEL MAY BE SUBJECTED TO.
- INSTALLATION OF A RELATIVELY INEXPENSIVE OPERATOR'S STATION MAY IMPROVE NOISE CONDITIONS.
- THE INSTALLATION OF ACOUSTICAL TILE, DOUBLE WINDOWS, AIR CONDITIONING EQUIPMENT ETC CAN HELP THE OPERATOR PROTECT FROM THE NOISE AND WATCH THE EQUIPMENT IN OPERATION.
- THESE WORKING CONDITIONS RESULTS IN A MORE EFFICIENT OPERATION AND REDUCE POTENTIAL INJURIES FROM NOISE.
- AN OPERATORS TOWER RAISED AT AN ELEVATION ABOVE MOST OF THE EQUIPMENT PROVIDES SAFETY FEATURES FOR THE OPERATOR AND INCREASED EFFICIENCY IN PLANT OPERATION.

**DANGER****VIBRATION LEVELS**

- FEEDERS ARE NOT INTENDED TO BE INSTALLED AT WORKSTATIONS OR REQUIRE CONTINUAL CLOSE CONTACT BY OPERATIVES. IF THIS IS UNAVOIDABLE, THEN THE FREQUENCY AND LEVEL OF EXPOSURE TO VIBRATION BY ANY OPERATIVE SHOULD BE ASSESSED. ATTENTION SHOULD BE GIVEN TO VIBRATIONS TRANSMITTED THROUGH THE FEET TO FEEDER FRAME SUPPORT STRUCTURE.



	 WARNING FEEDER • The feeder is designed to feed and transfer rock, recycle concrete, and also a wide range of other products and materials. The use of this machinery in any other way is prohibited and contrary to its intended use. • When the equipment is used for any other purpose, the user is responsible for any injuries or damage that may occur. • Weir Trio recommends that the equipment must be operated only by the personal who has undergone training.
	 WARNING FEEDING PLANT DUST • Feeding and auxiliary equipment such as chutes, transfer stations, cone crusher etc. creates dust. • Breathing this dust is hazardous to personnel health. Personnel working at a high level of dust in the air are prone to lung disease, depending on the concentrations of dust and the type of material being used. • Weir Trio recommends personnel wear dust-protection device, such as a respirator, to prevent dust inhalation. It is the responsibility of the operator to determine the necessity of the protective devices and make sure it is used appropriately. • Installation of a dust collection system or the use of water spray bars at the feed and /or discharge conveyor of the feeding circuit can reduce the airborne dust. • Feeding with a choke level also reduces the dust from the feeder. • Configuration of each rock feeding installation is different. Weir Trio, recommends the operator to consult the local representative of Weir Trio or a dust consultant for possible alternative methods of dust reduction.
	 WARNING WELDING • Never perform hot work on any feeder with out Weir Trio approval. • Take precautions while torch cutting and/or welding, as this may cause health hazards posed by many metals. Avoid breathing the fumes while performing this procedure. Perform this procedure at outdoor with separate clean air supply provided to the mechanic or in a well-ventilated area with the local exhaust of fumes. Follow OSHA or MSHA standards as appropriate. • The maintenance personnel who uses torch cutting equipment must be informed if there are hydraulic components in the immediate area in which they are working. Adequate precautions must be taken to avoid these components. Rubber lined components that are used in normal plant maintenance are not immune to the torch cutting equipment. • Any hot work on feeder could lead to structural failure, which could cause injury or death.

	 WARNING
	LIGHTNING PROTECTION • When the feeder is installed outside of an enclosed building or is connected to the outside by conductive parts, the feeder should be included within the site's lightning protection system assessment. • Where it has been assessed that the feeder must be protected from a lightning strike by bonding to the site's lightning strike grounding system, the connection should be made to meet or exceed the requirements of the governing standard such as the EU standard EN 62305 in Europe. • As part of the user's scheduled maintenance program, the connection should be visually inspected and tested in accordance with that standard and its exposed parts protected from corrosive environments.

	 CAUTION
	ELECTROSTATIC CHARGES • If the feeder is used in dry or dusty environments or its operating conditions causes the machine or its components to become electrically isolated from earth, for example by mounting on an ungrounded foundation such as anti-vibration rubber matting, this may lead to a build-up of static electricity leading to the risk of electrical discharges, (potential ignition source in hazardous environments) and electric shock. • To prevent this, a bonded connection from the feeder to the site's earth grounding system should be made to meet or exceed the requirements of the local governing standards such as the international standard IEC 60364. • As part of the user's scheduled maintenance program, the connection should be visually inspected and tested in accordance with that standard and its exposed parts protected from corrosive environments.

	 CAUTION
	RISK DUE TO CONVEYOR MACHINERY • Do not use a conveyor belt as walkways. • Always provide handrails along the conveyor belt walkway as an additional safety precaution. • Be sure that all parts of your body are kept clear of moving conveyor belts, especially at pinch points. • Do not lubricate machinery while it is in motion. • Never lean over a moving conveyor belt, or ride on a conveyor belt intended only for material transport. • Always keep clear of feed or discharge points of all material conveyor systems. • Be sure that all guards are securely fastened in place on sheaves, pulleys, and belts before and during equipment operation.

2.6 General Safety

This equipment is designed expressly for processing rock and material. When equipment is used for purposes other than those for which it was designed, the user takes on sole responsibility for any injuries or damage that may result.

2.7 Qualified Workers

All personnel involved in the operation, installation, inspection, and maintenance of the equipment must be qualified to carry out the work involved. If your personnel does not have the necessary knowledge and skill, appropriate training and instruction must be provided.

If required, the operator may engage the manufacturer/supplier to provide applicable training at a cost to the site.

Always coordinate repair activities with operations and workplace health and safety personnel. Follow all plant safety requirements and workplace health and safety laws and regulations, where applicable.

2.8 Markings and Approvals

It is a legal requirement that machinery and equipment put into service within certain regions of the world must conform to the applicable marking directives and standards such as CE marking directives covering machinery and, where applicable, low-voltage equipment, Electromagnetic Compatibility (EMC), Pressure Equipment Directive (PED), and Equipment for Potentially Explosive Atmospheres (ATEX).

Where applicable, the directives and any additional approvals cover important workplace health and safety aspects relating to machinery and equipment and the satisfactory provision of technical documents and workplace health and safety instructions.

Where applicable, this document incorporates information relevant to these directives and approvals. To confirm the applying approvals, and if the product is CE marked, check the serial number plate markings and the certification.

2.9 Safety Equipment

Do not operate the feeder without properly installed v-belt and coupling guards in place. If guards are removed during maintenance, they must be replaced prior to operating the feeder.

For auxiliary equipment (motors, belt drives, couplings etc.), standard workplace health and safety precautions must be followed and appropriate instruction manuals consulted before and during installation, operation, adjustment, and maintenance.

2.10 Fire-Fighting

- Make sure you are familiar with the site and local procedures and requirements for fire prevention and fire-fighting.
- Do not smoke while using cleaning solvents.
- Do not allow greasy and oily rags to accumulate in a poorly ventilated area. Store oily rags and other combustible material in a safe place.
- Never use an open flame to check coolant levels or to look for oil leaks anywhere on the equipment. Use a flashlight.
- Know where fire extinguishers are kept and how they operate and for what type of fire. Check regularly, at least monthly, to make sure it is in the working area.

2.11 Emergency Procedures

Make sure you are familiar with the site and local procedures and requirements. If there is an emergency, follow the site and local procedures.

2.12 Lock Out, Tag Out and Try Out

Lock Out, Tag Out and Try Out is a procedure that's designed to prevent the unexpected or accidental start-up of equipment and to alert all workers whenever it is unsafe to operate any piece of equipment. When used as intended, Lock Out, Tag Out and Try Out also protects personnel from energy stored in devices such as

springs, accumulators, batteries, hydraulic systems, etc. Ensure that the entire process energy is locked out and tagged out when working on feeders. Prevent material from falling onto feeders while people attend to it's maintenance.

2.12.1 When is Lock Out, Tag Out and Try Out required?

- When the equipment is under maintenance.
- When the equipment is broken or unsafe to operate.
- While clearing blocked or jammed mechanisms.
- Whenever the equipment is left unattended.

2.12.2 Who must apply a Lock Out, Tag Out and Try Out?

- Any person that is working on the equipment.
- Foreman or another person responsible for the work being done.
- If several people are working on a machine at the same time, each person must apply his or her own Lock Out, Tag Out and Try Out.

2.12.3 When can a Lock Out, Tag Out and Try Out be removed?

After performing the below steps:

1. All safety guards are back in place.
2. All work is complete.
3. All workers are notified that a lock has been removed.
4. All workers are positioned safely for a start-up.
5. Controls are positioned for a safe start-up.
6. The machine is ready for safe operation, and a walk around has been performed, ensuring that no one is in an unsafe position around the equipment.

2.12.4 Who can remove a Lock Out, Tag Out and Try Out?

- Only the person who applied a Lock Out, Tag Out and Try Out is permitted to remove them.
- Lock Out, Tag Out and Try Out rules laid out here are generic. To get instructions for a particular workplace, consult employers for Lock Out/Tag Out/Try Out procedure.

2.13 Installation Safety

	! DANGER LIFTING DEVICE SAFETY • USE LIFTING DEVICES TO LIFT HEAVY OR AWKWARD COMPONENTS. • THE LIFTING DEVICES MUST BE IN GOOD CONDITION, CERTIFIED, AND TAGGED. • THE LIFTING DEVICES MUST BE OF ADEQUATE CAPACITY AND MUST BE USED WHENEVER THEY ARE REQUIRED. • PERSONNEL MUST NEVER WORK UNDER SUSPENDED LOADS.
	! WARNING RISK OF INJURY DURING LIFTING Do not attempt to lift the feeder assembly by any other method, as this would not be strong enough to lift the entire feeder assembly's weight. This could result in serious personal injury or severe damage to the equipment.
	! CAUTION CAUTION WHILE USING CRANES • Cranes require normal periodic maintenance for efficient use. • One of the most feeder safety violation is the use of inadequate and unsafe lifting equipment. • The internal parts of the feeders are assembled and disassembled with crane facilities that are capable of lifting and lowering the various parts of a feeder. • Operate the crane within the rated capacity. The safe rated capacity includes weight of the hook, block and any material handling devices such as cables, slings, spreader bar etc. • To find the true weight of the load, which can be handled safely, subtract the weight of the hook, block and any material handling devices such as cables, slings, spreader bar etc. • Consider chain falls as the last option to assemble and disassemble the feeder.

	 CAUTION
	CAUTION WHILE USING MOBILE CRANES • When using a mobile crane, operate it within the rated capacity of the machine to avoid buckling the boom or tripping. • Safe ratings are based on operating the crane on firm, level ground. • Avoid fast swings, hoists or sudden braking as this can cause overloads. • Do not handle large, heavy loads in strong winds. • While moving the crane, check bridges before crossing. • Check clearance under bridges, for overhead electrical lines or any overhead obstruction.

	 CAUTION
	CAUTION WHILE USING LIFTING EQUIPMENT • Lift all feeder components by equipment with slow and even movements. • Always comply with maximum load regulations for the equipment. • Add the weights of the hoisting accessories such as chain, hooks etc. to the feeder weight. • Suitable lifting equipment must be used to lift the feeder using four lifting lugs provided on the feeder and the chains/slides must have an angle of no less than 60 degrees to the horizontal. • The feeder mass is indicated on the general arrangement drawing. • Always make sure that the lifting equipment is of sufficient capacity to lift the required weights. • Restriction of an operator who is not trained for lifting equipment and cranes. • Never walk or work under suspended loads.

2.14 Control System Safety

These instructions are not intended as a substitute for proper training or adequate experience in the safe operation of the control system. Only competent technicians who are familiar with this equipment shall install, operate and maintain the control system.

A competent technician has these qualifications:

- Is trained in industry-accepted high and low-voltage safe operating practices and procedures.
- Is trained and authorised to energise, de-energise, clear, and ground power distribution equipment.

	 DANGER
	RISK OF ELECTRIC SHOCK CONTACT WITH HAZARDOUS VOLTAGE CAUSES DEATH OR SEVERE PERSONAL INJURY. FOLLOW ALL LOCALLY APPROVED SAFETY PROCEDURES WHEN WORKING AROUND HIGH- AND LOW-VOLTAGE LINES AND EQUIPMENT.

	 DANGER
	RISK OF PERSONNEL INJURY OR DEATH POWER DISTRIBUTION AND TRANSMISSION EQUIPMENT MUST BE PROPERLY SELECTED FOR THE INTENDED APPLICATION. IT MUST BE INSTALLED AND SERVICED BY COMPETENT PERSONNEL WHO HAVE BEEN TRAINED AND UNDERSTAND PROPER SAFETY PROCEDURES. THESE INSTRUCTIONS ARE WRITTEN FOR SUCH PERSONNEL AND ARE NOT A SUBSTITUTE FOR ADEQUATE TRAINING AND EXPERIENCE IN SAFETY PROCEDURES. FAILURE TO PROPERLY SELECT, INSTALL OR MAINTAIN POWER DISTRIBUTION AND TRANSMISSION EQUIPMENT CAN RESULT IN DEATH, SEVERE PERSONAL INJURY, AND EQUIPMENT DAMAGE.

2.14.1 Electrical safe work

All relevant electrical work related to installation, operation and maintenance should be carried out according to the locally approved safety procedures and guidelines.

The operator should go through proper training before the manipulation of the control system.

A regular safety check should be done with the data recording.

	 DANGER
	NEED FOR NECESSARY REPAIRS IMMEDIATELY REPORT ANY SITUATION INVOLVING THE NEED FOR REPAIR, REPLACEMENT OR ADJUSTMENT OF AN EQUIPMENT COMPONENT IMMEDIATELY. REMEMBER, THE MINOR DEFECT COULD RESULT IN INJURIES TO YOURSELF OR OTHERS OR SERIOUS DAMAGE TO THE MACHINERY IF IT IS NOT CORRECTED PROMPTLY.

	 DANGER
	FIRE HAZARDS • NEVER WELD OR USE A CUTTING TORCH NEAR OPEN OR EXPOSED FUELS, LUBRICANTS, OR OTHER FLAMMABLE MATERIALS. FOLLOW YOUR COMPANY'S HOT WORK POLICY. • IF OXYGEN AND ACETYLENE MUST BE USED FOR CUTTING WORK, ASSESS THE WORK AREA TO ENSURE THAT FLAMMABLE MATERIALS ARE NOT PRESENT. • DO NOT LET GREASY OR OILY RAGS ACCUMULATE IN POORLY VENTILATED AREAS. STORE OILY RAGS AND OTHER COMBUSTIBLE MATERIALS IN AN APPROVED STORAGE CONTAINER AND DISPOSE OF THEM PROPERLY. • NEVER USE AN OPEN FLAME TO CHECK FOR FLUID LEAKS IN THE EQUIPMENT. USE AN APPROVED LOW VOLTAGE ELECTRIC LIGHT OR FLASHLIGHT. • NEVER SMOKE IN NON-DESIGNATED AREAS. FOLLOW YOUR COMPANY'S SMOKING RULES. ALWAYS EXTINGUISH SMOKING MATERIALS PROPERLY. KNOW WHERE THE NEAREST FIRE EXTINGUISHER IS LOCATED, HOW TO OPERATE IT, AND WHAT TYPES OF FIRES IT CAN BE USED TO EXTINGUISH. MAKE SURE THAT FIRE EXTINGUISHERS ARE IN PLACE, ARE IN PROPER WORKING CONDITION, AND HAVE A CURRENT, VALID INSPECTION TAG. IF A FIRE EXTINGUISHER IS NOT READILY AVAILABLE OR DOES NOT MEET THESE REQUIREMENTS, REPORT THE SITUATION TO YOUR SUPERVISOR.

	 WARNING
	HANDLE TOOLS, HEAVY PARTS, AND MACHINE COMPONENTS PROPERLY Do not toss parts or tools or drop them off platforms. Do not lower or swing electric or air power tools by their power cords or airlines. Use a suitable rope and/or container to lower the tools or parts to the floor. Review job activities and arrange to get help before you attempt to lift, carry, or move heavy parts. For heavy or awkward lifts, use proper hoisting devices.

	 CAUTION
	EQUIPMENT START-UP Be sure that control systems and alarms are working properly. Check gauges for proper readings. Check for any unusual noises that may indicate bearing failure or other mechanical problems. Check for sudden rises in lubricant temperature. Shut the equipment down immediately if problems are detected, and report the problem(s) to your supervisor. Always refer to the Operating Section of this manual if you are unsure about a start-up procedure or for additional information.

	 CAUTION
	LOCK-OUT/TAG-OUT/TRY-OUT (LOTOTO) The machine must be completely stopped with power off and locked-out with controls tagged before any cleaning, repairs, adjustments, or service is performed. All provisions of your company's Control of Hazardous Energy (Lock-Out/Tag-Out/Try-Out) program must be followed.

	 CAUTION DRESS PROPERLY FOR THE JOB AND USE APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT Follow your company's Personal Protective Equipment Policy. Discuss personal protective equipment requirements with your supervisor before operating equipment or performing service, repair or adjustment work. Don't wear loose clothes or jewellery, and restrain long hair.
	 CAUTION KEEP YOUR WORK AREA CLEAN Become fully familiar with the work site where the equipment is installed. Inspect the area daily and see that unsafe conditions are corrected. Make sure that all equipment guards are securely fastened in place, that steps, stairways, work platforms and barrier rails are sound and free of spills or obstructions. Keep the floor/work surface clean and free of oil, greasy rags, cables, chains, buckets and other obstructions. Keep tools and loose parts in a toolbox or appropriate storage areas. Use high flashpoint/low toxicity solutions for cleaning.

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3 Technical Data

This section describes the:

- Feeder serial number
- Feeder dimensions

3.1 Feeder Serial Number

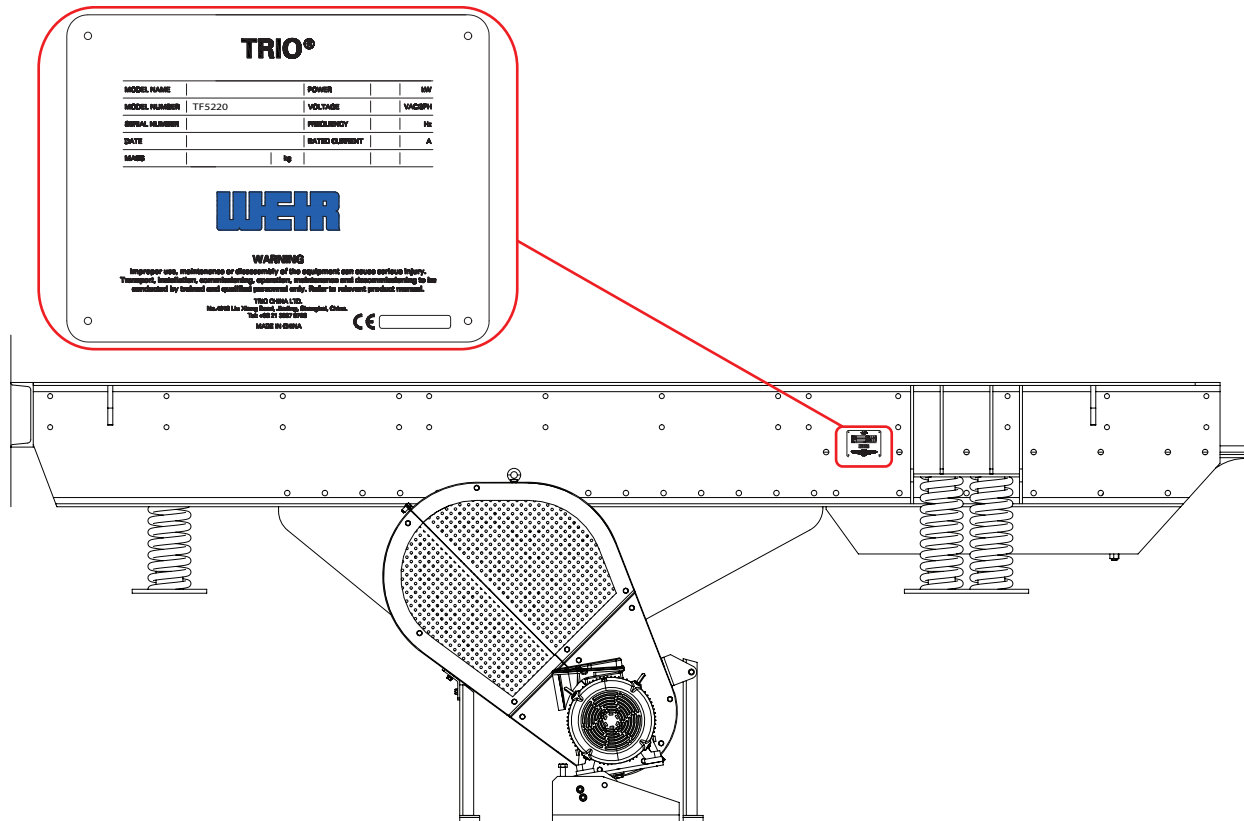


Figure 3-1: Feeder Identification

Feeders are assigned a serial number stamped on a plate on the machine (refer to [Figure 3.1](#)). Other components on the feeders may have separate serial numbers or identifying numbers. Always give the model and serial number or identifying the number of your feeder when ordering parts or requesting information. If parts are required for the other components, you should supply that serial number or identifying the number as well as the feeder numbers. The GA drawings and the Parts list are supplied along with the IOM manual. For more details, refer to the corresponding GA drawing.

3.2 Feeder Dimensions

Dimensions for the feeders are found in the serial specific parts manual and GA drawings.

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4 Description

This section explains about:

- General specification and features
- Feeder components

4.1 General Specification and Features

For general specifications, please refer to the corresponding Parts manual and GA drawings of the specific feeder. The Vibrating Grizzly Feeders are generally used to convey rock or gravel from the loader to the primary crusher. They can also be used under a crusher to protect expensive conveyor belting from large chunks and to eliminate loads on conveyors. The TF series feeders are oil lubrication configurations. The sizes range from 40"x10' to 72"x24' are available.

The feeders have the following features:

- Two eccentric shafts geared together
- Four spherical roller bearings
- The grizzly section has adjustable tapered openings in both dimensions preventing clogging or packing
- Robust steel coil springs are mounted on both ends.
- Replaceable pan liners
- Pivoting motor bases or adjustable motor bases

Vibrator Assembly and Components

The vibrator unit with its counter-rotating eccentric shafts produces the vibrating action which moves material along the feeder. It has double-row, spherical, self-aligning roller bearings, fully enclosed and protected against loss of lubricant and the entrance of sand and grit. Changing the position of the two eccentric shafts by re-timing, the gears shift the line of action or stroke angle toward the feed end, centrally or toward the discharge end. The factory setting produces a line of action through the center of gravity.

The vibrator is a one-piece shaft casing that ties the vibrator side plates together. It has an oil reservoir that runs completely across the vibrator and the oil is common to all parts. As a result, inner oil seals are not required.

The shaft casing and bearing housings are locked together through high-tensile bolted connections at the side plates. A special composition gasket, with predetermined compression, serves as a positive seal at these connections.

As an additional feature, the seal assembly consists of two seals separated by a small space that is charged with grease. The lip of the outer seal is turned toward the outside, to exclude dust, and the lip of the inner seal is turned toward the inside, to retain oil.

Grizzly Section

The grizzly section located at the discharge end of the feeder permits the loosened fines and small material to bypass the crusher. It has tapered bars spaced at intervals across the width of the feeder. The widest opening between the bars is at the discharge end to reduce the chance of rock wedging between the bars and obstructing the flow of material. Grizzly bars are peaked to prevent fines from riding into the crushing chamber. Angle Bar end supports minimize spillage, feeding discharge chute or plates. The nominal grizzly bar opening is determined by measuring the spacing halfway down the length of the grizzly bar. This spacing is normally determined by the crusher discharge setting.

Main Frame and Pan Section

The pan of the feeder is heavily reinforced to withstand the shock loads of dumped material. The feeder sides and pan are protected from wear by liner plates. All liners are easily accessible for installation and replacement.

Springs

The feeder is mounted on heavy-duty springs, which cushion against shock loads, provide flexible support and prevent vibration from being transferred to the rest of the plant. The number of springs varies with feeder size. The bolted outboard spring bracket permits individual replacement of coil springs without removing the hopper.

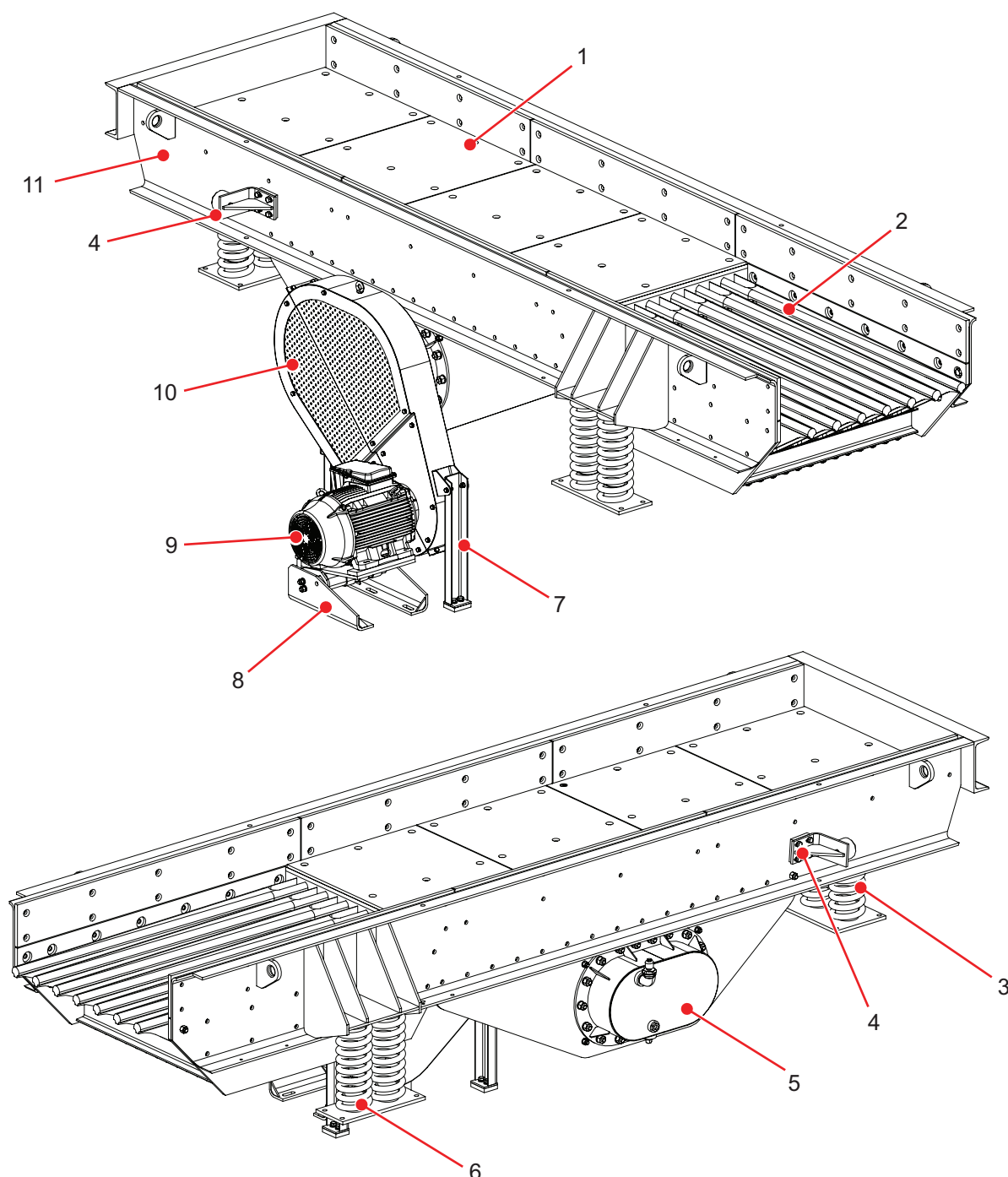
4.1.1 TF Series Feeders Specification

Feeders	Sizes		Machine Weight		Maximum Feeding		Production Capacity	Operating Speed	Power	
Models	in x ft	mm x mm	lbs	kg	inch	mm	stph	rpm	hp	kw
TF4010	40"x10'	1020x3050	7000	3170	24"	600	350-550	650-800	15	11
TF4012	40"x12'	1020x3650	7780	3530	24"	600	350-550	650-800	20	15
TF4016	40"x16'	1020x4877	10690	4859	24"	600	400-600	650-800	25	18.5
TF4616	46"x16'	1170x4877	11000	5000	30"	750	450-650	650-800	25	18.5
TF4620	46"x20'	1170x6070	14100	6402	30"	750	450-650	650-800	30	22
TF4620-2	46"x20'	1170x6070	14100	6402	30"	750	450-650	650-800	30	22
TF5217	52"x17'	1300x5200	14014	6370	32"	800	455-680	650-800	40	30
TF5220	52"x20'	1321x6070	16400	7440	36"	914	500-750	650-800	40	30
TF5220-2	52"x20'	1321x6070	16400	7440	36"	914	500-750	650-800	40	30
TF5224	52"x24'	1321x7316	20170	9148	36"	914	500-750	650-800	50	37
TF5224-2	52"x24'	1321x7316	20170	9148	36"	914	500-750	650-800	50	37
TF5820	58"x20'	1473x6070	18500	8383	36"	914	550-850	650-800	60	45
TF5820-2	58"x20'	1473x6070	18500	8383	36"	914	550-850	650-800	60	45
TF6420	64"x20'	1626x6070	22180	10082	36"	914	650-1000	650-800	60	45
TF6420-2	64"x20'	1626x6070	22180	10082	36"	914	650-1000	650-800	60	45
TF6424	64"x24'	1626x7316	24550	11135	36"	914	650-1000	650-800	60	45
TF6424-2	64"x24'	1626x7316	24550	11135	36"	914	650-1000	650-800	60	45
TF7220	72"x20'	1830x6070	25000	11342	40"	914	730-1500	650-800	75	55
TF7220-2	72"x20'	1830x6070	25000	11342	40"	914	730-1500	650-800	75	55
TF7224	72"x24'	1830x7316	30480	13830	36"	914	730-1500	650-800	75	55
TF7224-2	72"x24'	1830x7316	30480	13830	36"	914	730-1500	650-800	100	75

Table 4-1: TF Series Feeders Specification

4.2 Feeder Components

This section describes the major components of the feeder. For more details refer the corresponding parts manual.



1	Pan Liners	7	Guard Mounting Support
2	Grizzly Bars	8	Motor Mounting Base
3	Rear Spring Pad and Springs	9	Electric Motor
4	Snubber Assembly (Applicable for 20 ft and above length models)	10	Drive Guard
5	Vibrator Assembly	11	Feeder Frame
6	Front Spring Pad and Springs		

Figure 4-1: Feeder Components

4.3 Feeder Drive

The TRIO VGF feeders are equipped with Adjustable motor base tensioner drives. Refer to [“Feeder Drive” on page 6-5](#) for more details.

5 Transport and Storage

This section provides the instructions to follow during:

- Transportation
- Tie-down instructions
- Lifting instructions
- Storage of the feeder

5.1 Transportation

This section describes the following:

- Transport safety
- Chain of Responsibility (COR)
- Load restraint commandments and tools
- Transportation of the feeder

5.1.1 Transport Safety

The transport vendor must pack the freight such that it:

- Make sure any instructions required to lift or re-tether the load are fully visible.
- Does not cause any hazard to the supply staff, transport personnel, and the general public.
- Complies with all legal requirements, including mass, dimension, and load restraint requirements.
- Can withstand high-speed road transport over rough terrain.
- Can be safely loaded/unloaded from the transport vehicles.
- Minimises the risk of injury to all personnel involved in the freight transportation.
- Minimises the risk of damage to freight.
- Minimises the risk of damage to other freight and the general public.
- Has packing labels that specify the weight of the items and details of the recipient.
- Has ratchet-strap tie-downs over painted areas to minimise paint damage.
- Secures the loose items.

5.1.2 Chain of Responsibility

Under the Chain of Responsibility (COR) regulations, all parties with some control in the supply chain have legal responsibilities to make sure that the compliance with relevant heavy vehicle road laws, including mass, dimension, and load restraint laws. This includes people involved in consigning, loading, packing, and receiving freight as well as drivers of those vehicles (for example, in relation to speeding and fatigue management).

5.1.3 Load Restraint

The systems that secure the feeder must withstand the forces associated with the following four deceleration/acceleration, applied separately.

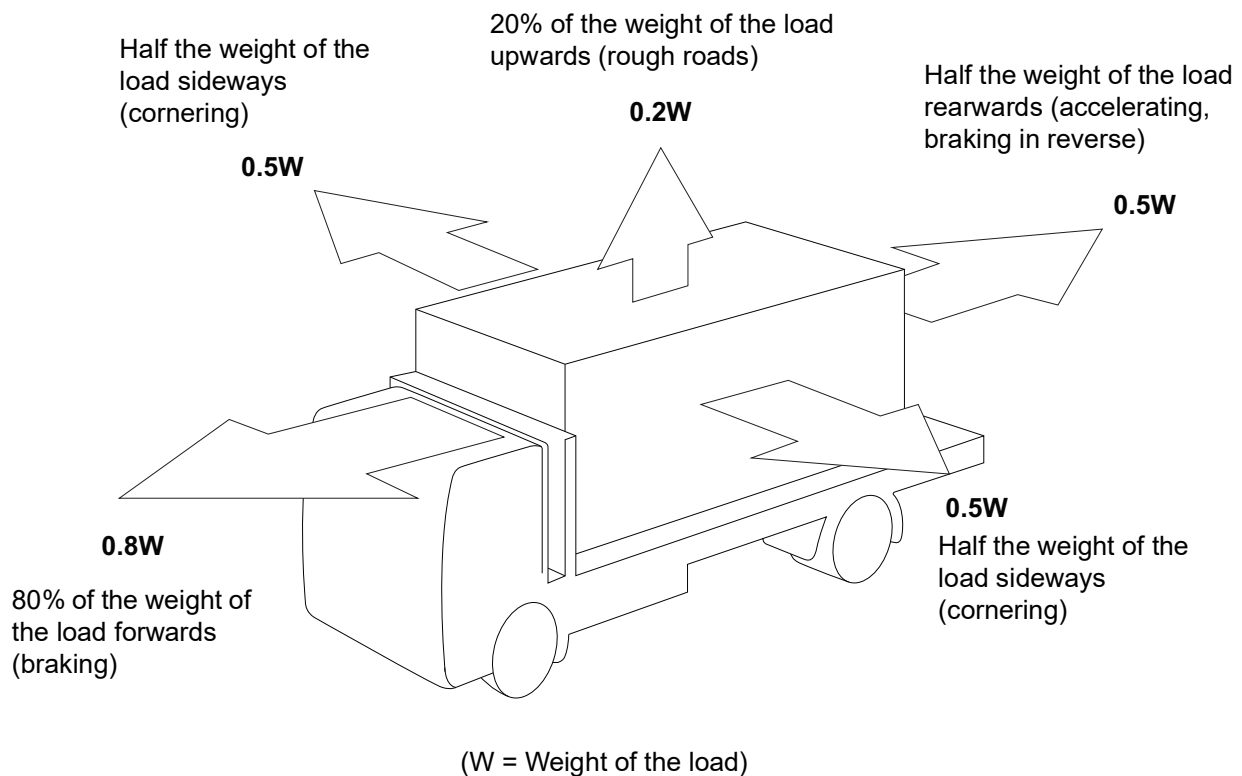


Figure 5-1: Load Restraint

Key requirements for load security are:

1. Make sure that the load platform, bodywork, and any load securing equipment are in a serviceable condition before loading the vehicle.
2. Secure the cargo so that it cannot move, roll-over, wander because of vibrations, fall off the vehicle, or make the vehicle tip over.
3. Determine the securing method(s) best adapted to the characteristics of the cargo (locking, blocking, direct lashing, top-over lashing, or a combination of these).
4. Make sure that the vehicle and blocking equipment manufacturers' recommendations are adhered to.
5. Make sure that the cargo securing equipment is commensurate with the constraints it encounters during the journey. Make sure that the securing equipment can withstand the following conditions, which are considered as normal circumstances that may occur during the journey:
 - Emergency braking
 - Strong cornering to avoid an obstacle
 - Bad road or weather conditions
6. Inspect the cargo and check for overload and/or poorly balanced weight distribution before starting when cargo is loaded, unloaded, or redistributed each time. Make sure that the cargo is distributed in such a way that the centre of gravity of the total cargo lies as close as possible to the longitudinal axis and is kept as low as possible (heavier goods below, lighter goods above).
7. Use equipment which supports the cargo securing such as friction mats, walking boards, straps, edge beams, etc. wherever possible.
8. Make sure that the securing arrangements do not damage the goods transported.

5.2 Lifting Instructions



DANGER

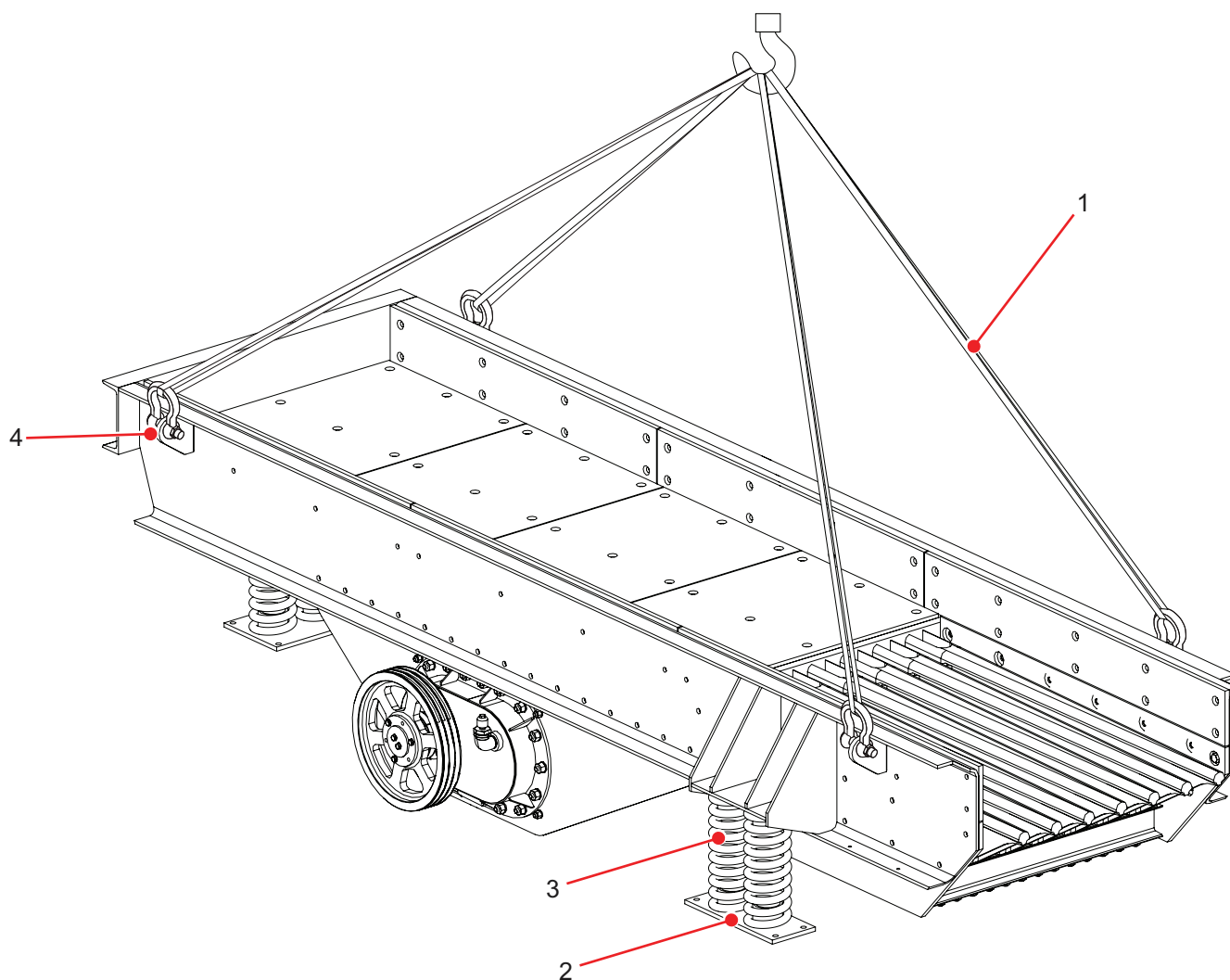


LIFTING EQUIPMENT SAFETY

- LIFTING EQUIPMENT MAY ONLY BE OPERATED OR MAINTAINED BY TRAINED PERSONNEL, WHO ARE CERTIFIED TO DO SO SAFELY.
- ALWAYS CREATE A LIFTING PLAN BEFORE LIFTING IS PERFORMED.
- ONLY USE RATED LIFTING EQUIPMENT TO LIFT HEAVY OR UNEVEN COMPONENTS.
- THE LIFTING EQUIPMENT MUST BE IN GOOD CONDITION, CERTIFIED AND TAGGED.
- THE LIFTING EQUIPMENT MUST BE OF ADEQUATE CAPACITY AND MUST BE USED WHENEVER THEY ARE REQUIRED.
- PERSONNEL MUST NEVER WORK UNDER SUSPENDED LOADS.

For the lifting precautions, refer to “Installation Safety” on page 2-13.

5.3 Lifting Equipment



1	Lifting Slings	3	Springs
2	Spring Pads	4	Lifting Lugs

Figure 5-2: Rigging Configuration

**DANGER****RISK OF SERIOUS INJURY AND DAMAGE TO THE FEEDER**

- LIFT ALL FEEDER COMPONENTS WITH SLOW AND EVEN MOVEMENTS TO PREVENT DAMAGE TO THE FEEDER SIDE PLATES.
- SUITABLE LIFTING EQUIPMENT MUST BE USED TO LIFT THE FEEDER USING FOUR LIFTING LUGS PROVIDED ON THE FEEDER. THE CHAINS/SLINGS MUST HAVE AN ANGLE AT LEAST 45 DEGREES TO THE HORIZONTAL.
- BELOW 45 DEGREE ANGLE CAUSE TOO MUCH STRESS ON THE FEEDER SIDE PLATES. IF AVAILABLE, USE A PROTRACTOR TO MEASURE THE ANGLE OF THE LIFTING CHAINS / SLINGS.
- MAKE SURE THAT COIL SPRINGS ARE ALL LOADED EQUALLY AND VERTICALLY. INCORRECTLY LOADED COIL SPRINGS CAN MISALIGN, CAUSING SERIOUS INJURY OR DEATH.

- [Figure 5-2 on page 5-3](#) shows the recommended rigging configuration.
- The feeder mass is indicated on the GA drawing or the nameplate.
- Please follow the site safety procedure for the lifting instructions.
- Set the lifting cables or chains in the correct position. Before lowering the feeder, make sure that the slope of the feeder matches with the slope of the support structure.

5.4 Recommended Storage Procedure for Equipment

All items supplied that are not fixed to the feeder are crated separately and labelled accordingly for their protection.

5.4.1 Storage of Motors

Motors should be stored indoors in a dry room that should conform to the following requirements:

- Clean and dry atmosphere.
- Vibrations free, to prevent damaging the bearing.
- Ambient temperature may not exceed 40 °C (104 °F).
- No high fluctuations of temperature in a short time as condensation could corrode the motor.
- When the fluctuations of temperature are higher than 20 °C (68 °F). Storage in an accustomed space is recommended.
- After long storage periods it is necessary to re-lubricate the motors after 100 operating hours.

5.4.2 Storage of Feeders

It is often necessary to store the feeder outside (12 months or less) while awaiting installation. If the feeder has not been prepared for storage prior to shipment, certain steps are necessary to avoid potential problems during and after installation and start-up.

- The feeder should be stored as level as possible and supported in the same manner as received on the carrier.
- Do not remove eccentric shafts from their containers. Store indoors if possible.
- The driven sheave should be rotated every 30 days. Store it in a place, protected from the weather. Keep coupling ends coated with a rust preventative
- All electrical and miscellaneous items should be wrapped in protective coverings, then boxed and stored indoors.
- Repair Immediately if there is any indication of rust or corrosion and any structural damage to any surfaces by repainting, welding or replacing parts.
- Do not store any other equipment on the feeder pan liners and grizzly bars.
- Apply rust preventative to all exposed machined surfaces and unpainted threads.
- The rubber surfaces of the components can be coated with a protective coating of transparent Neoprene based paint or another approved protective material.
- Cover the feeder in black plastic from direct sun and ozone attack. Rotate the assembly by hand to re-lubricate the bearing rollers.

5.4.3 Storage of Vibrator

When vibrators in service and it remain idle for a longer period more than two weeks should be given for storage treatment to prevent rusting of internal metal parts and drying of seals. Rust preventive oils such as MOBIL-KOTE 501 OR 503 (SOCONY MOBIL) is recommended because the parts need not be cleaned of this compound before placing the unit in operation. Drain the excess and refill to the recommended oil level with your operating lubricant.

If the vibrator is to be idle for an extended period and power is available to drive it while it is attached to the feeder, do the following before shutting down:

- Drain the operating oil and refill to the proper oil level with MOBILARMA 524 (or equivalent).
- It is recommended that the machine is marked to indicate that it is filled with long term storage oil to avoid future running in this condition.
- Run the unit for one hour before shutting down.
- After shut-down, coat the exposed portion of the oil flinger sleeve on the drive shaft with grease.
- Keep the rust preventive in the vibrator during the idle period and put the unit in operation every three months for one hour.
- When operations are resumed, drain the rust preventive and refill with the proper oil.

If power is unavailable or the vibrator cannot be put in operation, because it is not attached to the feeder (spare vibrator), do the following:

- Remove the end covers and spray all the internal parts with MOBILARMA 524 (or equivalent). Make certain the bearings and gears are thoroughly coated. On the gear side, it will be necessary to back off the gears to provide access to the bearings.
- An alternate method is to fill the vibrator oil compartment(s) completely with the rust preventive oil, then drain. Where multiple units are involved, this same oil can be re-used for the storage treatment on several machines.
- Coat the exposed portion of the oil flinger sleeve on the drive shaft with grease.
- Repeat this procedure at least once every three months.
- Before placing in operation, drain the excess of rust preventive and refill with proper oil.

5.4.4 Storage After Operation

Use the same instructions for the assembly, as stated in the above, with the following procedures:

1. Lockout and tag out the machine.
2. Relieve the tension on the V-belts by blocking the pivot motor base up. Place a paper between the belts and sheaves. This will prevent sticking.

6 Installation

This section describes the installation procedures of the feeder.

6.1 Installation Safety

During installation, ensure all safety warnings and instructions are followed, as mentioned in “Safety” on page 2-1.

6.2 Planning

- A service platform should be provided on both sides of the feeder for observation and servicing, if possible.
- To perform maintenance activity, provide provision for lifting device and removal/installation of the shaft on one side of the feeder.
- Allow room for grizzly bar removal, which generally can be provided at the discharge end by removing discharge chutes, if possible.

6.3 General Information

It is recommended that Weir Trio personnel be present when the installation takes place. All nuts must be of the Cleavelock type, Grade 10. Cleveloc nuts must not be re-used. All fasteners must be high tensile steel Grade 8.8. Rigging and handling of the feeder must be done by qualified personnel.

6.4 Feeder Receiving and Inspection

When the feeder is delivered, thoroughly check for damage that might have occurred during transport.

	i NOTICE
	INSTALLATION INSPECTION Inspect the feeder before proceeding with the installation.

Do the following during inspection:

- Note any damage on the shipper's bill of lading and have the deliverer sign your description of any problem(s).
- Keep a copy of the marked-up bill of lading for your files.
- Take photos if possible.
- File a claim with the carrier within 30 days of receipt of the equipment. Notify your dealer or Weir immediate upon receipt of damaged goods.
- Check loose parts, small parts boxes, and tools against the packing list to make sure that all items shipped are present.

6.5 Feeder Installation Requirements

When installing the feeder, be sure that the beams upon which it is mounted are level and in line. Provide sufficient rigidity to support the weight of the unit and its load and also to absorb the residual vibration from the live frame.

If feeder is installed on a plant, make sure the feeder is properly mounted to the structure.

	i NOTICE
	TORQUE SPECIFICATIONS Before tightening the fasteners, refer to “Torque Specifications” on page 15-1.

6.6 Suspension Components

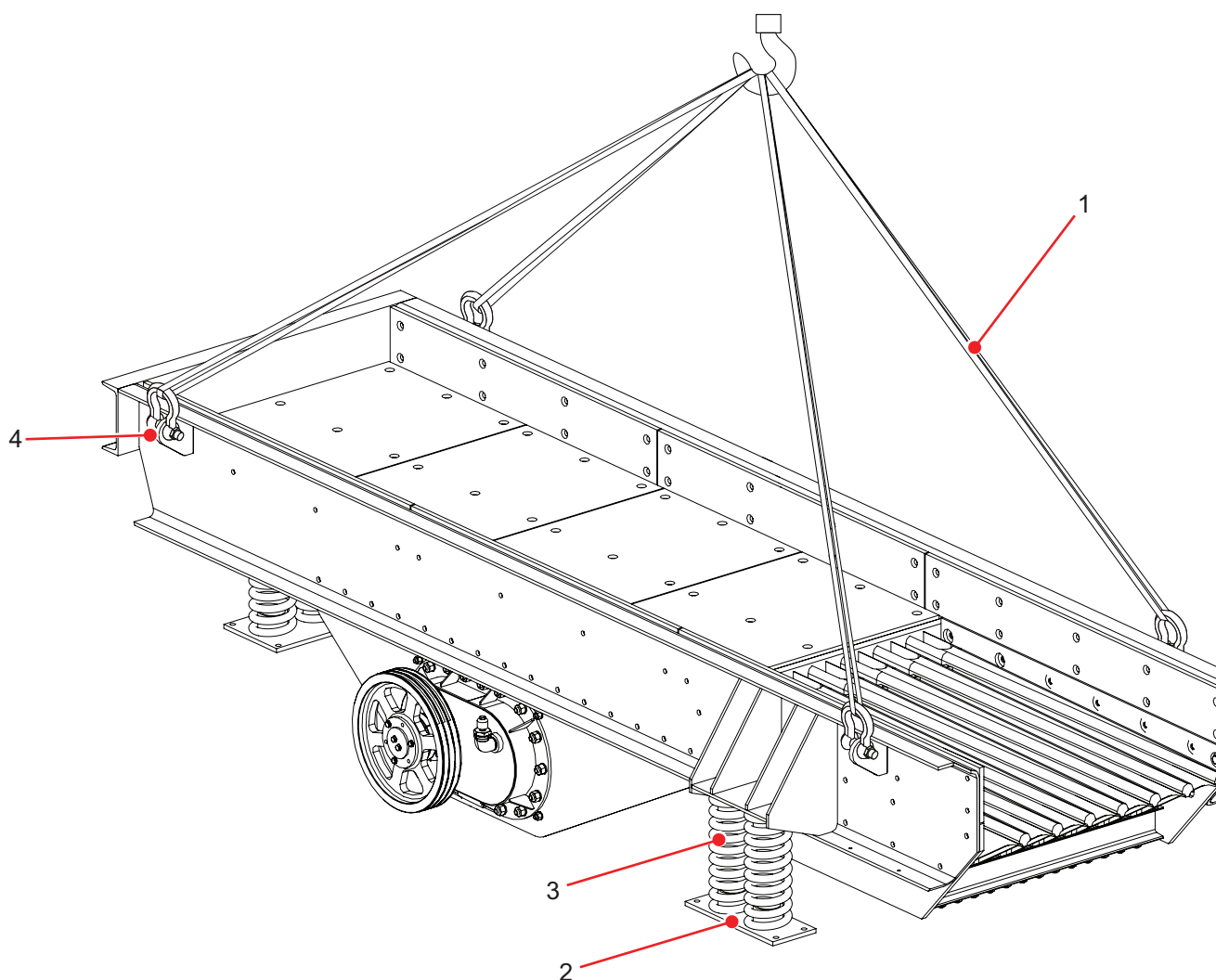
The suspension system (coil springs) are precisely matched to the machine. Broken springs must be replaced immediately. Details of the springs are given in the feeder GA drawing.

6.7 Lifting and Placing the Feeder

When installing the feeder, make sure that the beams on which it is mounted or from which it is suspended are level and in line. Provide sufficient rigidity to support the weight of the unit and its load and also to absorb the vibration residue from the live frame.

Make sure to provide space around the unit so that it may be inspected easily and maintenance operations facilitated.

- Lift the feeder horizontally above the suspension components.
- With a person at each support bracket guiding the feeder into the suspension components, lower the feeder slowly onto the suspension components.
- Check that the recommended minimum 75 mm (3 in) clearances between the feeder and surrounding steelwork are available.
- Feed chutes and hopper outer vertical sides should be within 12 mm (1/2 in) of the inner vertical sides of the feeder pan (or liner). This will help prevent material spillage or coarse material to wedge between the vibrating and stationary parts during operation.
- In order to accurately determine machine clearances, refer to the dimensions of your particular feeder.
- All structures are to be designed, verified and signed off by a qualified professionals.



1	Lifting Slings	3	Springs
2	Spring Pads	4	Lifting Lugs

Figure 6-1: Lifting Lugs

6.7.1 Initial Set-Up

1. Place steel coil springs onto spring pads.
2. Lift the mount feeder by carefully and place onto springs.



WARNING

LEVELING OF FEEDER

Follow proper rigging procedures to ensure the safety of personnel and prevent damage to equipment. Do not work under the feeder while it hangs from lifting device(s) support feeder with blocking or other fail-safe methods before going underneath the feeder.

3. Check to make sure all springs are vertical within plus or minus 3 mm (1/8 in.) when feeder assembly is mounted. If not, make necessary adjustments to spring mounting pads, base, or supporting structure to ensure this. See [Figure 6-2](#).

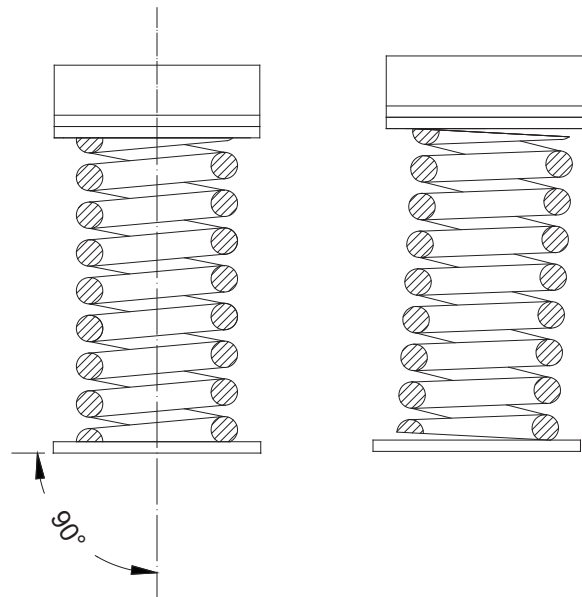


Figure 6-2: Feeder Springs Vertical Position



NOTICE

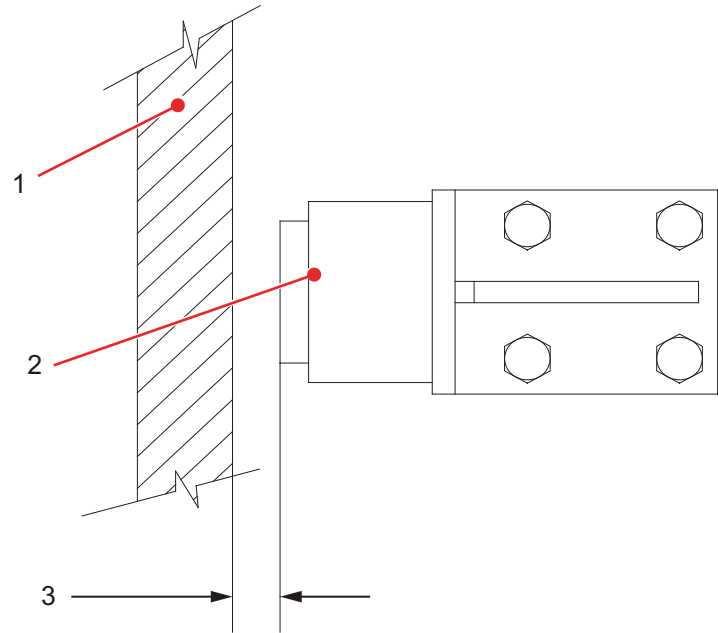
LEVELING

Feeder basket must be level both front-to-back and crosswise. Keeping unit level and stable is critical for optimum output, proper bearing lubrication, and minimum stress on all components.

4. Once the feeder is level and secured to supporting structure.

6.8 Snubber Assembly

Snubber Assembly option applicable for 20 ft and above length feeder models. The snubber assembly is mounted on the feeder frame and push against solid steel structure. Refer to [Figure 4-1 on page 4-3](#) snubber location on feeder. Snubber clearance value is shown in the [Figure 6-3](#), adjust it as per value.



1	Supporting Structure	3	25 mm to 30 mm (1 In to 1-3/16 In)
2	Snubber (Mounted on VGF)		

Figure 6-3: Snubber Clearance

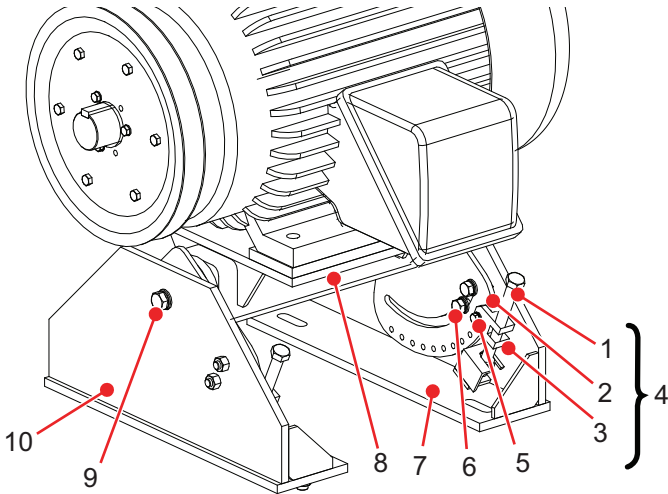
6.9 Shipping Braces and /or Blocking

Feeder’s components are equipped with shipping braces or some type of blocking to prevent damage during transport and shipping. Identify and remove all these braces and/or blocks and store them for future use.

6.10 Feeder Drive

The TRIO VGF feeders are equipped with adjustable motor base tensioner drives.

6.10.1 Installation Guidelines for Adjustable Motor Bases



1	Jacking Bolt	6	Hex. Screw
2	Friction Plate	7	Side Support Left
3	Adjusting Block	8	Base Plate
4	Pre-tensioning Device	9	Hex. Bolt
5	Hex. Set Bolt	10	Side Support Right

Figure 6-4: Adjustable Motor-base

i NOTICE

LOCATION OF GLIDE BEARING

The glide bearing has to be on the same side as the drive pulley.

6.10.2 Positioning of the Drive Motor on Adjustable Motor-base

The ideal angular positioning of the drive motor on the base is an important factor in utilizing the full compensation-arc of the pre-tensioned adjustable base element. The resulting lever arm between the element center axis and motor shaft shall be sufficiently dimensioned in order to compensate for the maximum possible belt elongation without additional re-tensioning of the base. The two sketches below are presenting the ideal positioning of the motor on the elastic base (ideal position blue marked). Each motor plate (position 2) is equipped with threaded bores for its “Centre” and/or “Off-set” mounting.

6.10.2.1 Centre Mounting

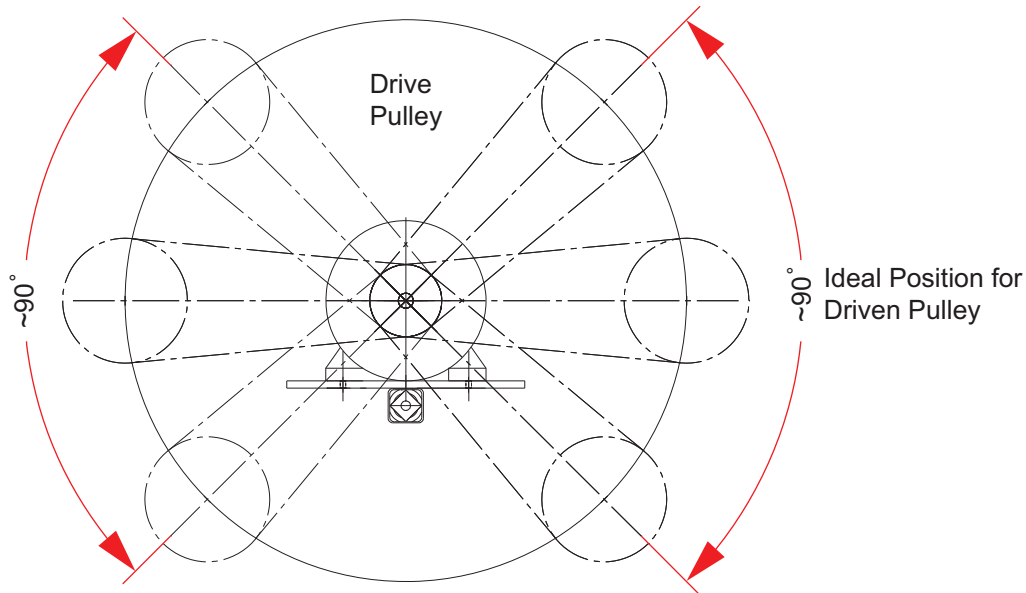


Figure 6-5: Centre Mounting

If the driven pulley is laterally positioned to the drive pulley (and/or ± 45 degrees below or above the center line), then the “Center” mounting of the plate with motor is recommended.

6.10.2.2 Off-set Mounting

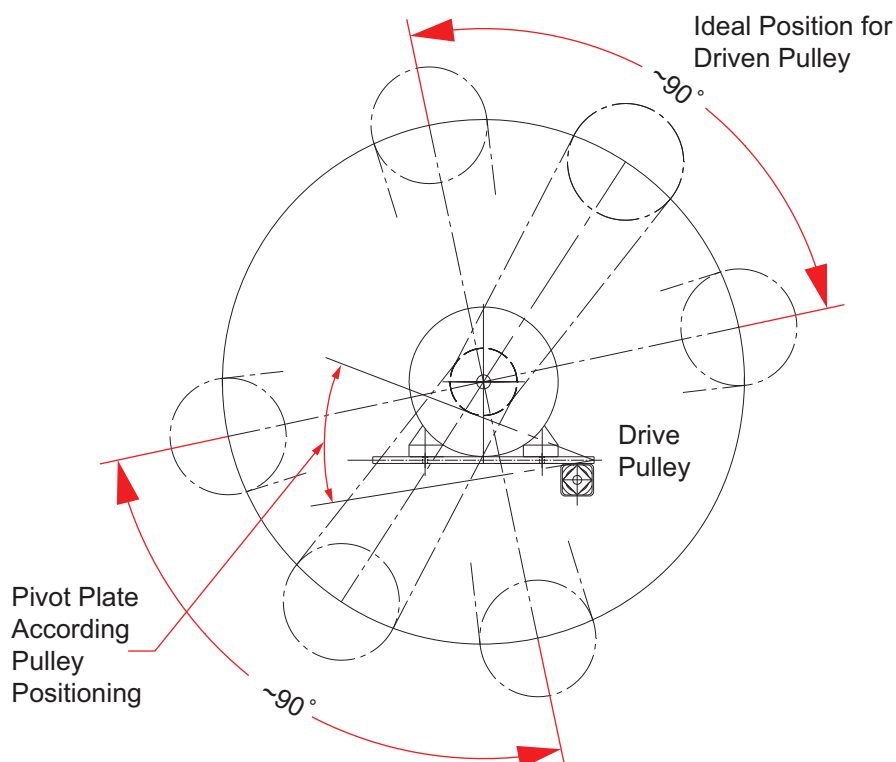


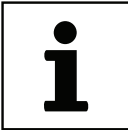
Figure 6-6: Off-set Mounting

If the driven pulley is above or below the drive pulley (see working angles on the sketch), then the “Off-set” mounting of the plate and motor is recommended. In addition, find the ideal plate inclination towards driven pulley in adjusting block (7) fixation on friction plate (6). Refer to [Figure 6-4 on page 6-5](#).

6.10.3 Motor-base Adjustable Fitting Instructions

It is important to note that on feeders the motor mount be positioned forward of the driven sheave towards the grizzly bars and near a 45° angle below the drive pulley. This will give the user a 90° angle between the center of the drive, center of the driven sheave and the center of the torsion tube on the mount.

The motor base arrangement is shown in [Figure 6-4 on page 6-5](#).

	i NOTICE
	SUPPLIER PART Below instruction is of external supplier part use it for reference for further information refer to respective supplier manual.

	! CAUTION
	EQUIPMENT DAMAGE Place the motor on the base plate only after bolting the adjusting block to friction plate.

	i NOTICE
	TORQUE INFORMATION For the mount bolt torque information, refer to Rotor Mount Supplier manual for different bolt sizes.

1. Bolt side supports and of motor-base on the machine part.
2. Check if the alignment of the base plate corresponds to the optimum working position, otherwise align with jacking bolt or select a different tensioning position on friction plate.
3. Bolt the motor to the base plate. The drive pulley must be placed on the glide bearing side of the motor-base.
4. Turn jacking bolt clockwise to raise the motor and place the belt on the drive pulley, tightening with torque.
5. Turn jacking bolt counter-clockwise to tension the belt. Check the belt tension with the data specified by the belt manufacturer.
6. Tighten all locking bolts of the friction plate after the tensioning process, tightening with a torque of nuts.
7. Tighten nuts for central hex. Shaft, tightening with torque.

If necessary, the adjusting block with jacking bolt could then be removed by loosening the hex set bolt (as protection from possible corrosion and dirt).

6.10.4 Belt Installation

Provide sufficient space around the belt drives for installation and maintenance activities. Do not force a belt over pulley flanges; this may result in internal belt damage.

Pulley Alignment

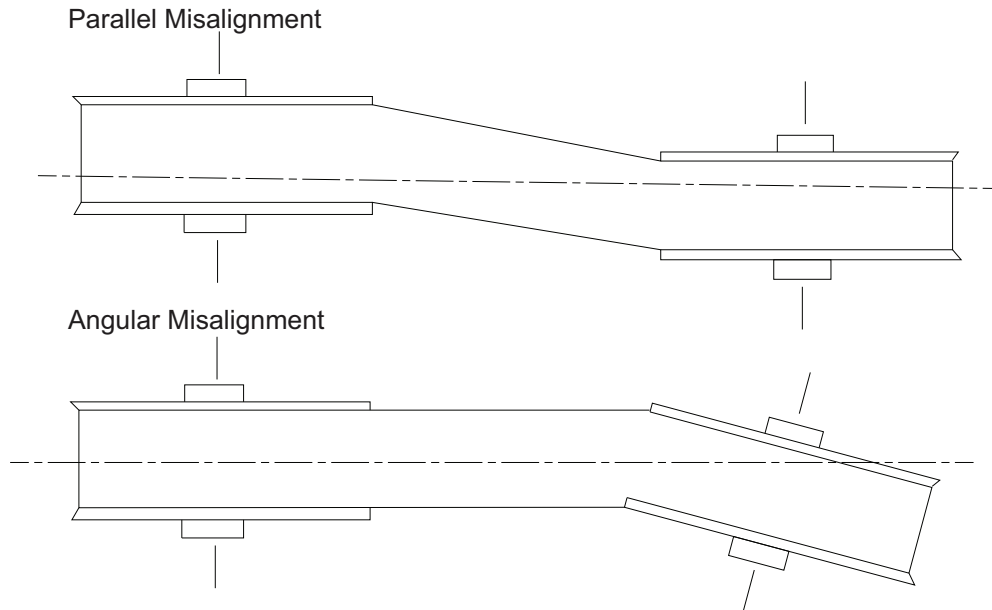


Figure 6-7: Misalignments

Misalignment of drive pulleys results in unequal tension across the belt width and extreme edge wear. Align the pulley alignment using a straight edge and check the shafts for parallelism. Misalignment on any synchronous drive should not exceed $\pm$ angular 5 mm/meter center distance parallel.

The drive support framework should be rigid. A flexible frame allows small variations in center distance, which affect belt tension. This can result in tooth jumping during high torque starts.

6.10.5 Drive Motor Rotation

Before installing the drive belt, check to make sure the drive motor is rotating in the proper direction. Since the feeder can have the drive assembly on either the right or the left side, it is important to check the motor for proper rotation. The motor should rotate so the belts are tight on the bottom side, having the belts tight on the top side will result in V-belts coming off of the feeder when loaded.

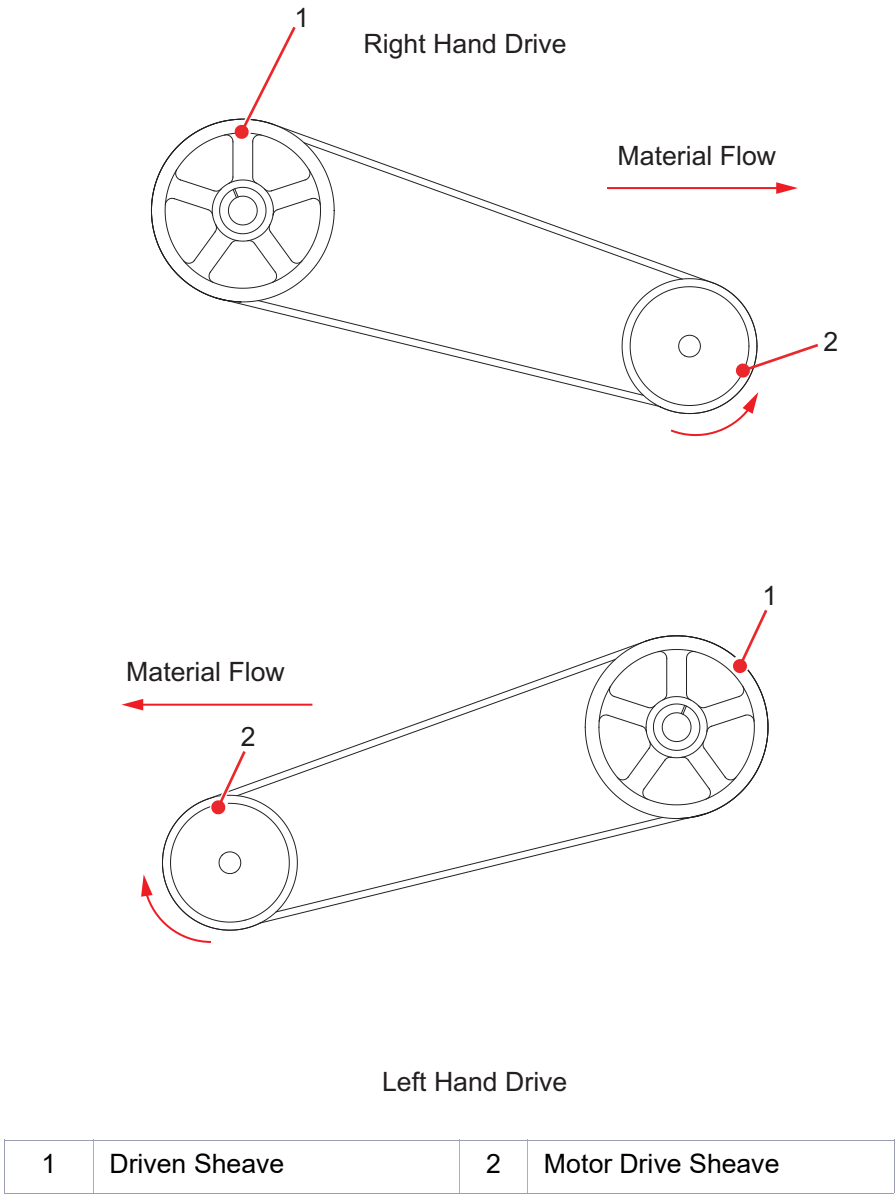


Figure 6-8: Right Hand Drive & Left Hand Drive Motor Rotation

6.10.6 Standard Drive Assembly

The standard drive assembly for the feeder is shown in Figure 6-9. To adjust the drive belt tension to maintain the minimum of 4.2 mm (1/6 in) of belt deflection:

- 1. Loosen the drive motor mounting bolts.
- 2. Slide the motor toward driven sheave enough to allow the belt to be installed without prying or stretching.
- 3. Slide the motor away from driven sheave until the tension is achieved to the 4.2 mm(1/6 in) of belt deflection.
- 4. Re-tighten the motor mounting bolts.
- 5. Make sure motor and feeder shafts are parallel and run true with the centerline of belt travel as shown in Figure 6-9.

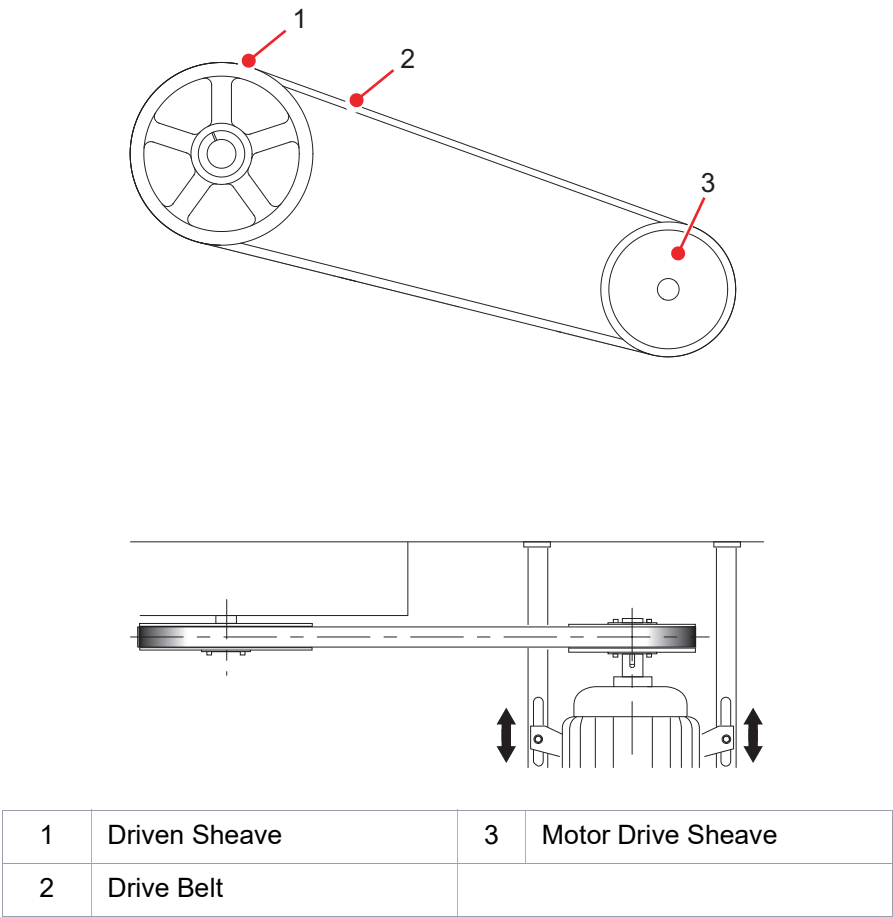
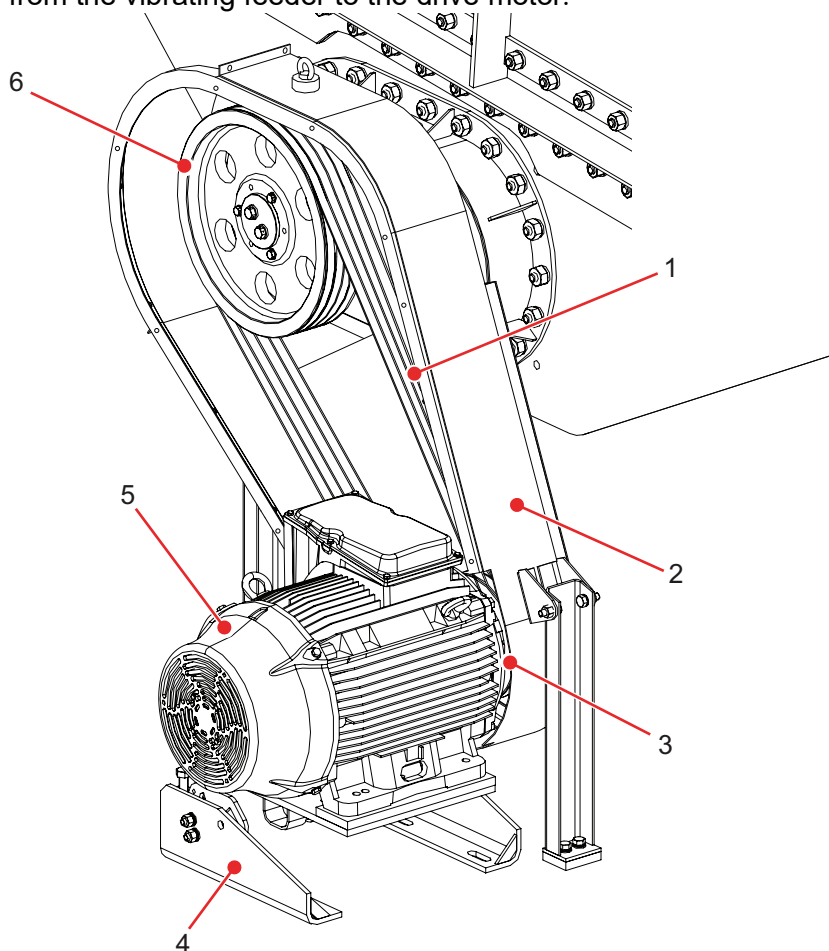


Figure 6-9: Standard Drive Assembly

6.10.7 V-belt Installation/Replacement

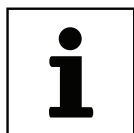
A V-belt drive system drives the Weir Trio vibrating grizzly feeders. The V-belt drive system prevents shock loads being transmitted from the vibrating feeder to the drive motor.



1	V-Belt	4	Motor Mounting Base
2	Drive Guard Main Cover	5	Motor
3	Motor Pulley	6	Driven Sheave

Figure 6-10: V - Belt Installation

Before working on any V-belt components always ensure:



i NOTICE

APPLY LOTO

- Isolate the machine and ensure that all rotating parts are locked.
- Apply LOTO before V-belt replacement.

1. Remove the drive guards that need to be removed for access to the V-belt using the correct tools. Refer to [“Drive Guard Installation/Replacement and Checks” on page 6-17](#) for drive guard replacement.
2. Clean the pulleys (motor pulley & driven sheave) to ensure that there is no debris, oil, grease, or rust on the pulley grooves. Replace pulleys with new ones, if they are damaged or worn. Refer to [“Driven Sheave Removal” on page 9-7](#) for removal and [Figure 8-15 on page 8-11](#) for installation.
3. Reduce the center distance between the drive and driven pulley by jacking the motor mounting base (where possible) or by adjusting the drive base to allow for easy removal or fitment of the V-belt. Refer to [“Motor-base Adjustable Fitting Instructions” on page 6-7](#) for more details.

4. Align the pulleys and ensure that the shafts are parallel.

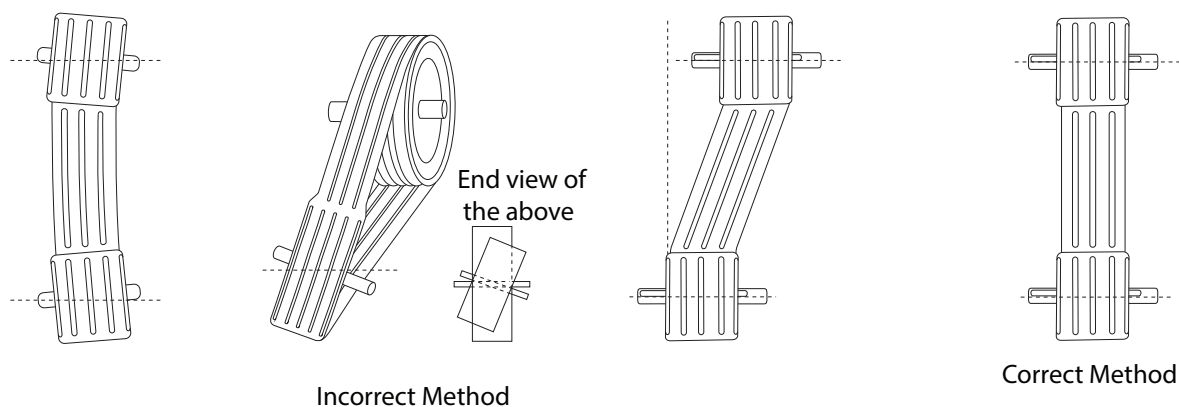


Figure 6-11: Pulley Alignment

5. Without using any force, place the V-belt around the pulleys (start by placing the V-belt on the drive pulley) and release the motor mounting base so that the belt is tensioned.

- Do not mix old and new belts.
- Do not mix belt from different manufacturers.
- Use belt of the same construction and rating.

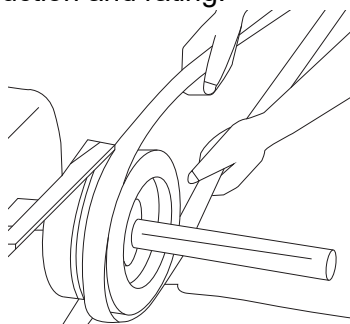
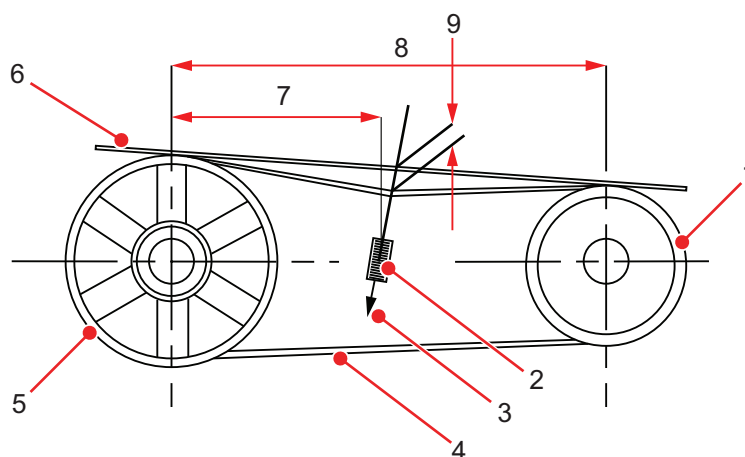


Figure 6-12: Fitting of a V-belt on a Drive Pulley

6. Use a belt tension indicator to set the V-belt to the maximum required tension value stated in the tension force table, refer to "[Table 6-1: V-belt Tension Force Requirements](#)".



1	Motor Driven Sheave	6	Straightedge
2	Spring Scale	7	1/2 Span Length
3	Force Measurement	8	Span Length
4	V-belt	9	Deflection Measurement - 1/6 Inch (4.2 mm) for each Inch of Span Length
5	Driven Sheave		

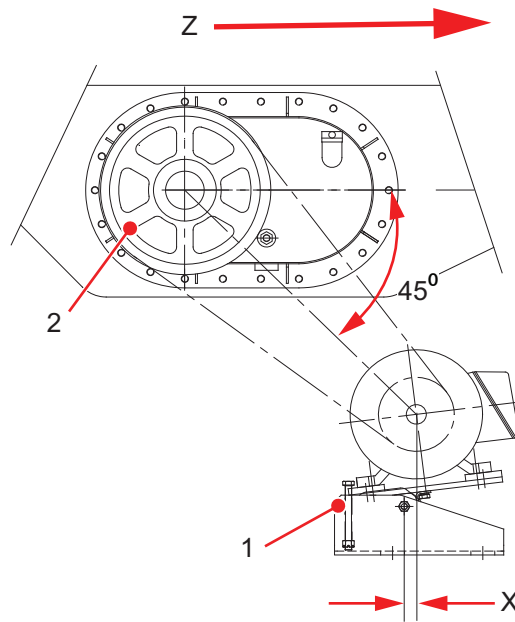
Figure 6-13: Tension-Deflection Diagram

Tensioning Forces				Take Up Allowance		
Belt Section	Force required to deflect belt 16mm per meter of span			Installation Allowance (mm)	Belt Pitch Length (mm)	Take Up Allowance (mm)
	Small Pulley Diam. (mm)	Newton (N)	Kilogram force (Kgf)			
22 X 14 (C)	200-400	40-60	4.5-6.5	50	8410-10670	145

Table 6-1: V-belt Tension Force Requirements

7. Put the guards back in place once the installation/replacement of the V-belt is done. The drive system should not be run without drive guards being in place. Refer to [“Drive Guard Installation/Replacement and Checks” on page 6-17](#) for drive guards installation.
8. Perform pre start-up checks
9. Run the drive system under load for 30 minutes. Stop the drive, check the belt tension, and reset it to the maximum required tension force.
10. After 10-14 days of continuous running, check the belt tension again and reset it to the maximum required value. If the drive is installed correctly, no further tensioning of the belt will be required.

6.10.8 Adjustable Motor Base Pre-tensioning Device



1	Pre-tensioning Device	X	Off-Set Mounting
2	Driven Sheave	Z	Material Flow Direction

Figure 6-14: Pre-tensioning Device Location

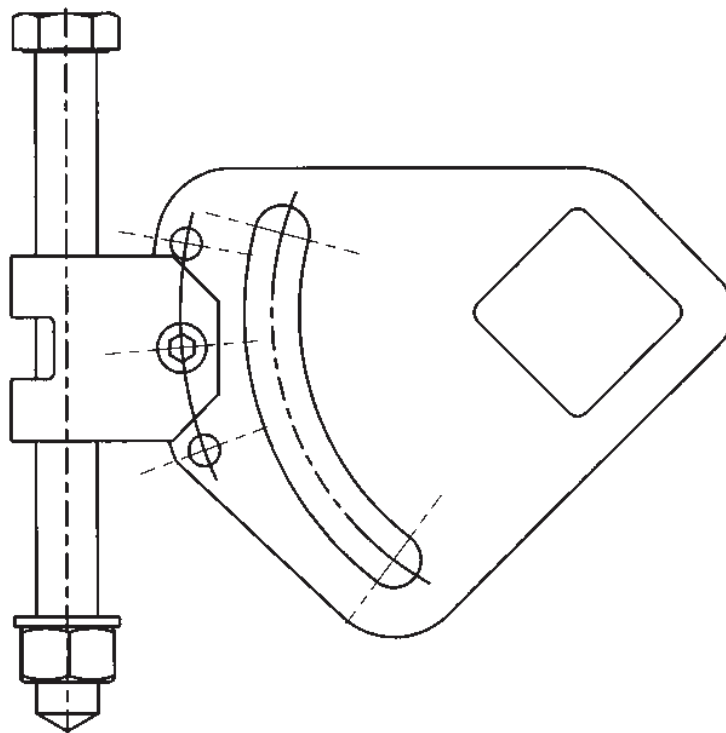


Figure 6-15: Pre-tensioning Device

The pre-tensioning device of the adjustable-motor-base affects the motion of rotation on the personal profile of the adjustable motor base. This rotation gives the required pre-tension effect on the belt and does compensate for their elongation.

Turning the threaded spindle increases or decreases the belt tension. On the friction plates of both pre-tensioning devices, there are several positions for fixing the adjusting block with the spindle.

By varying these positions, the change of plate-inclination can be effected to get the most compensation effective working angle between the drive pulley and driven pulley.

The motor bases are typically equipped with a cardanic glide bearing between the tube and the side of the motor mount. This bearing compensates the radial tensile forces of the belt and keeps the two components in a parallel position, even at extremely high tension forces.

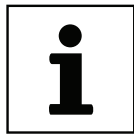
	i NOTICE
	MOTOR BASE ADJUSTMENT This fiber-reinforced plastic bearing has to be positioned on the pulley side of the belt drive. Check this by assembling the motor-base. If this bearing would be installed on the fan-side of the motor, a misalignment of the pulleys occurs due to the lack of parallelism.

6.10.8.1 Re-tensioning

Adjustable motor-bases are automatically re-tensioning drive mountings for friction belt drives. Regular re-tensioning is, therefore, unnecessary. It is only recommended in the case of very long center distances between driving and driven pulley to check the test pressure on the belt and re-tension occasionally if necessary.

6.10.9 Motor Pulley Installation

Inspect pulley grooves for wear using pulley gauges and worn pulleys should be replaced with new motor pulleys.

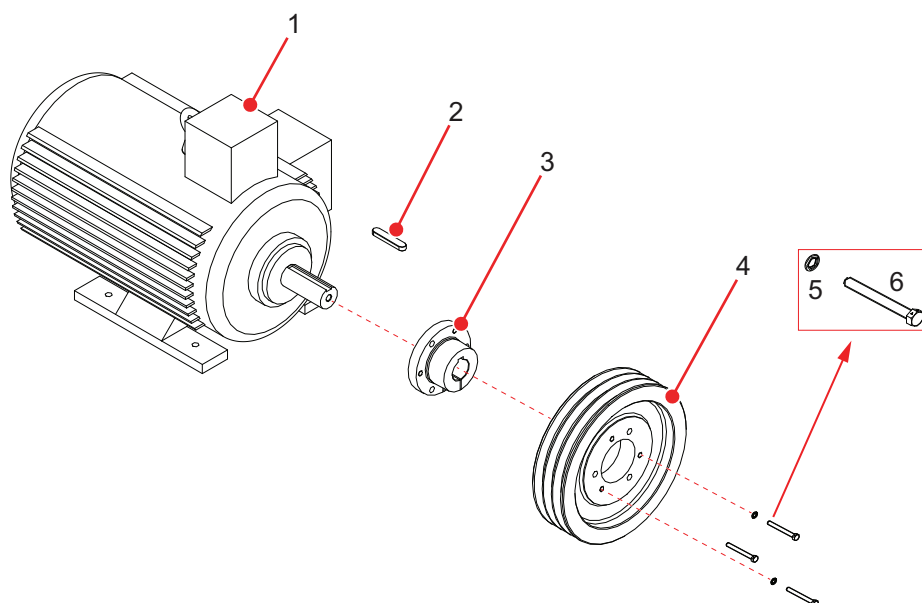


i NOTICE

LOTO

- Isolate the machine and ensure that all rotating parts are locked.
- Apply LOTO before motor pulley installation.

1. Insert a key in the motor shaft keyway.
2. Install the motor bushing to the motor shaft assembly. As close to the motor, as allowed by the guards and pulley alignment.
3. Use an appropriate lifting device to lift the motor pulley.
4. Align and install the motor pulley on the motor bushing by tightening the fasteners.



1	Drive Motor (Not Supplied)	4	Motor Pulley
2	Key	5	Washer
3	Motor Bushing	6	Bolt

Figure 6-16: Motor Pulley Installation

5. Remove the lifting device from the motor pulley.

6.11 Guards Installation and Checks

6.11.1 Drive Guard Installation/Replacement and Checks

When guards are not installed, always ensure:

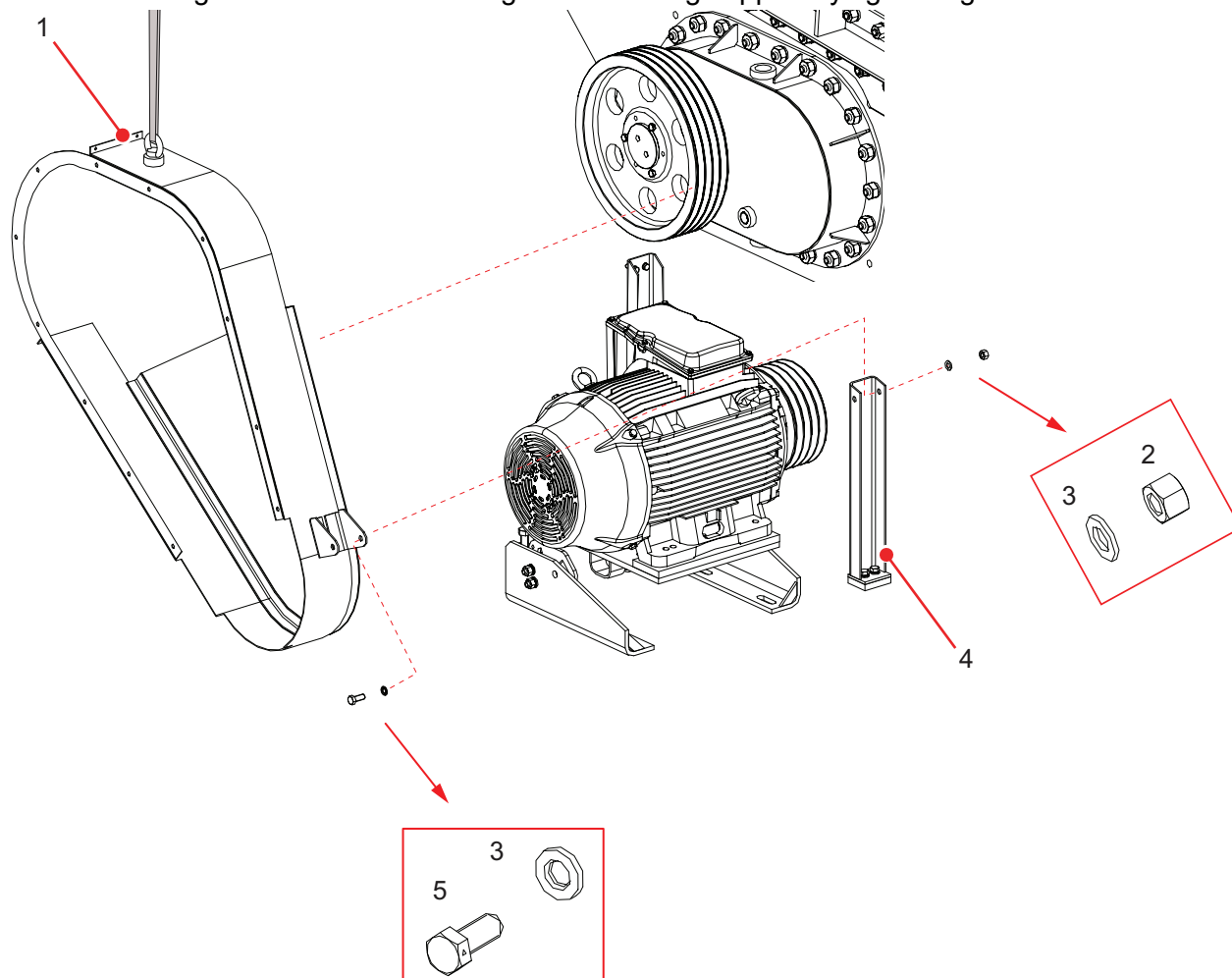


i NOTICE

APPLY LOTO

- Isolate the machine and ensure that all rotating parts are locked.
- Apply LOTO before drive guard installation.

1. Ensure the guard mounting channel is firmly fitted on the plant structure.
2. Use an appropriate lifting device to lift the drive guard main shell.
3. Install the drive guard main shell on the guard mounting support by tightening the fasteners.

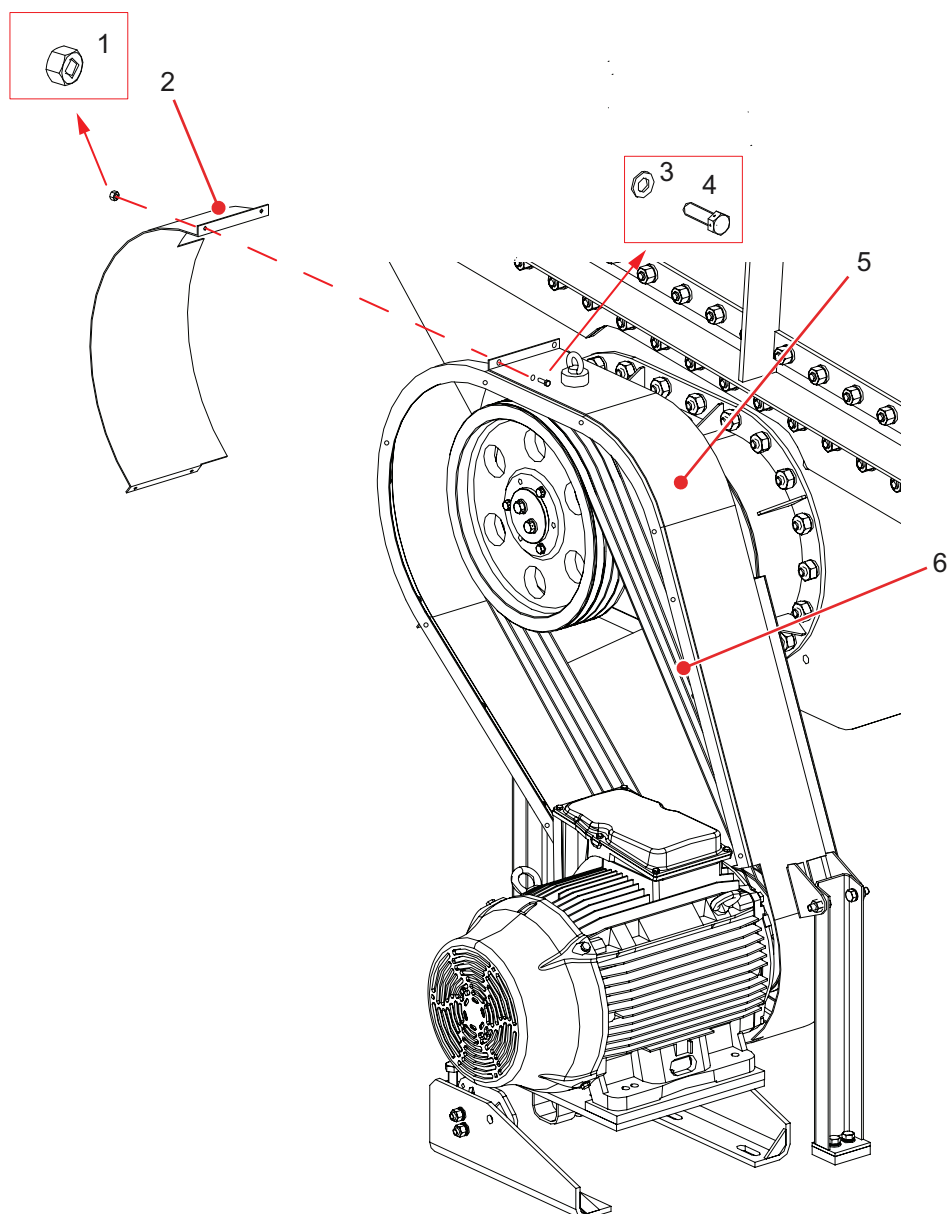


1	Drive Guard Main Shell	4	Guard Mounting Support
2	Nut	5	Bolt
3	Washer		

Figure 6-17: Drive Guard Main Shell

4. Remove the lifting device from the drive guard main shell.

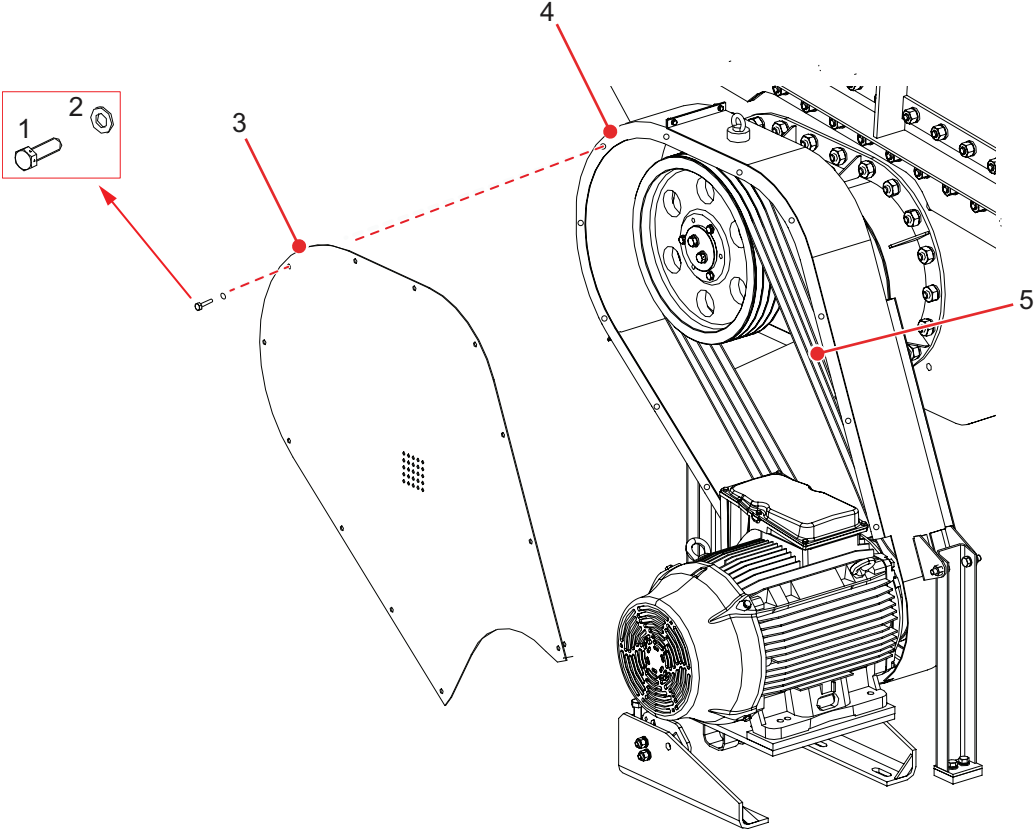
5. Install the V-belt, refer to “V-belt Installation/Replacement” on page 6-11 for V-belt installation.
6. Install the side enclosure to the drive guard main shell by tightening the fasteners.



1	Nut	4	Bolt
2	Side Enclosure	5	Drive Guard Main Shell
3	Washer	6	V-Belt

Figure 6-18: Side Enclosure Installation

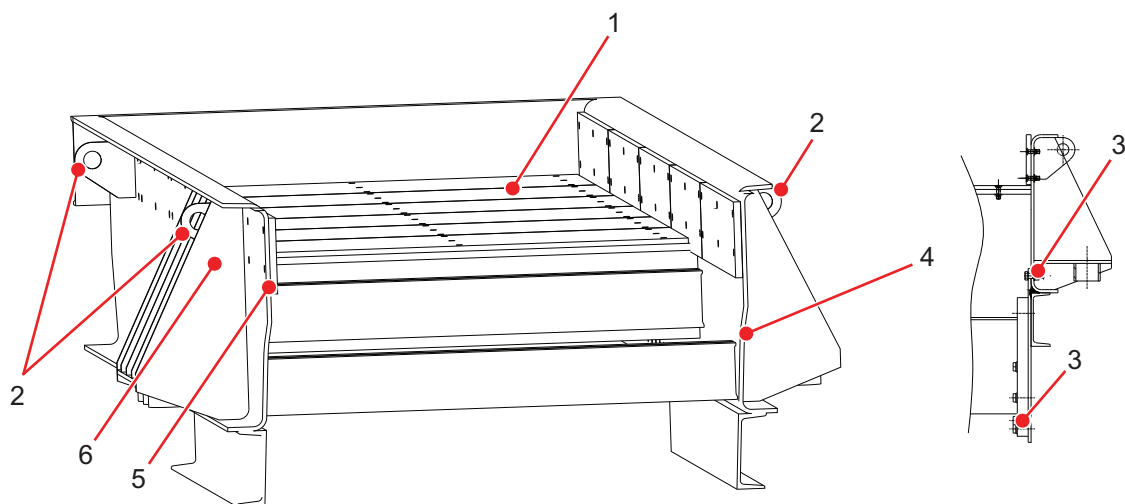
7. Install the drive guard main cover to the drive guard main shell by tightening the fasteners.



1	Bolt	4	Drive Guard Main Shell
2	Washer	5	V-Belt
3	Drive Guard Main Cover		

Figure 6-19: Drive Guard Main Cover Installation

6.12 Bolted Connections



1	Bottom Wear Plate	4	Feeder Frame
2	Lifting Lugs	5	Side Wear Plate
3	Bolts		

Figure 6-20: Typical Application of Bolt System

- All bolted connections on the frame must be checked after the first 24 hours operation, subsequently at monthly intervals. Refer to [Table 10-3 on page 10-4](#) for more details.
- Loose connections cause disturbing noises and damage to the feeder frame. Stop the feeder immediately, ascertain the cause, and fix it.
- Use a torque wrench to check and replace the bolts. [Table 6-2](#) gives the details of the torque ratings.

Bolt Size (Grade 10.9)	Torque (Nm)
M8	33.8
M10	60
M12	105
M14	168
M16	260
M20	507
M24	877
M30	1883

Table 6-2: Torque Ratings for Grade 10.9 Bolts

i NOTICE

TORQUING PERIOD

- All bolts should be re-torqued after a period of 24 hours when replaced.
- After the first weeks run, recheck the tightness of all bolted connections, refer to [“Torque Specifications” on page 15-1](#).
- Spot checks may be conducted at about 200-hour intervals following this initial re-tightening. Refer to for [“Gear Case Bolt Torque” on page 10-4](#) more details.

6.13 Feed Chutes

It is advantageous to have the outer vertical sides of the chute within 1/4 in (6 mm) - 1/2 in (12 mm) of the inner vertical sides of the feeder pan (or liner). This prevents the material spillage, as well as any tendency for coarse material to wedge between the vibrating pan and stationary chute during operation.

6.14 Operating Clearances

When the installation is complete, start and stop the machine checking clearances all around. Approximately 75 mm (3 inch) operating clearance should be allowed above and below the moving live frame and any stationary parts to prevent interference. This clearance accommodates the oval motion and stroke of the vibrating feeder basket as well as the additional motion during start-up and shut-down. Vertical clearance below the feeder is also very important. It prevents unnecessary stress and wears on feeder basket, discharge chutes, and feed box. Be sure to provide enough clearance for easy installation. Start and stop the machine several times to make sure that sufficient clearance exist.

6.15 Discharge Chutes

The chutes handling the oversized material must have a sufficient slope to prevent material hang-up. This is usually 60° from the horizontal. Enclosed chutes must have a minimum depth of four (4) times the maximum particle size. These could be hinged or easily lifted away.

6.16 Walkways

Walkways should be provided so your operators can observe the performance of the vibrating feeder. These walkways should be designed as work platforms to be used to perform routine maintenance, change feeder liners, grizzly bars, and to repair the assembly.

6.17 Check Fasteners for Tightness

Keeping all structural members tight is a very important part of all vibrating equipment since the forces generated will tend to loosen parts leading to eventual failure.

Due to the high initial tightening torque of fasteners used on-feeder, heat-treated bolts and lock nuts will not loosen as readily as ordinary ones, but their tightness should occasionally be checked using a torque wrench. If no torque wrench is available, correct tightness may be estimated by utilizing the turn of the nut method.

The steel structure, where the feeder is installed, requires maintenance. Please contact structure manufacturer for the maintenance process of the steel structure and follow the process accordingly.

6.18 Abrasion Protection

Grizzly bar members should be checked on a weekly basis for material accumulation. Any particles left in this space for a lengthy period of time will, under vibrating conditions of the machine, gradually abrade the member resulting in premature failure.

Regularly inspect the liners, if found defective, replace it with the same material.

Feeder might be supplied with an optional wear liner. Wear liners are provided according to specific feeder type/series. They are also provided according to the customer's special requirements.

The side plates of the vibrating feeder are protected against abrasion from the feed material by side wear liners. These liners run down inside both side plates of the vibrating feeder and held in place by countersunk bolts. The side wear liners are supplied with an abrasion-resistant steel plate.

Visually check the side wear liners on a monthly basis. If they are worn, replace them immediately.

If this is not done immediately, then the abrasion continues through to the main side plates resulting in premature failure of the feeder.

The snubber assemblies should be inspected monthly and worn parts replaced when necessary.

6.19 Electrical Connection

The cables should not touch any part of the machine or supporting structure. Check voltage requirements and ensure that motor is running in the correct direction.

Make certain all electrical equipment is properly grounded. Be aware of any wet spots located near electrical components that could cause a potential shock hazard. Attach an appropriate ground cable to the mainframe and sink an appropriate ground rod into the soil to the depth required by your local code. Check local codes to have the entire installation approved before operation.

Always provide an emergency stop control next to the vibrating feeder. Ensure suitable overload protection so that if one motor trips out, the second unit must shut down automatically.

7 Commissioning and Operation

The feeder is generally used as a component of an overall plant system. Specific equipment start-up procedures depends on the particular plant and the material flow as well as the type of controls used. Always refer to controls schematics and information received along with the equipment for start-up and operation. Safe work procedure, emergency shut-down procedure and isolation procedure needs to comply with the specific mine requirements.

7.1 Commissioning Safety

	WARNING LEVELING OF FEEDER Ensure that the feeder is levelled. Potential hand injuries can occur. Please be aware of possible nip points during installation. Nip points vary depending on the installation and configuration of the area the equipment is being installed.
	WARNING MOVING PARTS During machine start-up, injury can occur if a person comes in contact with moving parts in the machine. • Ensure the correct training of personnel operating the machine. • Ensure that all guards are fitted prior to start-up. • Ensure that no person is in the vicinity of the machine before start-up.
	CAUTION FIRE HAZARD Note that some of the materials used in the construction of the equipment could be flammable and emit hazardous fumes. Do not perform operations involving high temperatures in the vicinity of the feeder.
	CAUTION PORTABLE ELECTRICAL EQUIPMENT Refer to operation and safety manuals of portable electric equipment before use.
	CAUTION USING EQUIPMENT IN RESTRICTED SPACE • An upfront agreement on the installation space of the equipment is required. • Ensure that hearing protection is worn (noise level approximately 80dB). • Restricted or confined spaces can prove to be hazardous.
	CAUTION SLIPPING HAZARD When opening the feed supply, there is a risk of flooding the feeder. This causes the materials to spill and resulting in slipping of personnel.

7.2 Operation Safety

	 DANGER
	RISK OF PERSONNEL INJURY OR DEATH FOLLOW ALL SITE-SAFETY INSTRUCTIONS WHILE OPERATING THE FEEDER. FAILURE TO FOLLOW THE SAFETY INSTRUCTIONS CAN CAUSE SERIOUS INJURY LEADING TO DEATH. THE FOLLOWING GUIDELINES POINTS CAN ALSO BE USED AS ADDITIONAL SAFETY GUIDELINES. • NEVER PERFORM ANY MAINTENANCE UNLESS THE POWER SUPPLY IS LOCKED OUT. • NEVER OPERATE THE FEEDER UNLESS ALL GUARDS AND OTHER PROTECTIVE DEVICES ARE IN PLACE. • NEVER CLIMB OR STAND ON THE FEEDER WHILE IT IS OPERATING. • NEVER ATTEMPT TO CHECK BEARING TEMPERATURES WHILE THE FEEDER IS OPERATING. • NEVER OPERATE THE FEEDER WITHOUT PROPER LUBRICATION. • NEVER OPERATE THE FEEDER IF AN EXCESSIVE OR UNUSUAL NOISE EXISTS. • NEVER OPERATE THE FEEDER WITH THE VIBRATOR COUNTERWEIGHT PLATES REVOLVING IN THE WRONG DIRECTION. • NEVER INCREASE OR DECREASE THE VIBRATOR RPM WITHOUT WEIR TRIO APPROVAL. • NEVER FEED MATERIAL TO THE VIBRATING FEEDER UNLESS IT IS RUNNING AT OPERATING SPEED. • ALWAYS STOP THE FEED AND CLEAR THE PAN AND GRIZZLY AREA BEFORE STOPPING FEEDER. • NEVER CHANGE THE VIBRATOR OUTPUT SETTING WITHOUT WEIR TRIO APPROVAL.

	 WARNING
	RISK OF PERSONNEL INJURY Before the start-up, make sure that there is no personnel near the feeder. Read and understand the operating instructions before proceeding with the start-up. Do not perform any check, adjustments or repairs while the feeder is in operation.

	 WARNING
	ROTATING OUT-OF-BALANCE WEIGHTS FAILURE • Failure to rotate out-of-balance weights could cause debris to projected from the machine. • Follow cold commissioning check sheet. • All personnel to be out of the line of fire of exciters/motors OB weights at all times.

	WARNING MOVING PARTS During machine start-up, injury can occur if a person comes in contact with moving parts in the machine. • Ensure correct training of personnel operating the machine. • Ensure that all guards are fitted prior to start-up. • Ensure that no person is in the vicinity of the machine before start-up.
---	--

	CAUTION EXCESSIVE NOISE Ensure that hearing protection is worn when approaching the machine (noise level approximately 80dB).
---	--

7.3 Preparations Before Starting

A periodic inspection ensures the successful operation of the feeder. Maintain accurate written records and checklist of the inspections.

Regular pre-start and maintenance inspections eliminate costly downtime and improve the life of the feeder.

1. Check and make sure that the lubricating oil described in [“Lubrication” on page 11-1](#) is used in the lubricating system.
2. Do a final inspection of the feeder to make sure that there are no tools or mechanical obstructions in the feeder.
3. Check and make sure that all fasteners are tight.
4. Leave small amount of material on the pan to protect liners.
5. Use the checklist to inspect all items of the feeder.
6. Inspect the feeder body for any damage.
7. Inspect the condition of the springs.
 - Inspect springs are leaning or not.
 - Inspect springs are free of dirt or not.
 - Inspect springs broken coils.
 - Inspect springs coils paint missing between the coils, indicating a weak spring or that the feeder is overloaded.
8. Inspect the condition of the grizzly bars.
9. Check grizzly bars for correct spacing, damage and loose or missed hardware.
10. Make sure that all guards are fixed correctly.
11. Check the vibrator oil level on both sides and add oil if necessary.
12. Check breather caps on both gear cases. Keep dirt and material build-up away from breathers.
13. Check drive belt tensioner to make sure the correct tension is on belt and belt is in good condition.
14. Check around the unit for any oil leaks.

7.4 Feeder Start Up



CAUTION

RISK OF PERSONNEL INJURY OR EQUIPMENT DAMAGE

Always use a horn or a warning device to inform all the personnel about the equipment start-up. Make sure that there is no personnel in dangerous position prior to start-up.

1. Check the surrounding area to ensure no personnel are in a dangerous position prior to start-up.
2. Always sound a horn or other warning device to warn others that you are about to start up the equipment.
3. The feeder is part of a plant system. Make sure it is started in proper sequence with other components.
4. Start all the equipment in the plant as per the plant operation sequence prescribed. If no prescribed sequence is available, start with the last piece of equipment to discharge material and work back toward the initial feed point until all equipments are operating empty.
5. Make sure the feeder is running at the proper RPM before the feed material is introduced. Start the feeder at 100% RPM, once up to full speed reduce the feeders speed to an acceptable level for the plant.
6. Begin material flow to the plant. Keep safety in mind.
7. Record any observation about plant operation conditions in the plant logbook.

7.5 Normal Operation

1. During daily operation, check for unusual sounds or signs of excessive vibration. Make sure you know the cause of any abnormality. Correcting problems when they first appear saves downtime later.
2. Watch for product spillage due to pan liners damage or material build up between the flashing and the pan.
3. Become familiar with the amount of heat radiating from gear cases when the system performs normally.
4. Record any observations in the plant logbook.

7.6 Checks During Operation

1. Check product spillage due to pan liners damage or material build up between the flashing and the pan.
2. Check abnormal movement of the feeder.
3. Check for the smell of over heating lubrication oil.
4. Check excessive V-belt slippage.
5. Check the drive shaft speed.
6. Check grizzly bar or pan plugging.
7. Check the bearing temperatures regularly.

7.7 Feeder Shut-down

1. Stop the material feed to the feeder.
2. Make sure that the feeder has a bed of material.
3. Stop the feeder.

7.8 Checks After Operation

1. Clear any rocks that may be stuck between the grizzly bars.
2. Clean the pan section if mud is stuck in the pan.
3. Inspect the feeder pan for wear.
4. Inspect the feeder for damage.

7.9 Check List for Starting New Feeder

7.9.1 Cold Commissioning Check List

Client:	Project:
Plant:	Equipment:

1. Visually check for obvious damage during transport and installation.

Comments:

Description	Remarks	Action
2. General		
Compare the left and the right foundation plates for height difference. It should be within 4 mm.		
Check 75 mm (3 in) clearance between stationary structures and all parts of the moving live frame, including the pivoted motor base and motor assembly		
Check 12.5 mm (1/2 in) side clearance between hopper and feeder.		
Check feed chutes and hopper outer vertical sides should be within 6 mm(1/4 in) to 12.5 mm (1/2 in) of the inner vertical sides of the feeder pan (or liner).		
Check vibrating frame and snubber clearance 25 mm (1 in) to 30 mm (1-3/16 in)		
Check that the suspension is correct		
The minimum inspection and maintenance clearance around the feeder should be 750 mm (30 in)		

Description	Remarks	Action
3. Vibrator Assembly		
Check oil level and top up if necessary to the correct oil level plug or sight glass indicator		
Note: If oil level low, investigate for oil leaks.		

Description	Remarks	Action
4. Drive		
Check oil levels and do initial fill		
Remove locking system. (if fitted)		
Check the belt alignment between the driven pulley and the motor shaft pulley.		
Check the belt tension		
Check the clearance of the V-belt guard		

Description	Remarks	Action
5. Start-up		
Start the motor and check for correct rotation of drive.		
Note: If incorrect call responsible person and have electrical connections corrected.		
Check for interference with surrounding equipment / structure		
Test machine with vibration analyser at prescribed points as shown on the analysis sheet and the acceleration (g') as well as stroke (mm)		
Note: Record and attached plot to this report.		
Check the operating speed of the machine using strobe light or tachometer		

Description	Remarks	Action
6. Machine Run		
Allow machine to run between four and six hours		
Check for interference with surrounding equipment / structure		
Check bearing temperatures and record on inspection sheet		
Check for oil leaks.		

Checked By: _____

Signature: _____

For Client: _____

Date: _____

Table 7-1: Cold Commissioning Check List

7.9.2 Hot Commissioning Check List

Client:	Project:
Plant:	Equipment:

1. Observe free flow of material over feeder at design tonnage.

Comments:

2. Feed presentation for maximum use of screening area satisfactory?

Comments:

Description	Inquiry Specification	Actual	Action
3. Process Parameters			
Feed (tph or m ³ /h)			
Cut point / aperture			
Top size			
Material bulk density (ton/m ³)			

Description	Under Load	Without Load
4. Feeder Dynamics		
Acceleration (G's)		
Stroke angle (mm)		

Checked By: _____

Signature: _____

For Client: _____

Date: _____

Table 7-2: Hot Commissioning Check List

7.10 Feeder Gear Timing

The standard feeder vibrator gear timing is set at the factory at 45 degrees. This timing is satisfactory for most of the materials used in the feeder (refer to [Figure 7-1](#)).

Some materials may feed better at different gear timing angles.

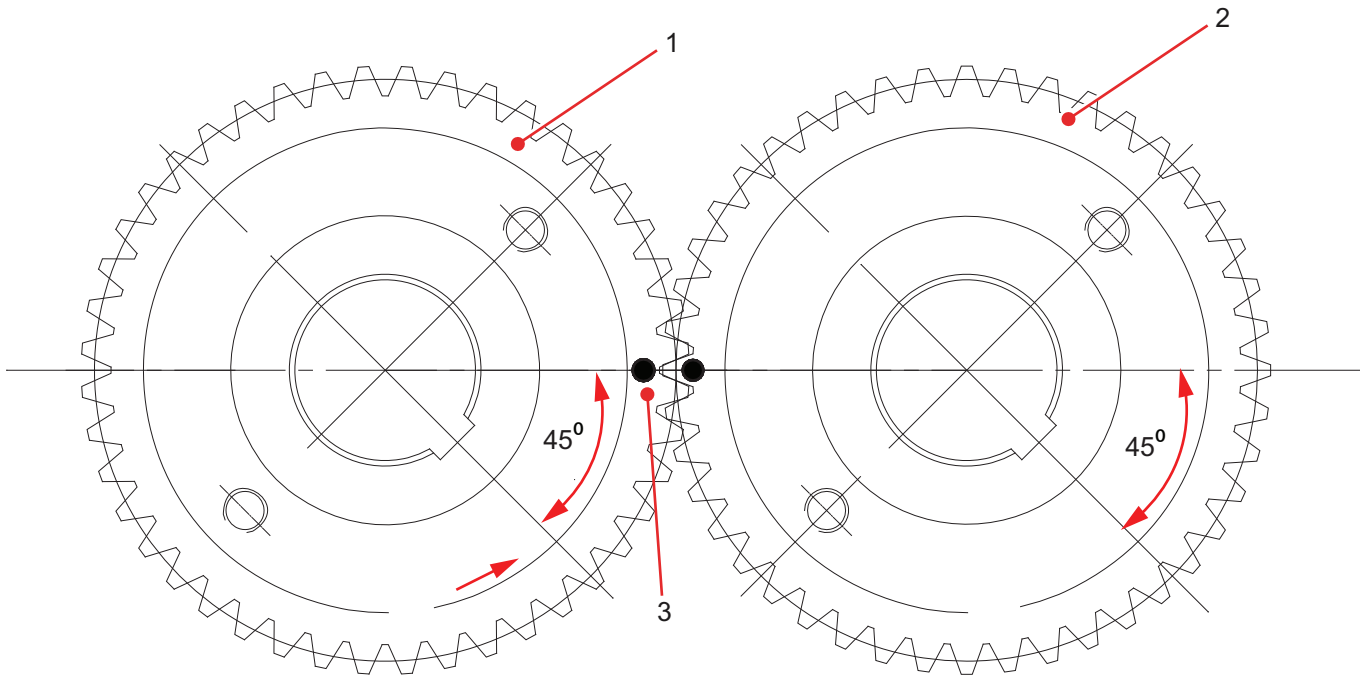


CAUTION

RISK OF DAMAGE TO VIBRATOR SHAFTS

Changing the feeder gear timing angles may cause the vibrator shafts to contact each other. This can lead to damage to the shaft components.

Contact Weir Trio representative to change the feeder factory settings.



1	Gear 1	2	Gear 2
3	Assembling/Disassembling Angle Marking		

Figure 7-1: Feeder Gear Timing



NOTICE

MATERIAL STUCK

Rocks stuck in grizzly bar, 30° timing may solve the issue.

Material stuck in the rear of the pan slowing down the feed rate, 60° timing may solve the issue.

Certain materials may feed better at different gear timing angles. While it is possible to change the gear timing, certain feeder sizes set at different timing angles may cause vibrator shafts to come into contact with one another. Refer [Figure 7-2](#) to for feeder gear timing action examples.

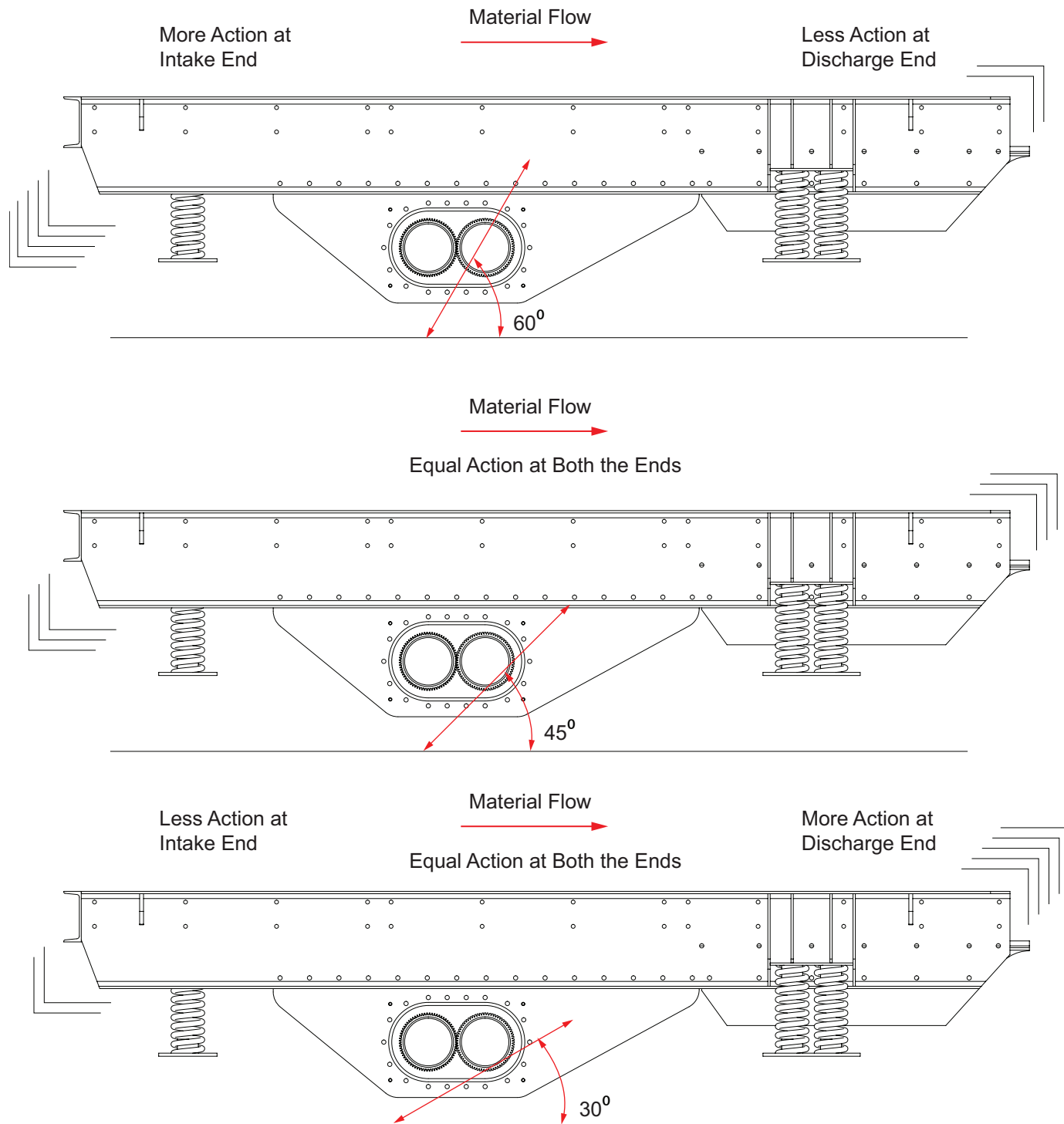


Figure 7-2: Feeder Gear Timing Action Examples

7.11 Changing the Vibrator Gear Timing

Contact Weir Trio to get the information about the specific timing information for the particular size of the feeder.

The below procedure is an example of changing the TF4012 feeder to 33° timing. The tooth count in gears varies with the type of feeders.

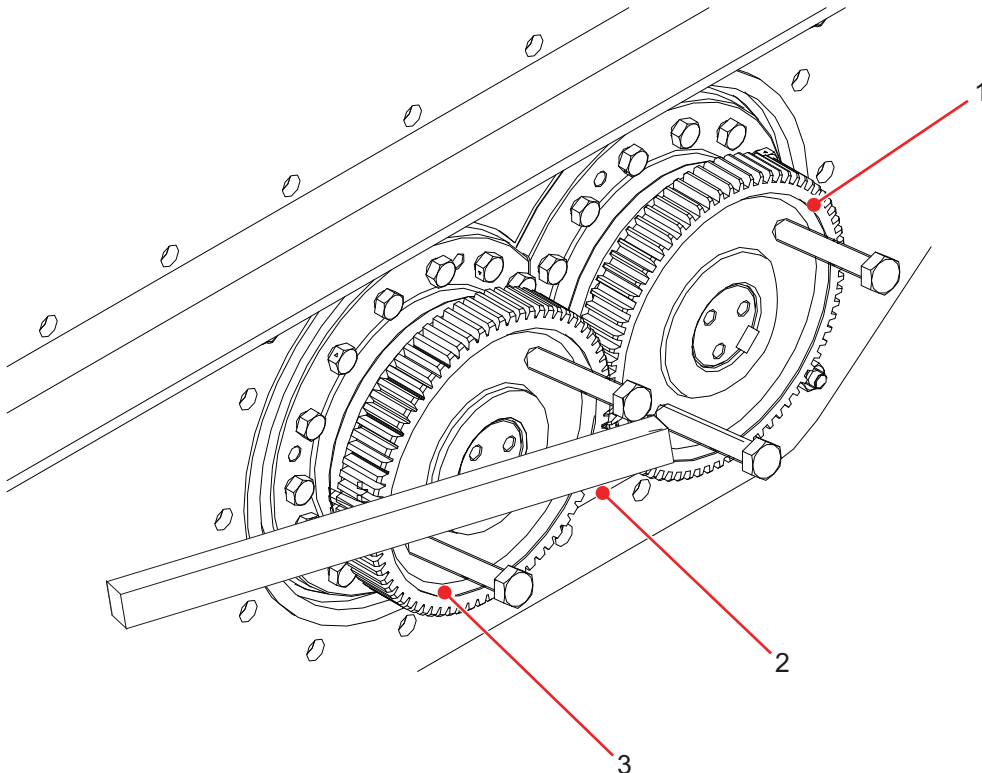



DANGER

LOCK OUT AND TAG OUT THE WORK AREA

- THE ELECTRICAL, HYDRAULIC, PNEUMATIC OR MECHANICAL POWER SOURCE OF THE FEEDING EQUIPMENT MUST BE LOCKED OUT AND TAGGED OUT WHENEVER ANY MAINTENANCE ACTIVITY IS CARRIED OUT ON IT.
- ACCIDENTAL START-UP OF THE EQUIPMENT CAN RESULT IN FATAL ACCIDENT.

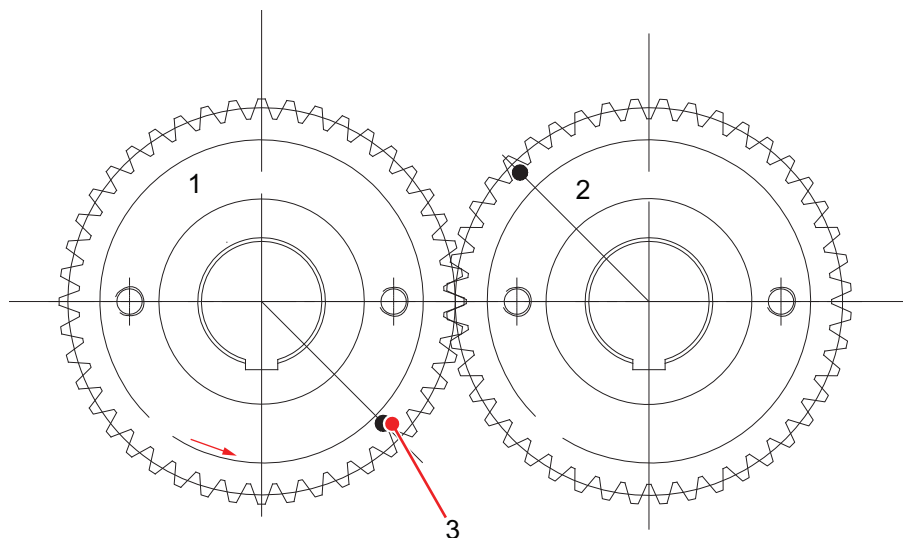
1. Follow the plant shut-down procedure and apply LOTO.
2. Drain the lubricant oil from the vibrator assembly, refer to [“Lubrication Oil Change Procedure” on page 11-4.](#)
3. Remove the gear cover from the non-drive side. Refer to [Figure 9-1 on page 9-1](#) for gear cover removal.
4. Refer to [Figure 7-1 on page 7-8](#) for factory fitting feeder gear timing at 45°
5. On the gear 1:
 - a.Remove the lock wire from the gear shaft cover bolt.
 - b.Remove the gear shaft cover.
6. Install pusher bolts into gear 1 and gear 2. Insert a bar on the bolts, to hold and turn the gears and shafts. See ([Figure 7-4](#)).



1	Gear 2	2	Bar
3	Gear 1		

Figure 7-3: Pusher Bolts and Bar Installation

7. Move gear 1 until no longer meshed with gear 2. When gears are unmeshed, the gears will rotate downward and cause injury if not properly supported with a pry bar as shown in [Figure 7-3](#) on page 7-10. The resting position is shown in [Figure 7-4](#).



1	Gear 1	2	Gear 2
3	Assembling/Disassembling Angle Marking		

Figure 7-4: Feeder Gear Timing Rest Position

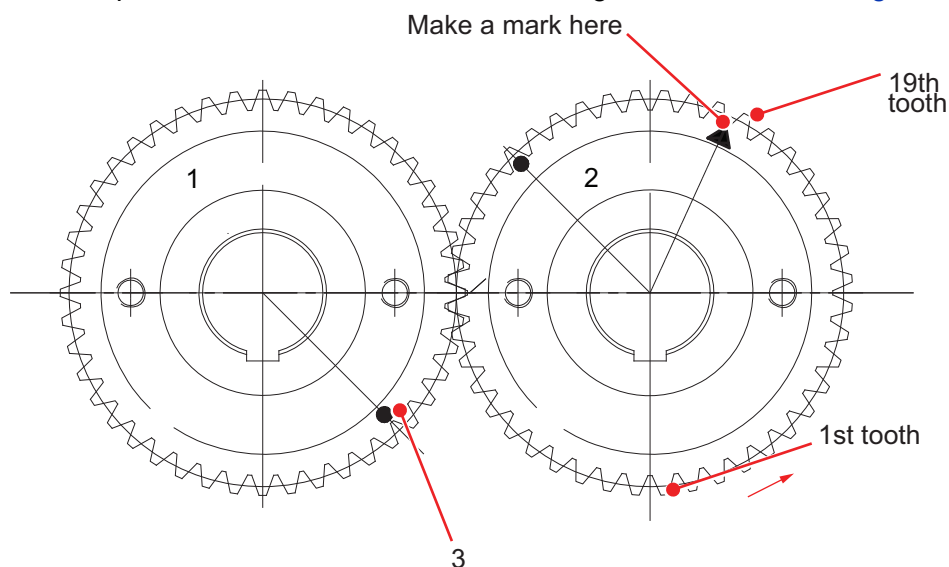


CAUTION

RISK OF EQUIPMENT DAMAGE OR PERSONNEL INJURY

Gravity will cause the shafts and gears to rotate when gears are unmeshed.

8. Make a mark at the place beside the root of the tooth on gear 2 as shown in [Figure 7-5](#).



1	Gear 1	2	Gear 2
3	Assembling/Disassembling Angle Marking		

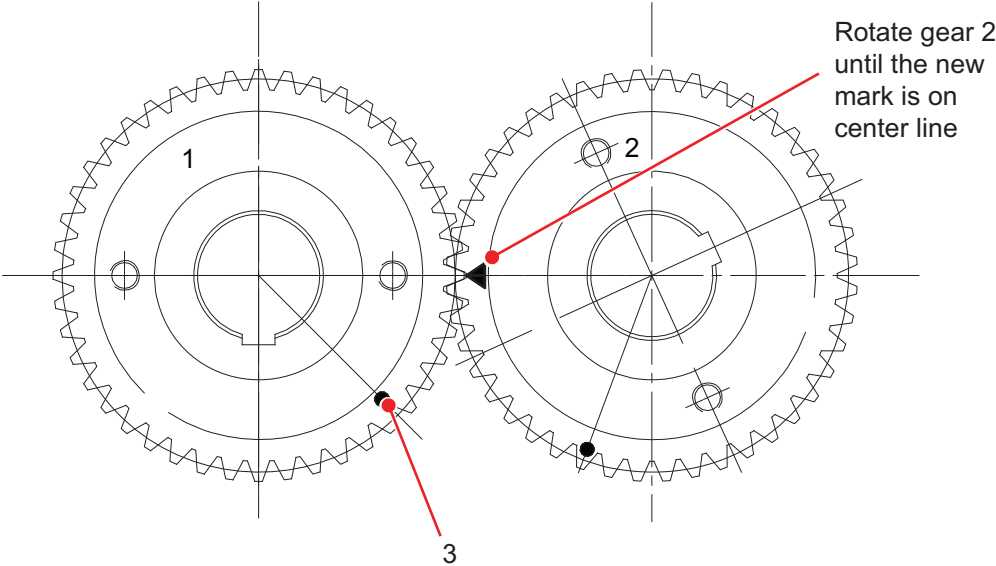
Figure 7-5: Feeder Gear Timing Marking



 **NOTICE**

TOOTH COUNT
The number of teeth on gears varies on the Feeder model variant and may differ from that shown.

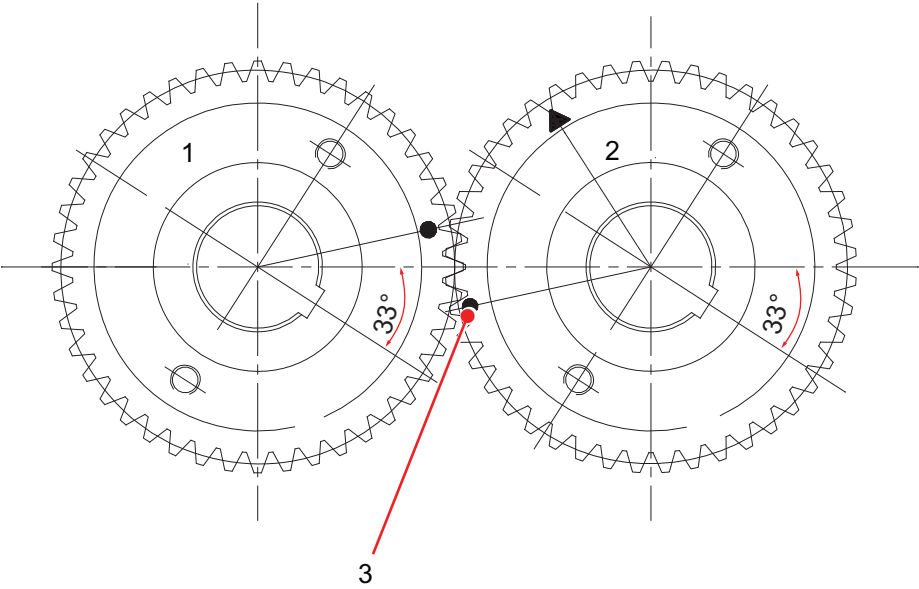
9. Rotate the gear 2 in a counter-clockwise direction, until the new mark is at the centerline as shown in Figure 7-6.



1	Gear 1	2	Gear 2
3	Assembling/Disassembling Angle Marking		

Figure 7-6: Feeder Gear Timing (Mark at Center)

10. Hold gear 2 and move gear 1 back in to mesh together with gear 2.
11. Release the gears and the timing angle should be 33°. This can be verified with the angle of the keys that lock the gear to the shaft as shown in Figure 7-7.



1	Gear 1	2	Gear 2
---	--------	---	--------

3	Assembling/Disassembling Angle Marking		
---	---	--	--

Figure 7-7: Feeder Gear Timing (At 33° position)

- 12. Install the gear shaft cover on gear 1 and tighten the bolts.
- 13. Insert the lock wire into the gear shaft cover bolt heads and tie it.
- 14. Install the gear side cover and tighten the bolts. Refer to [Figure 8-24 on page 8-17](#) for more details.
- 15. Fill the vibrator assembly to the recommended level.
- 16. Remove LOTO.

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8 Assembly

8.1 Assembly Safety

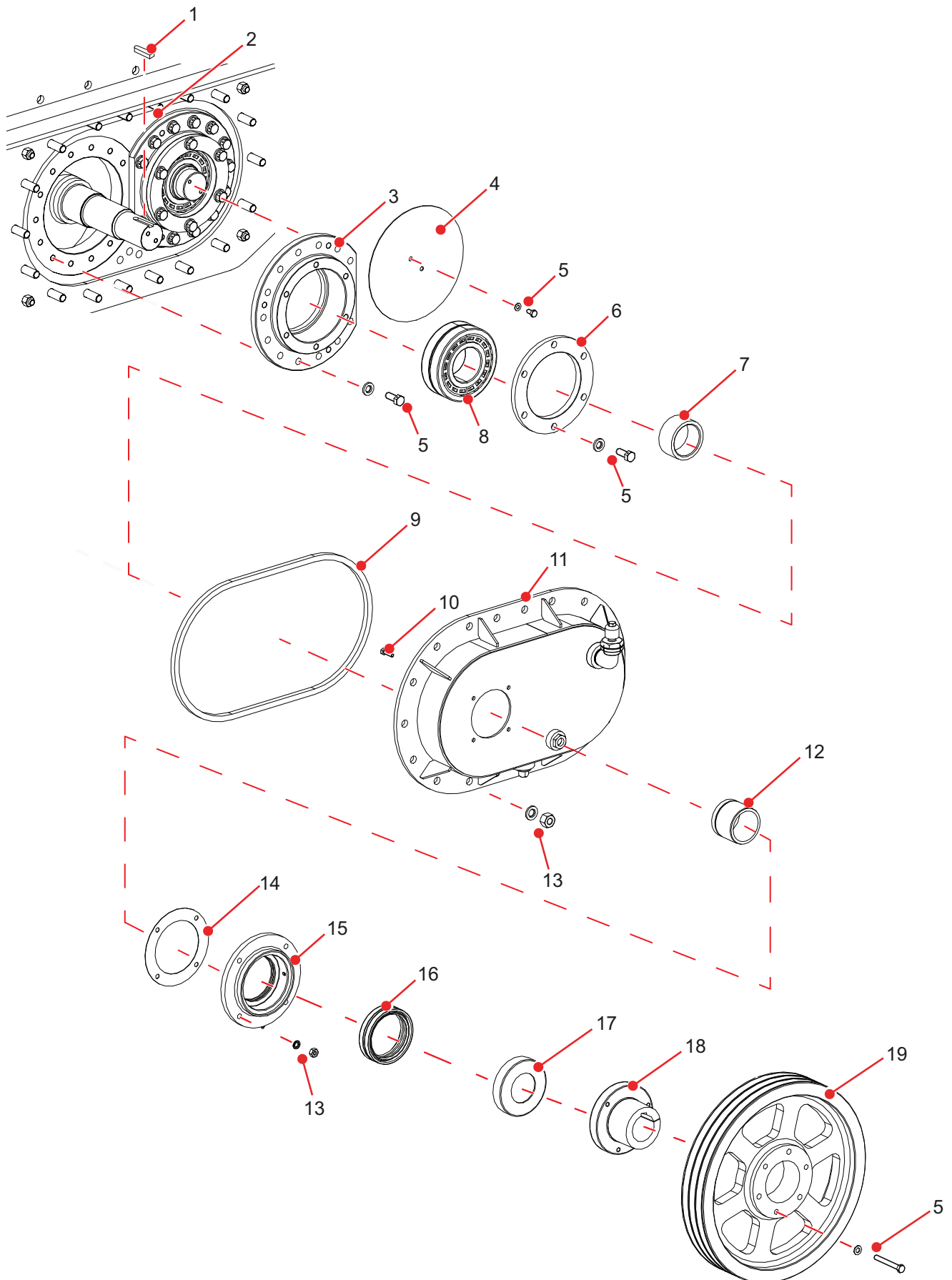
During assembly operations, ensure all safety warnings and instructions are followed, as mentioned in "Safety" on page 2-1.

8.2 Assembly Instructions

	i NOTICE TOOLS AND EQUIPMENT CONDITION Make sure that tools and devices required for assembly are not damaged, worn out, and ready for assembling the feeder.
	! CAUTION FOLLOW SAFE WORKING PRACTICES Follow safe working practices during all assembly and maintenance work.
	! CAUTION GUIDELINES FOR CORRECT ASSEMBLY • When the assemblies are dismantled for a complete overhaul, all parts must be closely inspected, and new parts checked for correct identification. • Used parts that are being reused must be thoroughly cleaned and repainted where required. Mating surfaces must be free of rust, dirt, and burrs and given a coat of grease before they are fitted together to assist future overhaul. • It is preferable to renew small bolts and set screws during overhaul and all threads must be coated with graphite grease before assembly.
	! CAUTION EQUIPMENT DAMAGE Check and ensure that all parts have no burrs or sharp edges. This ensures that the components are assembled correctly with full surface contact.

8.3 Vibrator Assembly

8.3.1 Drive Side Vibrator Assembly



1	Key	11	Sheave Side Cover
2	Vibrator Tube	12	Shaft Sleeve
3	Bearing Housing	13	Nut and Washer
4	Oil Flinger	14	Gasket
5	Bolt and Washer	15	Drive Shaft Oil Seal Housing
6	Bearing Cover	16	Drive Shaft Seal
7	Spacer	17	Shaft Enclosure
8	Bearing	18	Sheave Bushing
9	Rubber Seal	19	Driven Sheave
10	Bolt		

Figure 8-1: Drive Side Vibrator Assembly

To assemble the drive side vibrator assembly components, proceed as follows:

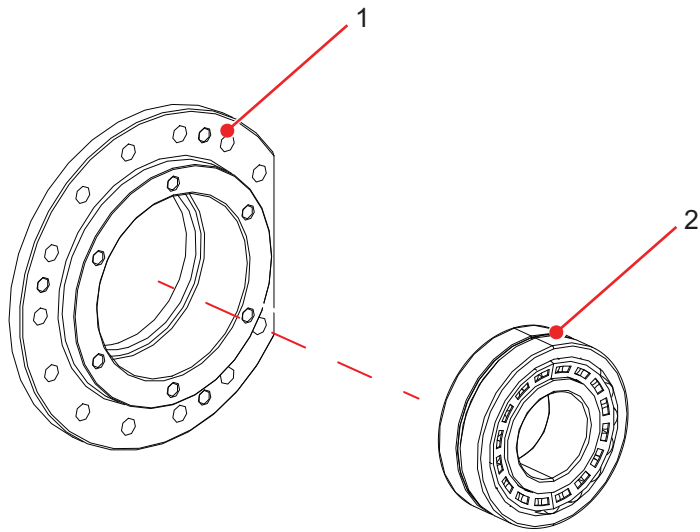
- 1. Ensure bearing housing is clean and free of burrs.
- 2. Insert the bearing into the bearing housing, as shown in [Figure 8-2](#).
- 3. Use the dead blow hammer and punch, tap bearing down until it seats against the bottom of the bore.
- 4. This is a slip fit, not an interference fit. If properly aligned with the bore, the bearing slides in with very little force.



i **NOTICE**

BEARING RACE

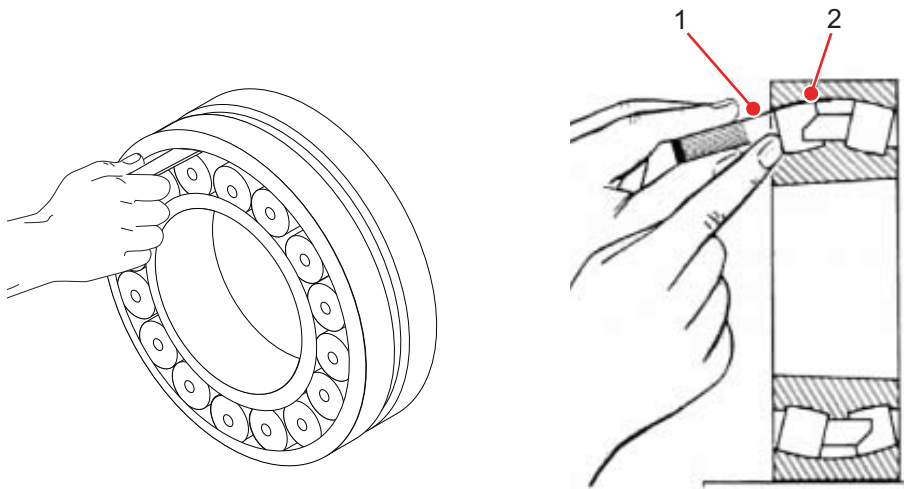
- Do not strike the bearing race with a steel hammer.
- Bearings should be tapped on the outer race. Tapping the inner race may cause bearing damage.



1	Bearing Housing	2	Bearing
---	-----------------	---	---------

Figure 8-2: Bearing Insertion

- 5. Note the internal clearance between the uppermost roller and the inner surface of the bearing using a feeler gauge for future reference.



1	Feeler Gauge	2	Bearing
---	--------------	---	---------

Figure 8-3: Bearing Clearance

6. Attach the eye bolt and lifting sling to the bearing assembly.

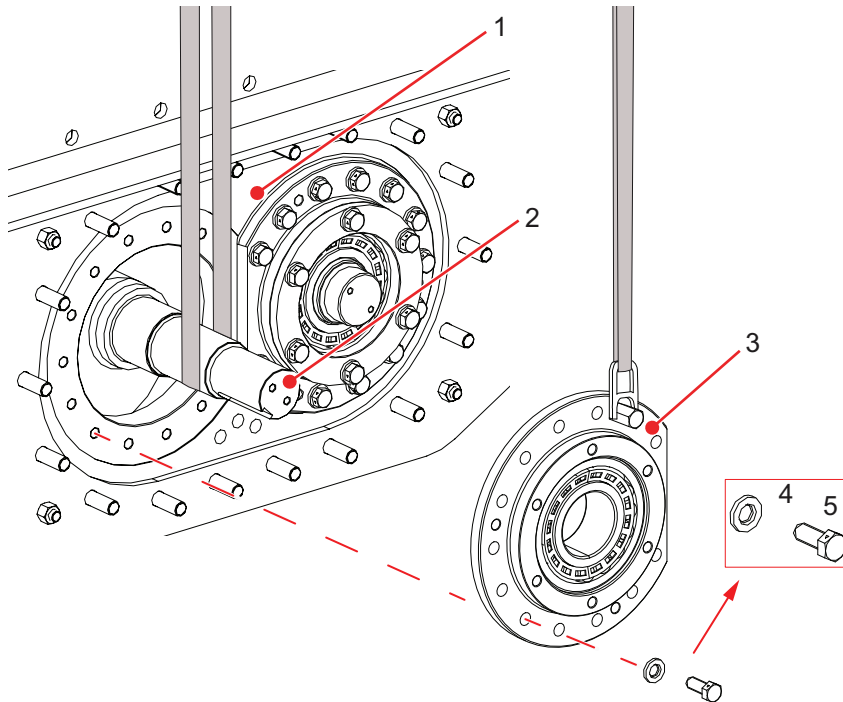


CAUTION

EQUIPMENT DAMAGE

Carefully move the bearing assembly towards the shaft, unless the shaft and bearing assembly may damage.

7. Slide the bearing assembly onto the shaft and support the shaft near the fixed height on both sides. This will prevent the shaft from causing a bearing to tilt.
8. Install the bearing assembly on the shaft by tightening the bolts and washers, refer to "[Torque Specifications](#)" on page 15-1. Insert the lock wire into the bearing housing bolt heads and tie it.
9. Install the bearing assembly onto the shaft until fully seated inside the mechanism and against the bearing journals.

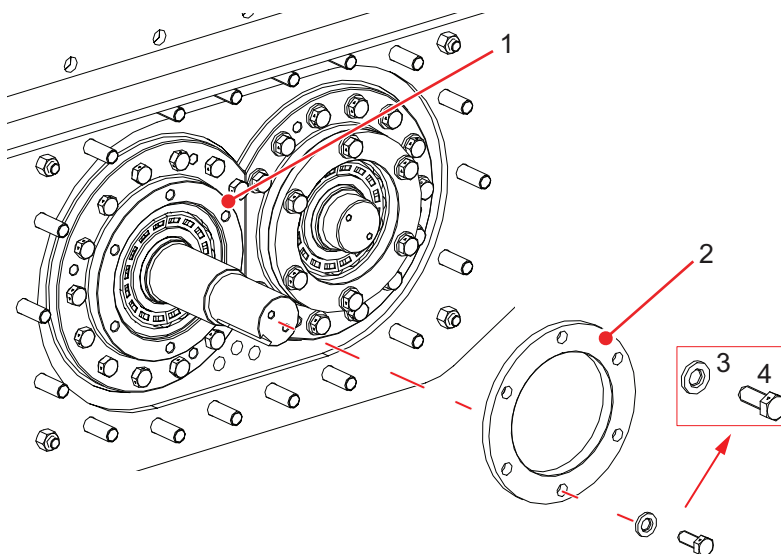


1	Vibrator Tube	4	Washer
2	Shaft	5	Bolt
3	Bearing Assembly		

Figure 8-4: Bearing Assembly

10. Remove the eye bolt and lifting sling from the bearing assembly.
11. Follow the same procedure to install the remaining bearing assembly's.

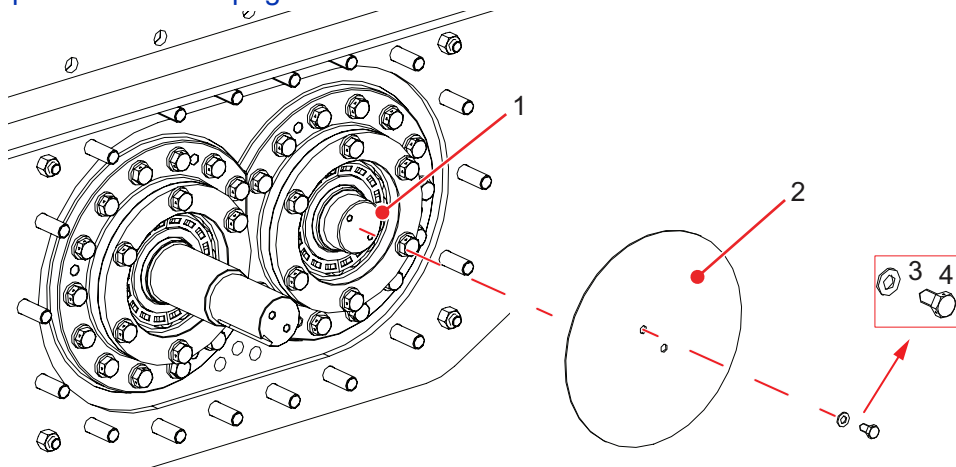
12. Install the bearing cover on the bearing housing by tightening the bolts and washers, as shown in [Figure 8-5](#), and refer to “[Torque Specifications](#)” on [page 15-1](#). Insert the lock wire into the bearing cover bolt heads and tie it.



1	Bearing Housing	3	Washer
2	Bearing Cover	4	Bolt

Figure 8-5: Bearing Cover Assembly

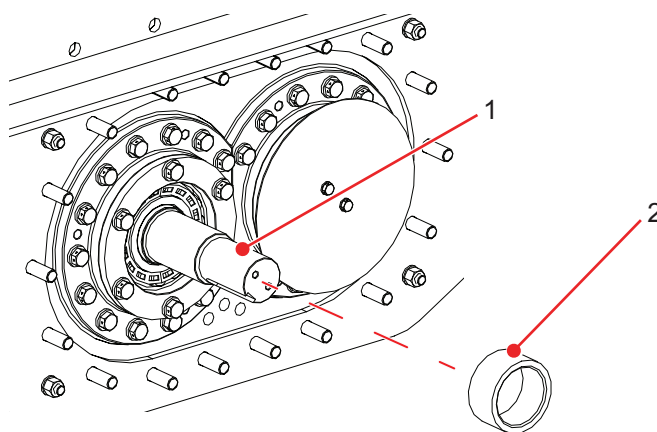
13. Install the oil flinger on the shaft by tightening the bolts and washers, as shown in [Figure 8-6](#), and refer to “[Torque Specifications](#)” on [page 15-1](#).



1	Shaft	3	Washer
2	Oil Flinger	4	Bolt

Figure 8-6: Oil Flinger Installation

14. Insert the spacer onto the shaft, as shown in [Figure 8-7](#).



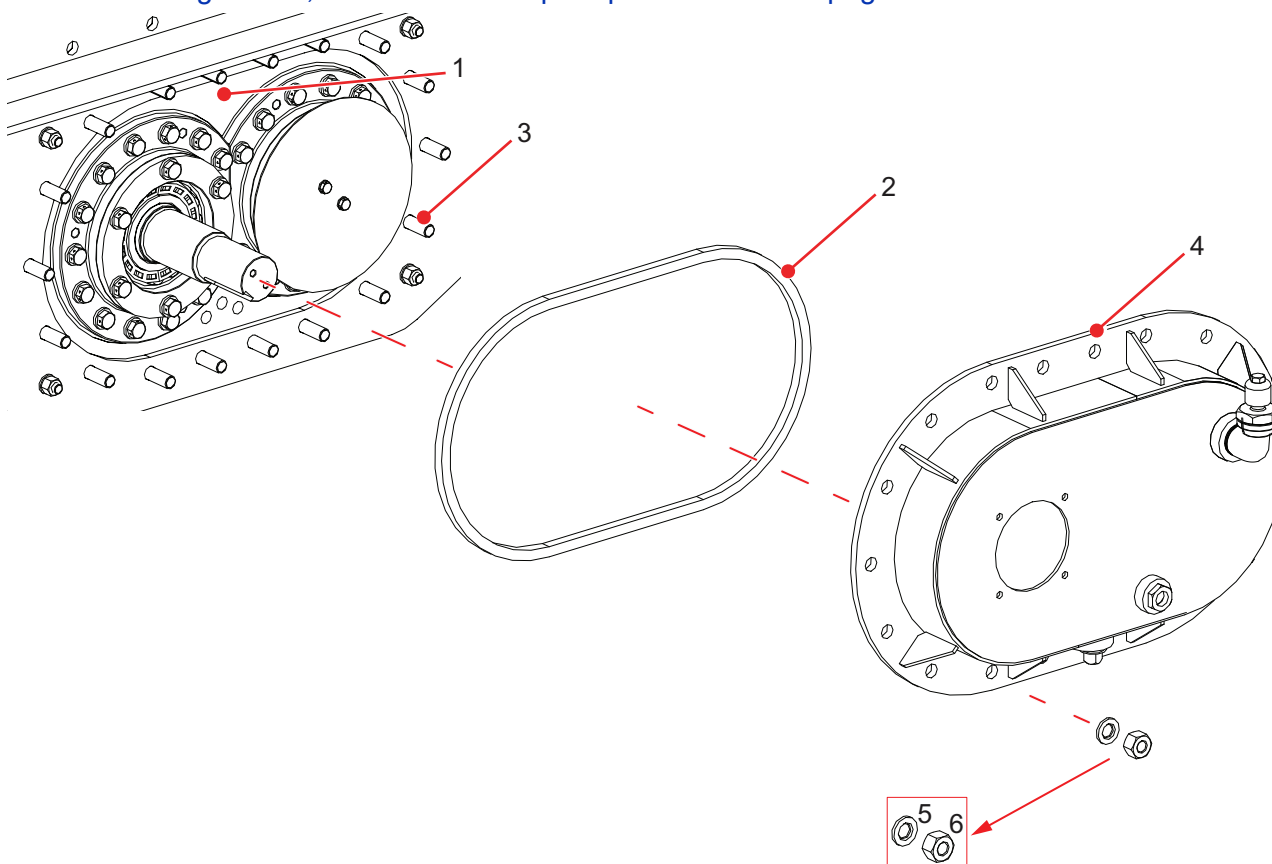
1	Shaft	2	Spacer
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Figure 8-7: Spacer Insertion

15. Install the rubber seal on to the vibrator tube.

16. Attach an appropriate lifting device to the sheave side cover.

17. Align and install the sheave side cover to the vibrator tube assembly by tightening the nuts and washers, as shown in [Figure 8-8](#), and refer to “[Torque Specifications](#)” on [page 15-1](#).



1	Vibrator Tube	4	Sheave Side Cover
2	Rubber Seal	5	Washer
3	Bolt	6	Nut

Figure 8-8: Sheave Side Cover Assembly

18. Remove the lifting device from the sheave side cover.

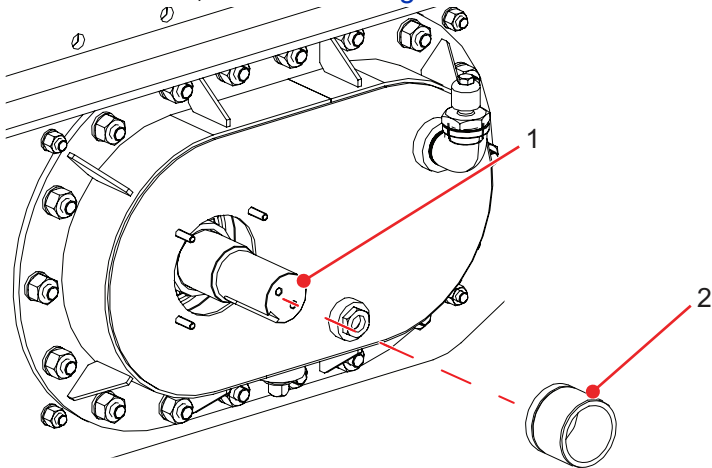


i NOTICE

SHAFT DETAILS

Shaft No.1 (shaft nearest discharge end of feeder) is longer on the driven side to accommodate the driven sheave and oil-seal assembly. Otherwise, two shaft assemblies are identical.

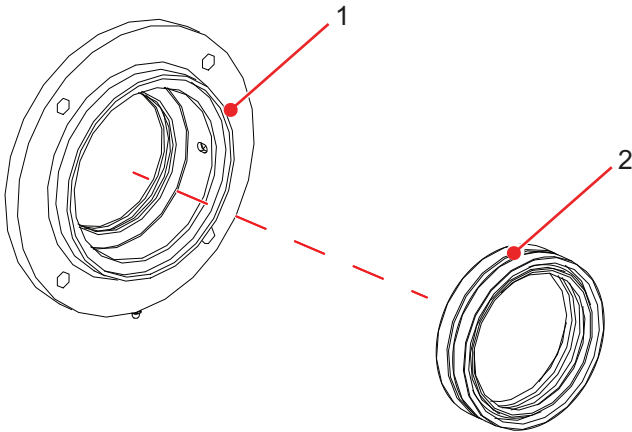
19. Insert the shaft sleeve onto the shaft, as shown in [Figure 8-9](#).



1	Shaft	2	Shaft Sleeve
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Figure 8-9: Shaft Sleeve Insertion

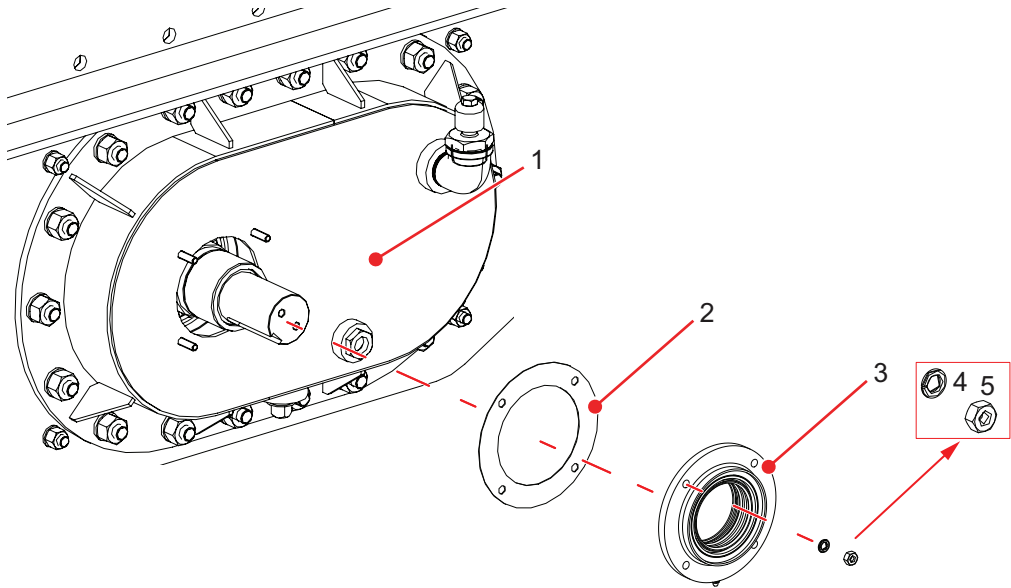
20. Insert the drive shaft seal into the drive shaft oil seal housing.



1	Drive Shaft Oil Seal Housing	2	Drive Shaft Seal
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Figure 8-10: Drive Shaft Seal Insertion

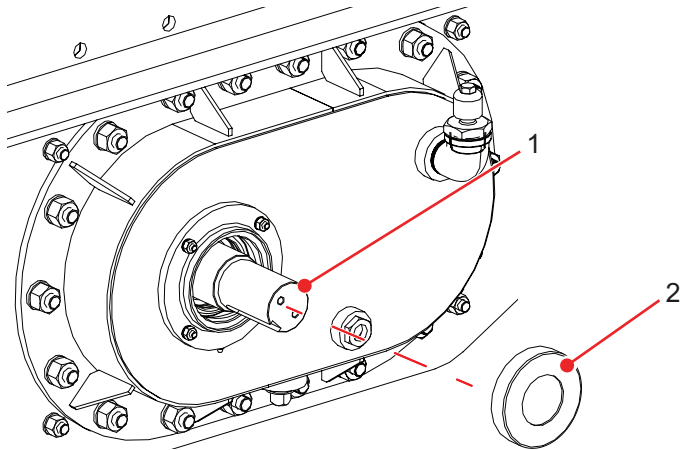
21. Install the drive shaft oil seal housing and gasket to the sheave side cover by tightening the nuts and washer, as shown in [Figure 8-11](#), and refer to “Torque Specifications” on page 15-1.



1	Sheave Side Cover	4	Washer
2	Gasket	5	Nut
3	Drive Shaft Oil Seal Housing		

Figure 8-11: Drive Shaft Oil Seal Housing

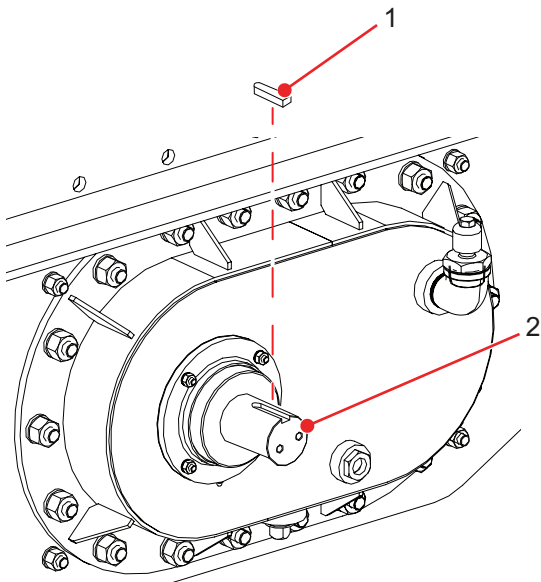
22. Insert the shaft enclosure onto the shaft, as shown in [Figure 8-12](#).



1	Shaft	2	Shaft Enclosure
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Figure 8-12: Shaft Enclosure Insertion

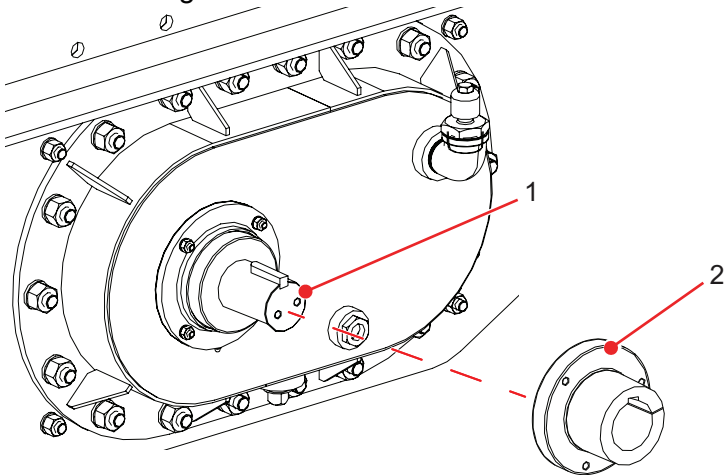
23. Insert the key into the shaft keyway.



1	Key	2	Shaft
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Figure 8-13: Key Insertion

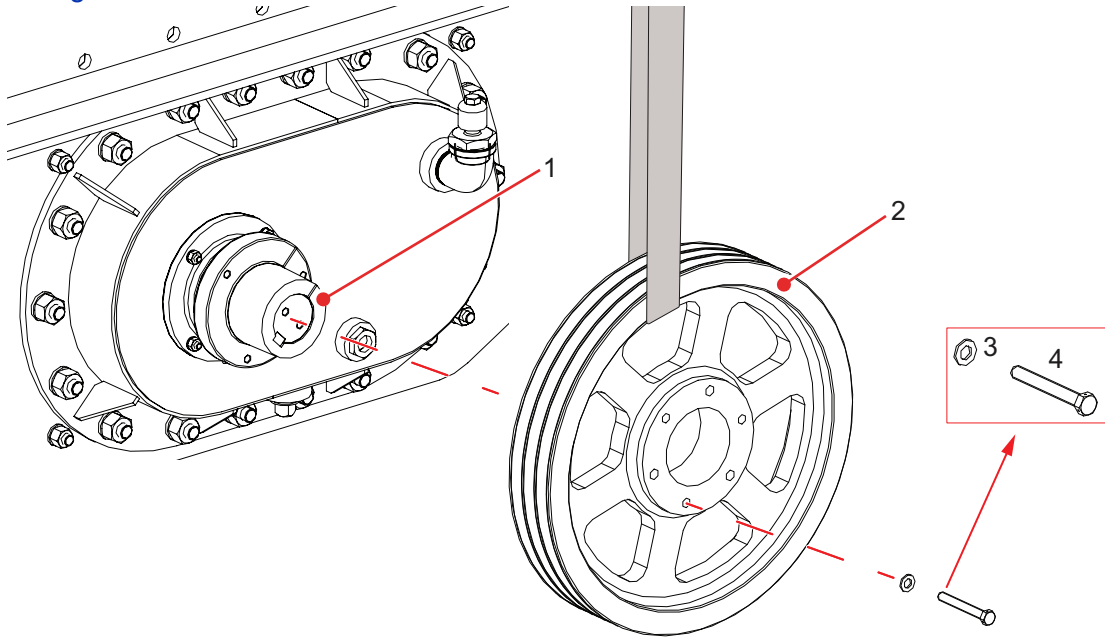
24. Align and Insert the sheave bushing onto the shaft.



1	Shaft	2	Sheave Bushing
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Figure 8-14: Sheave Bushing Insertion

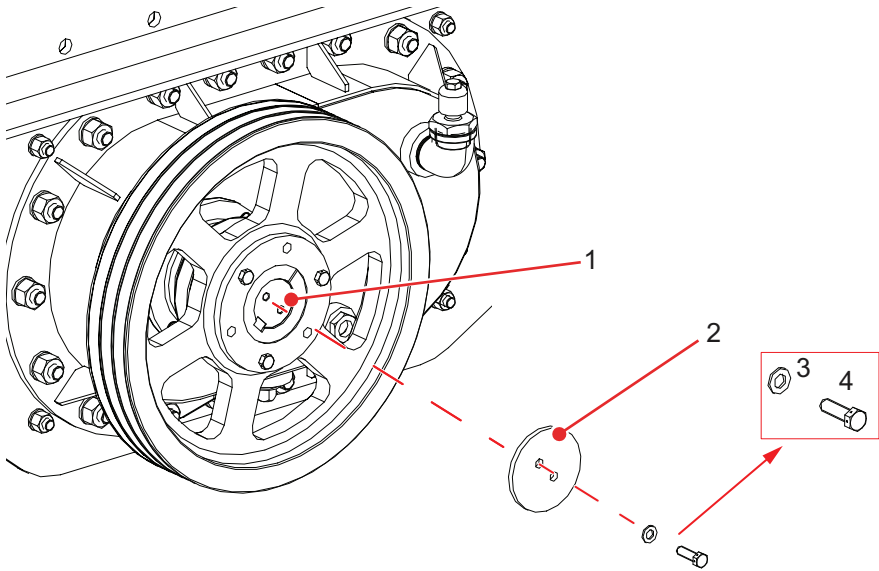
- 25. Attach an appropriate lifting device to lift the driven sheave.
- 26. Align and Install the driven sheave to the sheave bushing holes and tightening the bolts and washers, as shown in [Figure 8-15](#).



1	Sheave Bushing	3	Washer
2	Driven Sheave	4	Bolt

Figure 8-15: Driven Sheave Installation

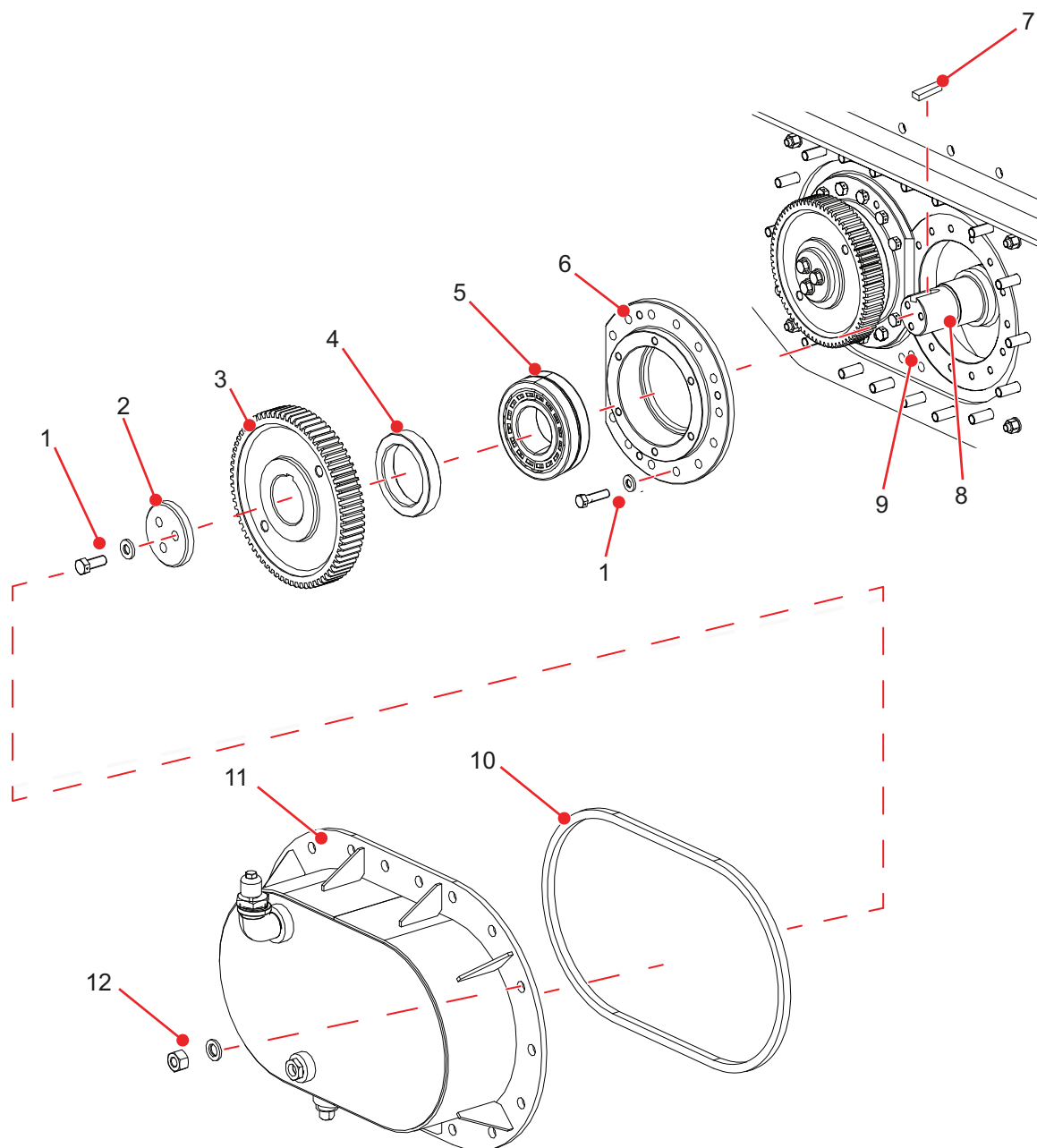
- 27. Remove the lifting device from the driven sheave.
- 28. Install the shaft cover to the shaft by tightening the bolts and washers, as shown in [Figure 8-12](#), and refer to “[Torque Specifications](#)” on page 15-1.



1	Shaft	3	Washer
2	Shaft Cover	4	Bolt

Figure 8-16: Shaft Cover Installation

8.3.2 Gear Side (Non Driven Side) Vibrator Assembly



1	Bolt and Washer	7	Key
2	Shaft Cover	8	Shaft
3	Gear	9	Vibrator Tube
4	Spacer	10	Rubber Seal
5	Bearing	11	Gear Side Cover
6	Bearing Housing	12	Nut and Washer

Figure 8-17: Non Drive Side Vibrator Assembly

To assemble the non driven side (gear side) vibrator assembly components, proceed as follows:

- 1. Ensure bearing housing is clean and free of burrs.
- 2. Insert the bearing into the bearing housing, as shown in [Figure 8-2 on page 8-4](#).
- 3. Use the dead blow hammer and punch, tap bearing down until it seats against the bottom of the bore.
- 4. This is a slip fit, not an interference fit. If properly aligned with the bore, the bearing slides in with very little force.



 **NOTICE**

BEARING RACE

- Do not strike the bearing race with a steel hammer.
- Bearings should be tapped on the outer race. Tapping the inner race may cause bearing damage.

- 5. Note the internal clearance between the uppermost roller and the inner surface of the bearing using a feeler gauge for future reference, as shown in [Figure 8-3 on page 8-4](#).
- 6. Attach the eye bolt and lifting sling to the bearing assembly.



 **CAUTION**

EQUIPMENT DAMAGE

Carefully move the bearing assembly towards the shaft, unless the shaft and bearing assembly may damage.

- 7. Slide the bearing assembly onto the shaft and support the shaft near the fixed height on both sides. This will prevent the shaft from causing a bearing to tilt.
- 8. Install the bearing assembly onto the shaft until fully seated inside the mechanism and against the bearing journals. Insert the lock wire into the bearing housing bolt heads and tie it.
- 9. Remove the eye bolt and lifting sling from the bearing assembly.
- 10. Insert the two spacers on the gear side shafts, as shown in [Figure 8-18](#).

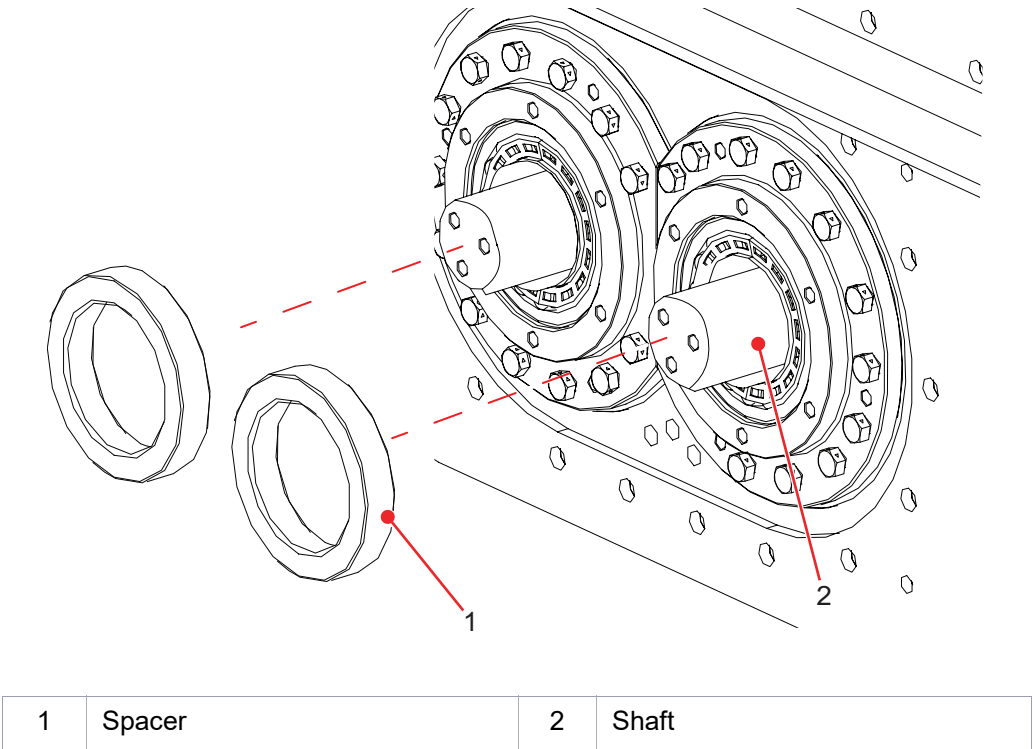
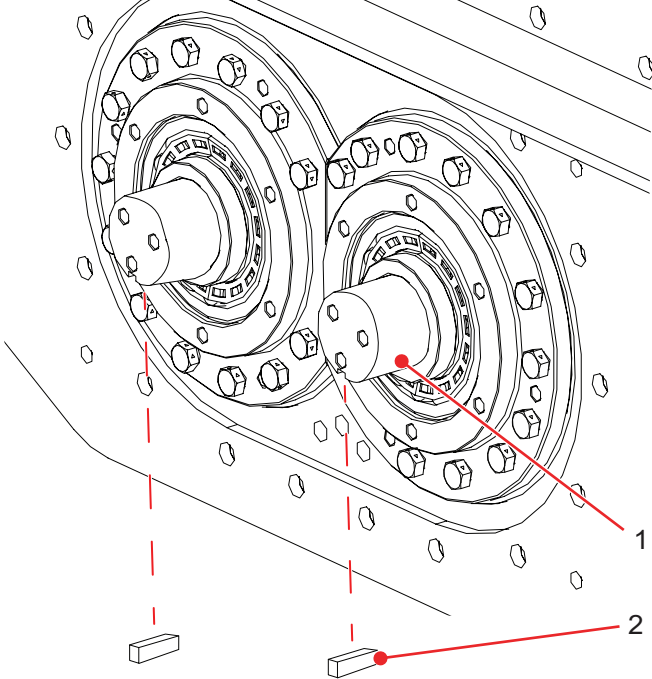


Figure 8-18: Spacer Insertion

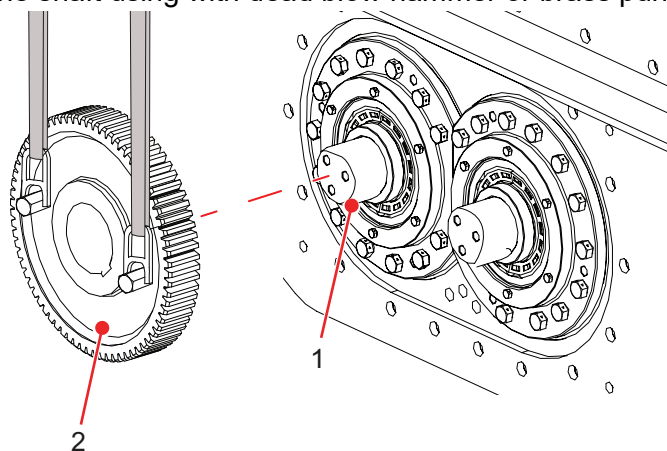
11. Insert a key in the shaft keyway, as shown in [Figure 8-19](#).



1	Shaft	2	Key
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Figure 8-19: Shaft Key Insertion

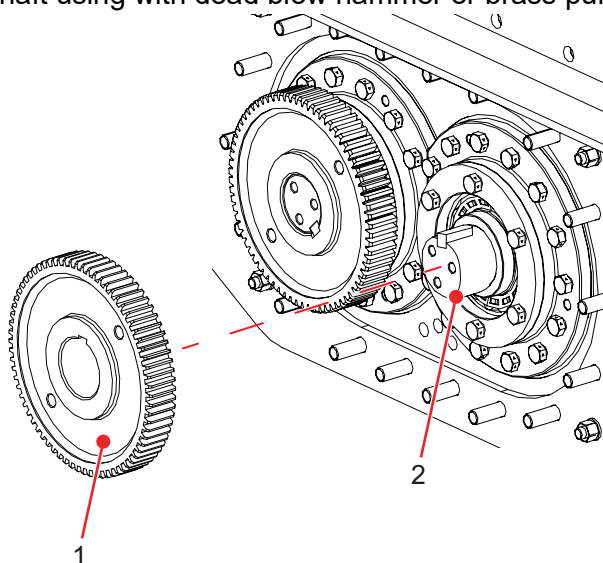
12. Install the bearing cover on the bearing housing by tightening the bolts and washers as shown in [Figure 8-5 on page 8-6](#), and refer to “[Torque Specifications](#)” on [page 15-1](#).
13. Insert the lock wire into the bearing cover bolt heads and tie it.
14. Use an appropriate lifting device to lift the gear 1.
15. Lift and gradually move the gear 1 towards the shaft.
16. Install the gear 1 on the shaft using with dead blow hammer or brass punch.



1	Shaft	2	Gear 1
---	-------	---	--------

Figure 8-20: Gear 1 Installation

17. Install the gear 2 on the shaft using with dead blow hammer or brass punch, but do not mesh the gears.

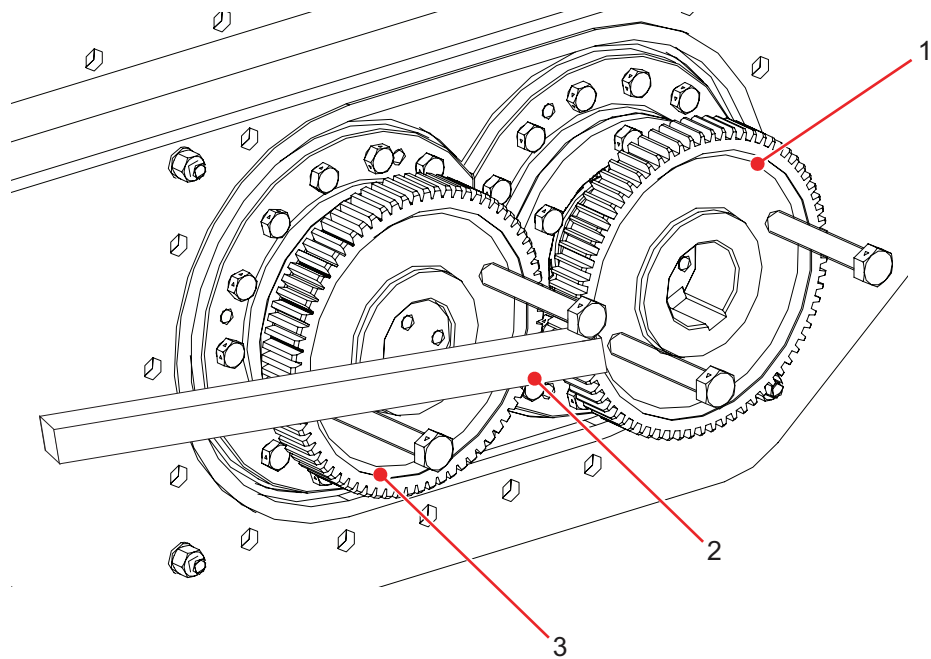


1	Gear 2	2	Shaft
---	--------	---	-------

Figure 8-21: Gear 2 Installation

18. Install the pusher bolts into gears 1 and gear 2 to allow a bar to be used to hold or turn gears and shafts.
19. Use a bar to hold gear 1 with the keyway facing downwards.

20. Rotate the shaft 2 with a bar until the desired amount of teeth are counted, then seat gear 2 until the teeth fully mesh.



1	Gear 2	2	Bar
3	Gear 1		

Figure 8-22: Gears Meshing



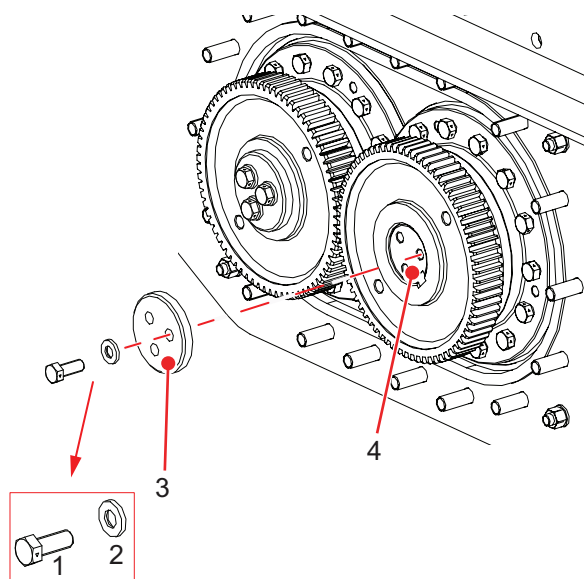
 **NOTICE**

GEARS ALIGNMENT

Gears are being reused use the marks made during disassembly for aligning the gear.

- 21. Slowly rotate the bars downwards to avoid injury.
- 22. Remove the pusher bolts from both gears.
- 23. Refer to “[Changing the Vibrator Gear Timing](#)” on page 7-10 for timing angle instructions.
- 24. Confirm the 45 degrees timing angle by checking the angle of the gears that lock the gears to the shaft.

25. Install the shaft cover to the shaft by tightening the bolts, and washers, refer to [“Torque Specifications” on page 15-1](#).



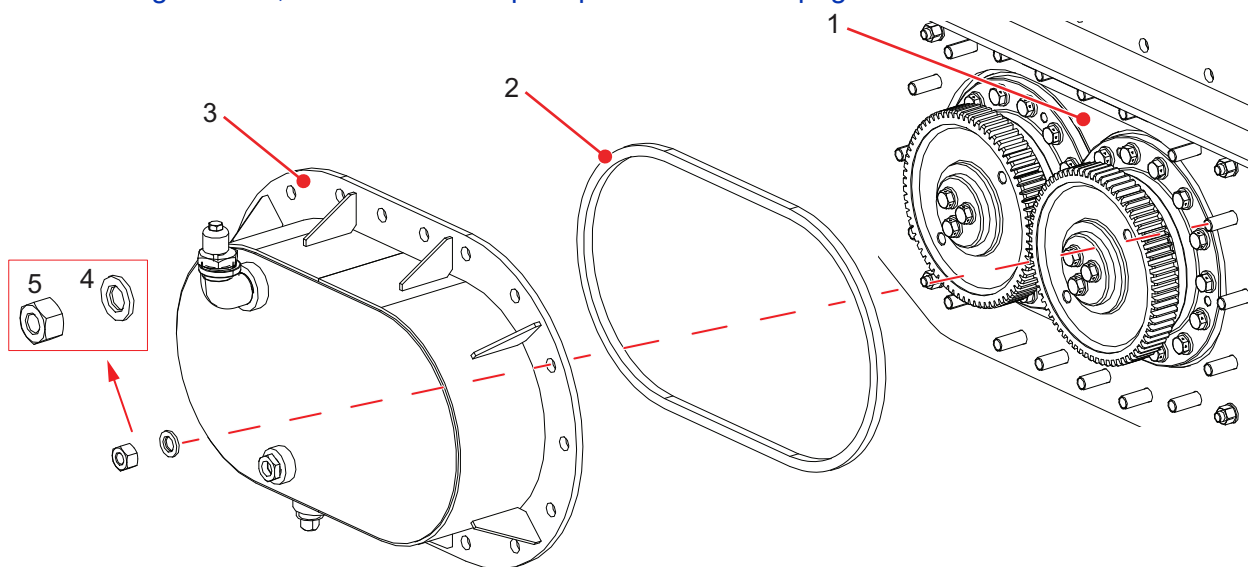
1	Bolt	3	Shaft Cover
2	Washer	4	Shaft

Figure 8-23: Shaft Cover Installation

26. Insert the lock wire into the gear shaft cover bolt heads and tie it

27. Attach an appropriate lifting device to the gear side cover.

28. Install the gear side cover and a rubber seal to the vibrator tube by tightening the nuts and washers, as shown in [Figure 8-24](#), and refer to [“Torque Specifications” on page 15-1](#).



1	Vibrator Tube	4	Washer
2	Rubber Seal	5	Nut
3	Gear Side Cover		

Figure 8-24: Gear Side Cover Assembly

29. Remove the lifting device from the gear side cover.

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9 Disassembly

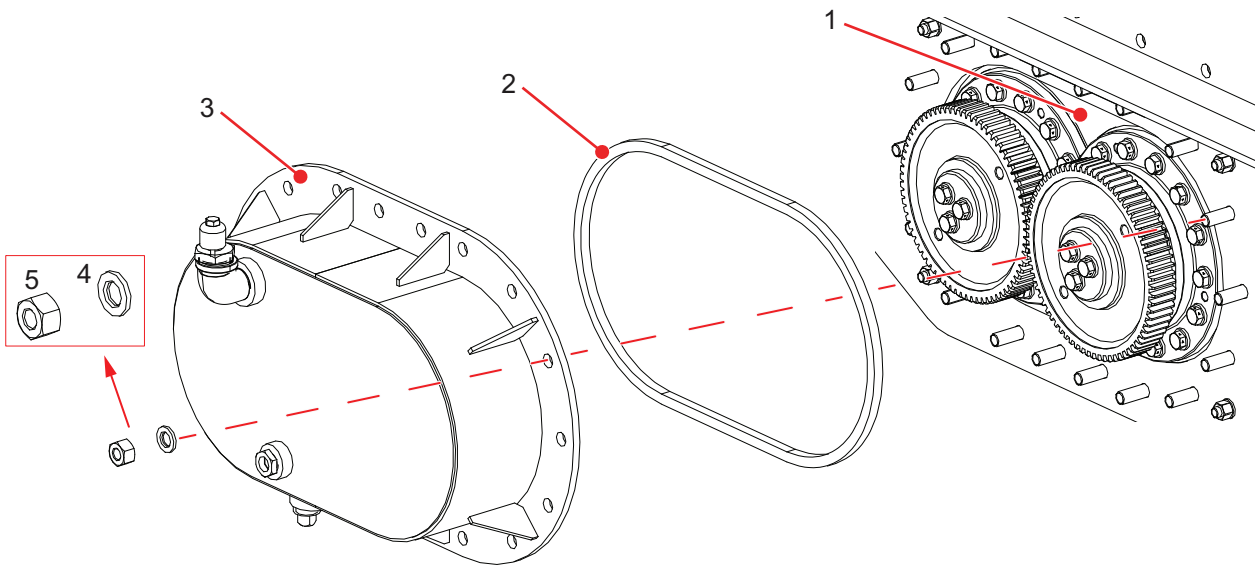
During disassembly operations, ensure all safety warnings and instructions are followed, as mentioned in “Safety” on page 2-1.

9.1 Vibrator Disassembly

9.1.1 Non-Drive Side Vibrator Disassembly

Follow the procedure for the gear side disassembly:

- 1. Follow the plant shut-down procedure and apply LOTO.
- 2. Remove the belt guard and drive belt from the feeder.
- 3. Drain the lubrication oil from the vibrating mechanism, refer to “Lubrication Oil Change Procedure” on page 11-4.
- 4. Attach an appropriate lifting device to the gear side cover.
- 5. Remove the gear side cover and rubber seal from the vibrator tube by removing the nuts and washers, as shown in Figure 9-1. Store the sheave side cover on a stable surface.

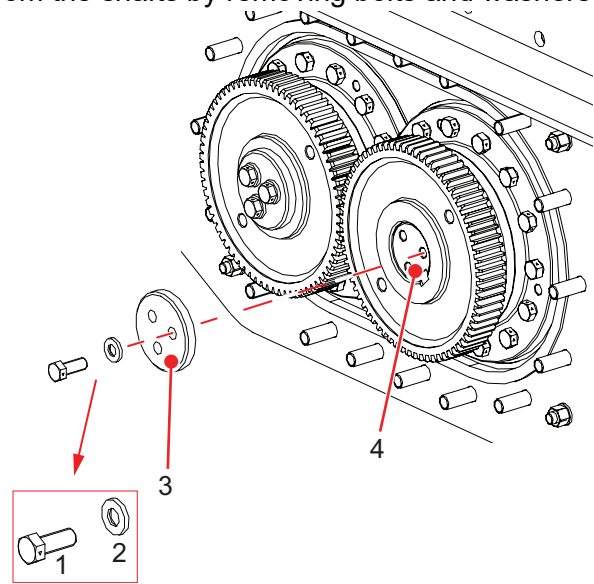


1	Vibrator Tube	4	Washer
2	Rubber Seal	5	Nut
3	Gear Side Cover		

Figure 9-1: Gear Side Cover Disassembly

- 6. Remove the lifting device from the gear side cover.

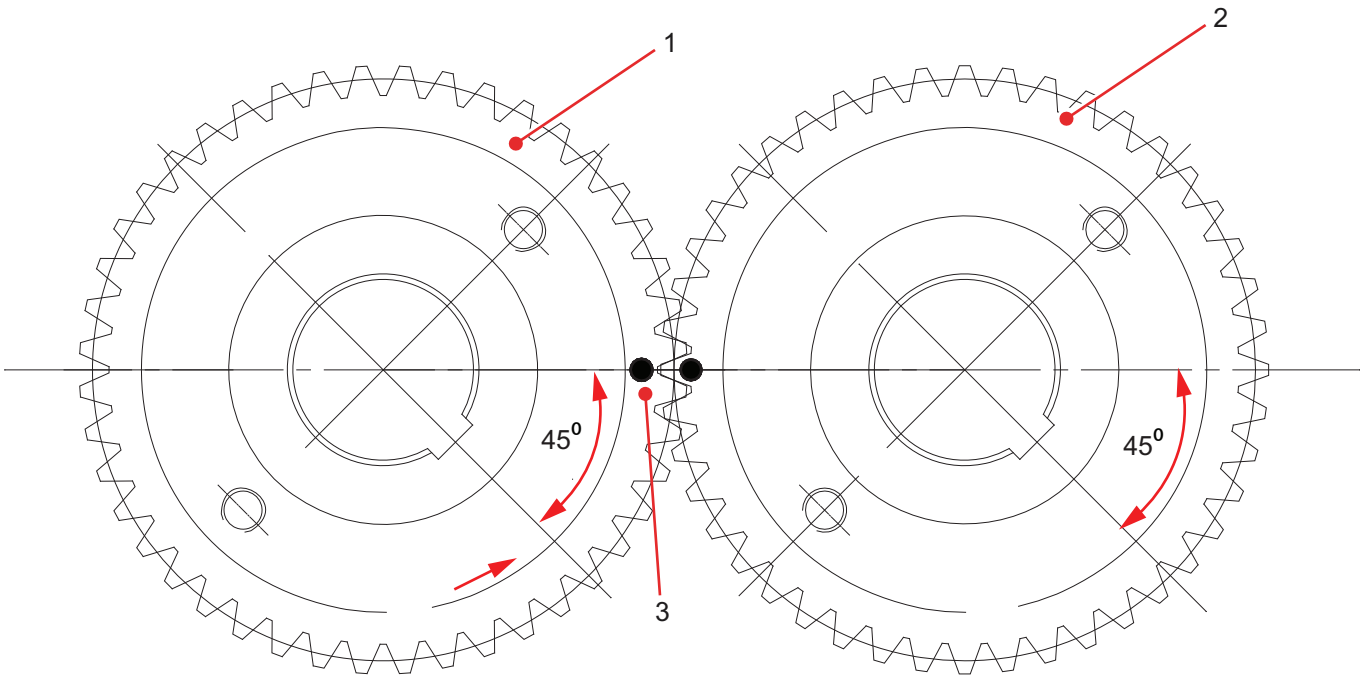
7. Remove the shaft cover from the shafts by removing bolts and washers and store it for reuse.



1	Bolt	3	Shaft Cover
2	Washer	4	Shaft

Figure 9-2: Shaft Cover Removal

8. Mark the gear timing angle before gears removal.



1	Gear 1	2	Gear 2
3	Disassembly Marking Angle		

Figure 9-3: Feeder Gear Timing

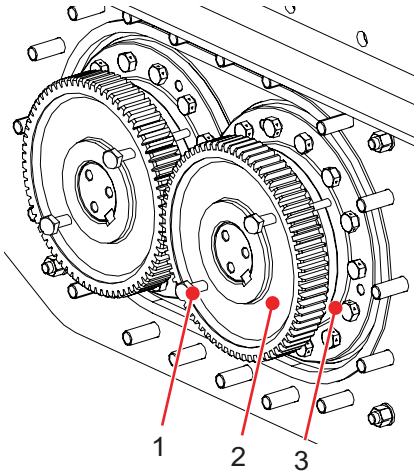


NOTICE

SECURE GEARS

Gravity will cause the shafts and gears to rotate when gears are unmeshed.

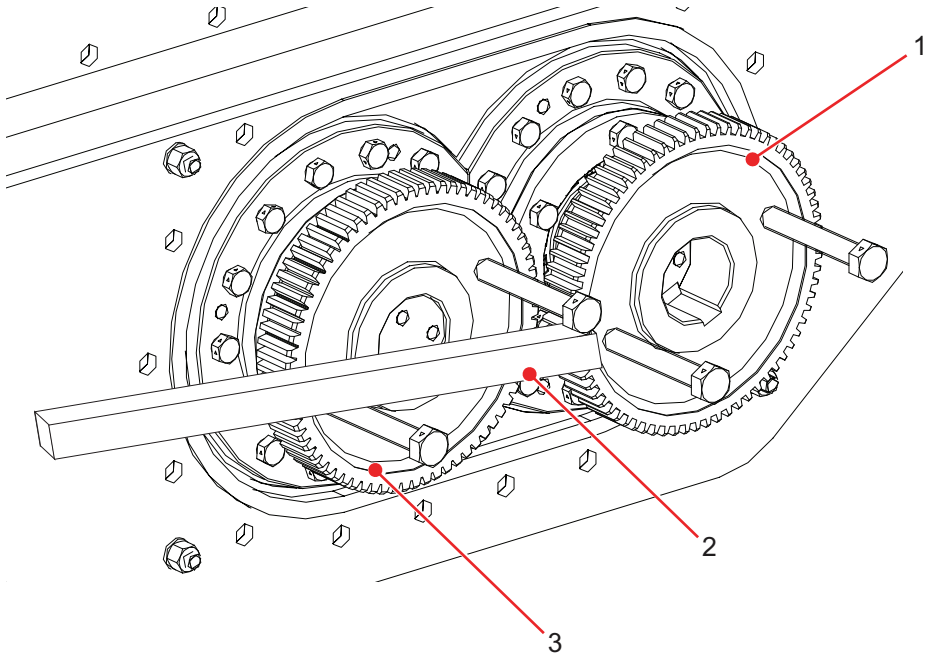
9. Install the two jacking screws to the gear 2, break the grip and disengage gear 2 from the shaft by tightening the jacking screws.



1	Jacking Screws	3	Bearing Cover
2	Gear 2		

Figure 9-4: Jacking Screws Installations

10. Lower the eccentric shafts down until resting in a neutral state. Continue pushing gear 2 off of the shaft.
11. Attach a lifting slings to the gear 2.
12. Remove the gear 2 from the shaft and store it.

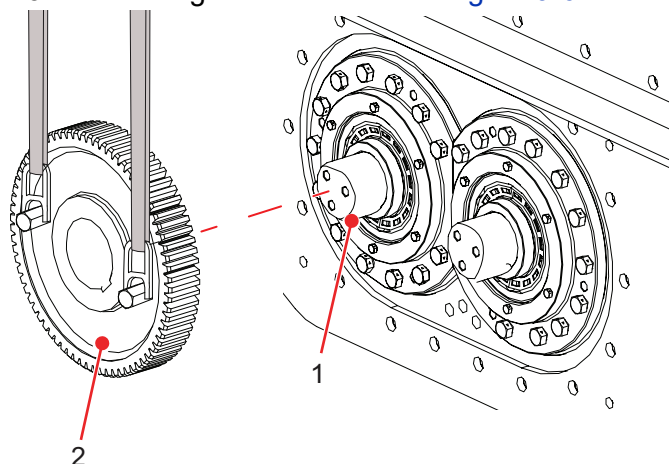


1	Gear 2	2	Bar
3	Gear 1		

Figure 9-5: Gear 2 Removal

13. Remove the appropriate lifting device from the gear 2.

14. Repeat the steps 9 to 13 to remove gear 1 as shown in [Figure 9-6](#).



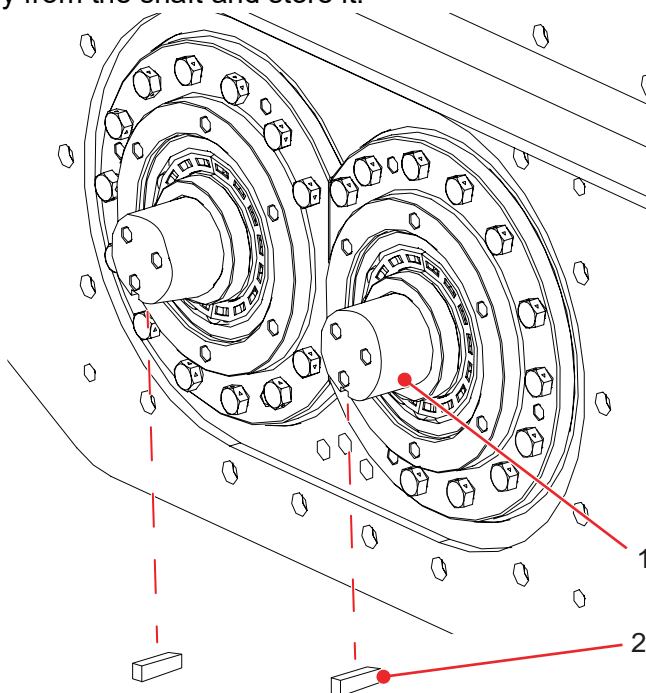
1	Shaft	2	Gear 1
---	-------	---	--------

Figure 9-6: Gear 1 Removal

15. Cut and remove the locking wires from the bearing cover bolts.

16. Remove the bearing cover from the bearing housing by removing the bolts and washers, refer [Figure 9-20 on page 9-12](#).

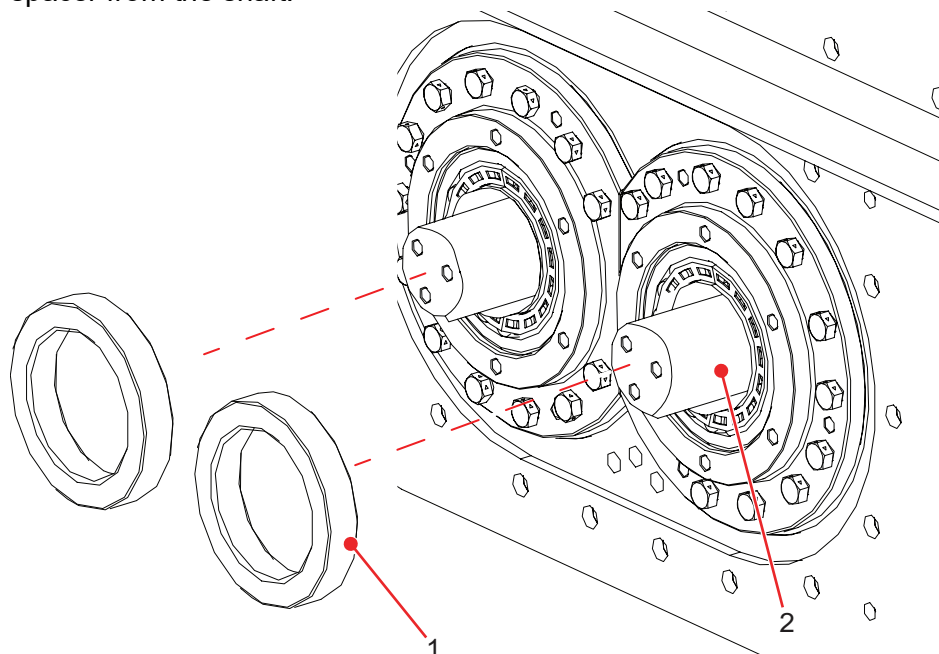
17. Remove the shaft key from the shaft and store it.



1	Shaft	2	Key
---	-------	---	-----

Figure 9-7: Shaft Key Removal

18. Remove the spacer from the shaft.



1	Spacer	2	Shaft
---	--------	---	-------

Figure 9-8: Spacer Removal

19. Cut and remove the locking wires from the bearing housing bolts.

20. Remove the bolts and washers from the bearing housing.

21. Attach an appropriate lifting device to the bearing housing.

22. Install the three jacking screws to the bearing housing, release and remove the bearing housing from the vibrator tube by tightening the jacking screws. See figure (Figure 9-21 on page 9-13).



CAUTION

DAMAGE TO SHAFT

Using a lifting sling or provide suitable support to maintain the shaft in the center position, to avoid bending and damage, as shown in Figure 9-21.

23. Place the bearing housing on the bench and remove the lifting device from the bearing housing.

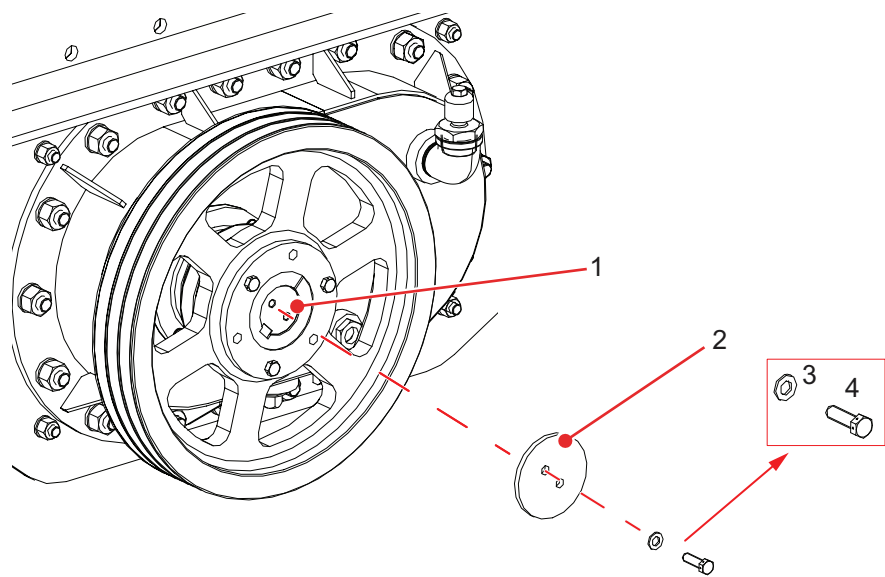
24. Remove and store the bearing from the bearing housing, by placing brass drift against outer race and striking drift with a hammer.

25. Use a hammer and drift in an alternating pattern around an outer race to ensure that it doesn't jam in the housing. See figure (Figure 9-22 on page 9-13).

9.1.2 Drive Side Vibrator Disassembly

Follow the procedure for the drive side disassembly:

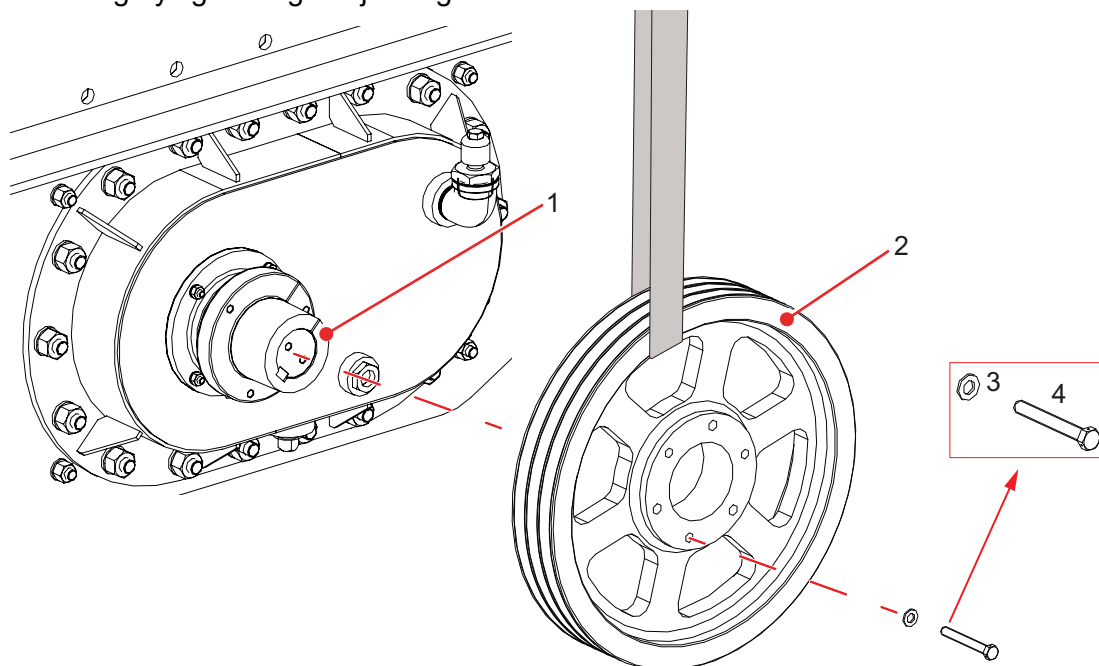
- 1. Follow the plant shut-down procedure and apply LOTO.
- 2. Remove the belt guard and drive belt from the feeder.
- 3. Remove the shaft cover from the shaft by removing the bolts and washers and secure it for reuse.



1	Shaft	3	Washer
2	Shaft Cover	4	Bolt

Figure 9-9: Shaft Cover Removal

4. Attach an appropriate lifting device to lift the driven sheave.
5. Remove the bolts and washers from the driven sheave.
6. Install the three jacking screws to the driven sheave, release and remove the driven sheave from the sheave bushing by tightening the jacking screws.



1	Sheave Bushing	3	Washer
2	Driven Sheave	4	Bolt

Figure 9-10: Driven Sheave Removal

7. Remove the lifting device from the driven sheave.

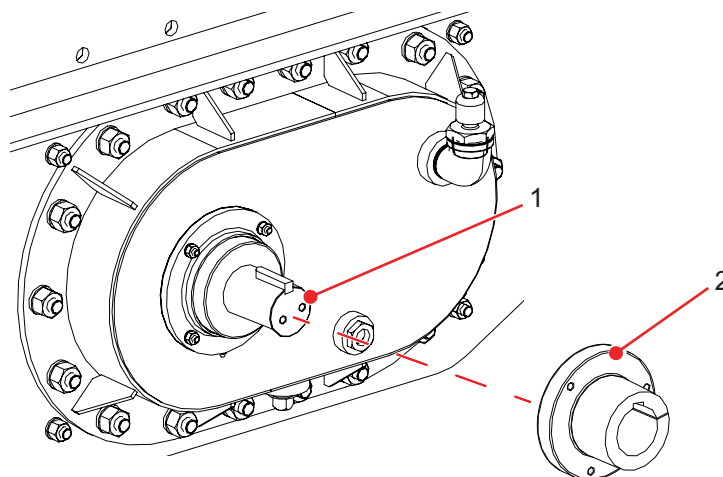


CAUTION

DAMAGE TO THE SHAFT

Carefully create a gap in the flange bushing split with the wedge in the flange portion of the bushing, not the seating portion.

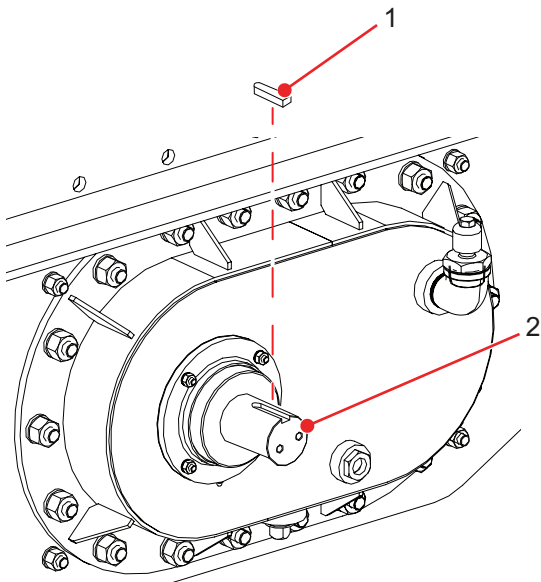
8. Insert a wedge into the flange bushing split and create a gap, remove the sheave bushing from the shaft and store it.



1	Shaft	2	Sheave Bushing
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Figure 9-11: Sheave Bushing Removal

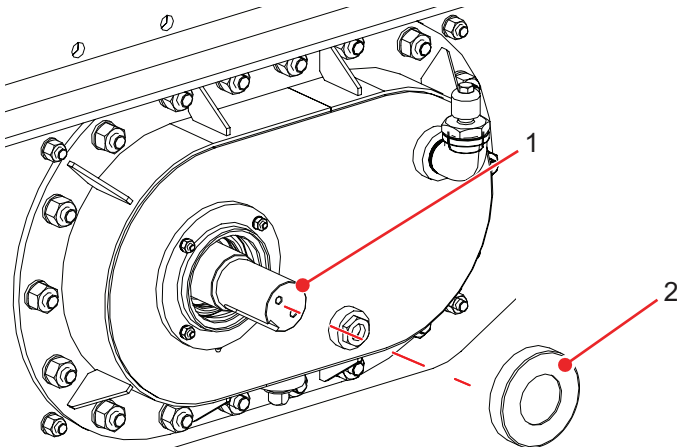
9. Remove the key from shaft keyway and store for reuse.



1	Key	2	Shaft
---	-----	---	-------

Figure 9-12: Key Removal

10. Remove the shaft enclosure from the shaft and store for reuse, as shown in [Figure 9-13](#).



1	Shaft	2	Shaft Enclosure
---	-------	---	-----------------

Figure 9-13: Shaft Enclosure Removal

11. Remove the drive shaft oil seal housing and gasket from the sheave side cover by removing the nut and washer, as shown in [Figure 9-14](#).

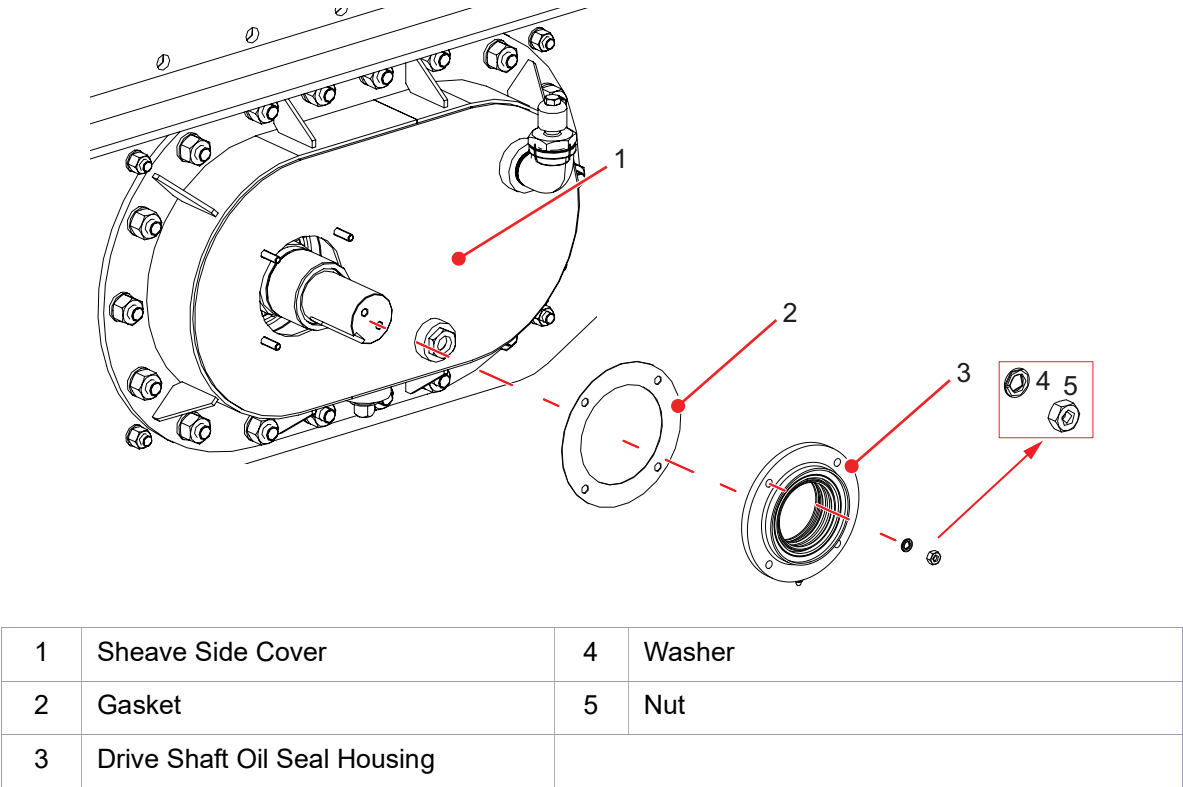


Figure 9-14: Drive Shaft Oil Seal Housing

12. Remove the drive shaft seal from the drive shaft oil seal housing and store it.

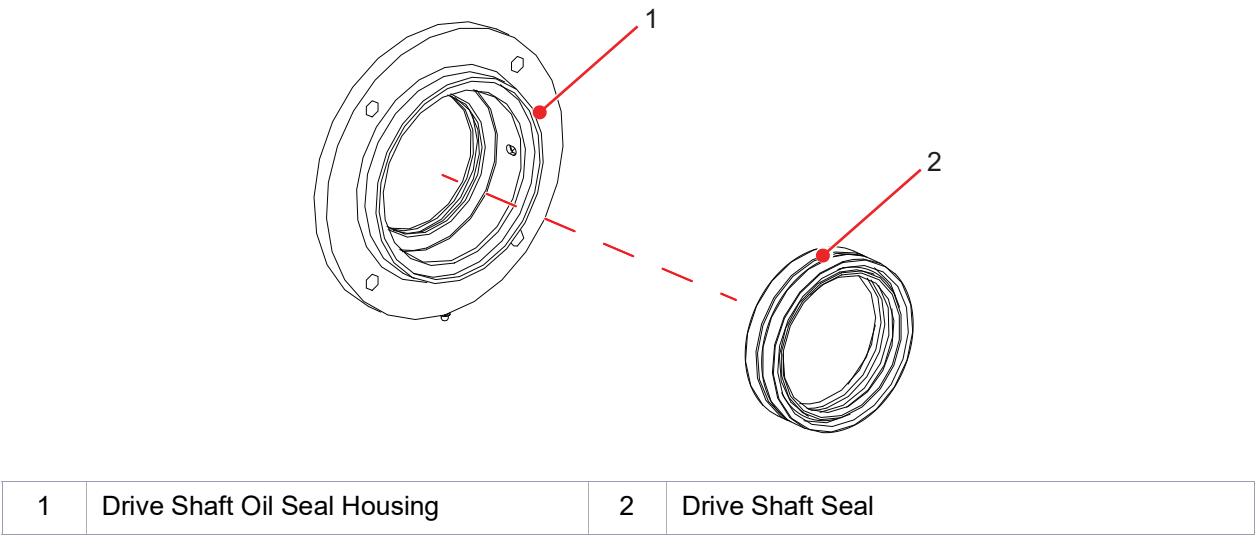
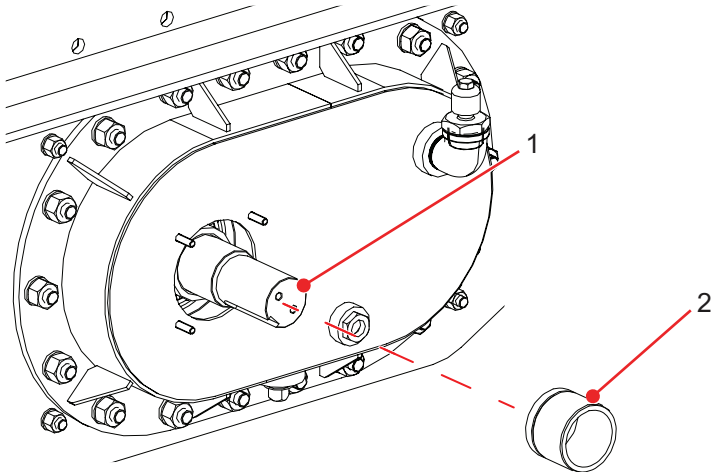


Figure 9-15: Drive Shaft Seal Removal

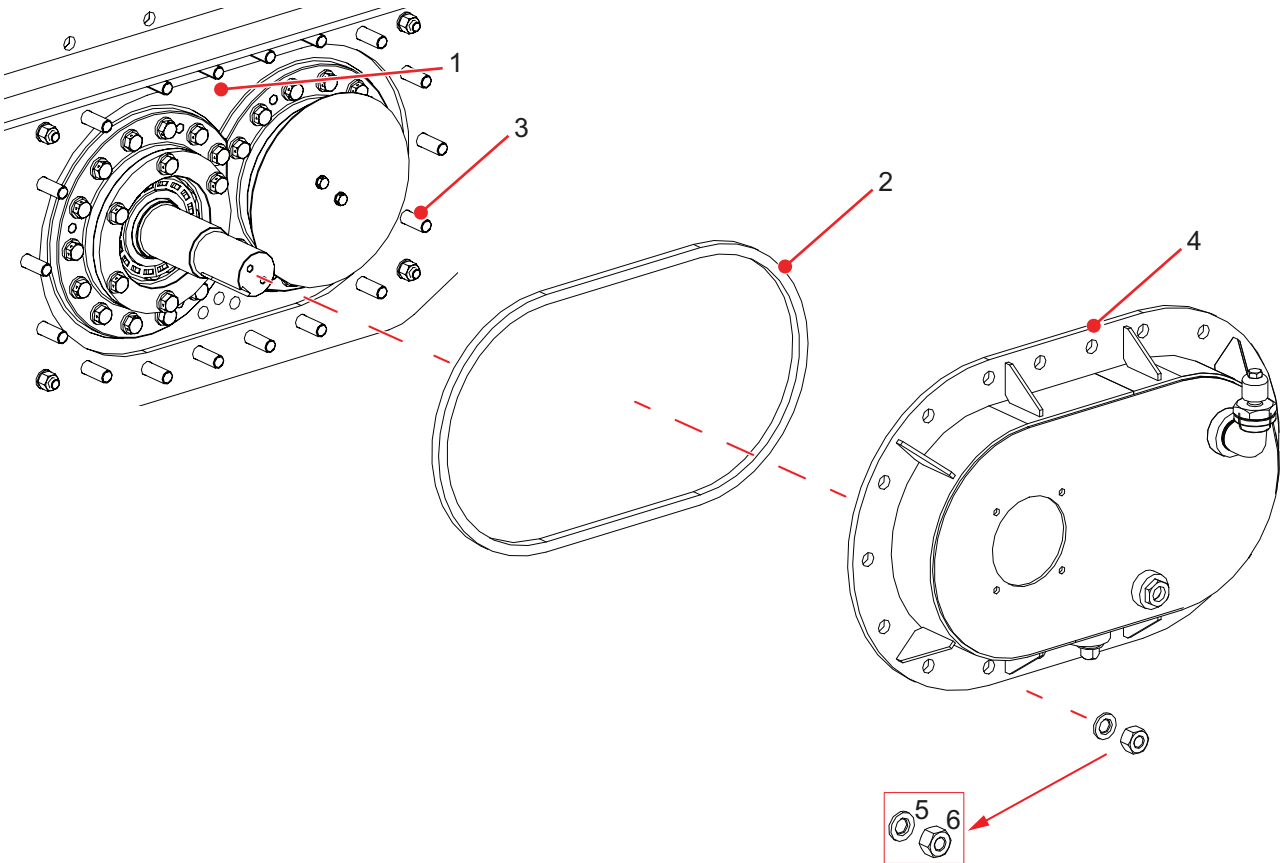
13. Remove the shaft sleeve from the shaft.



1	Shaft	2	Shaft Sleeve
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Figure 9-16: Shaft Sleeve Removal

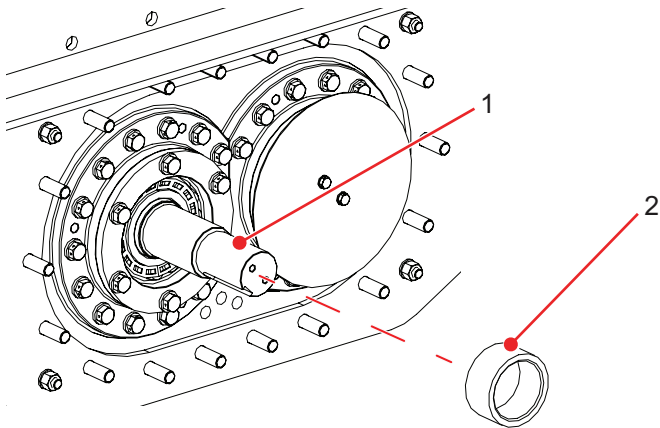
14. Attach an appropriate lifting device to the sheave side cover.
15. Remove the sheave side cover and rubber seal from the vibrator tube by removing the nuts and washers, as shown in [Figure 9-17](#). Store the sheave side cover on a stable surface.



1	Vibrator Tube	4	Sheave Side Cover
2	Rubber Seal	5	Washer
3	Bolt	6	Nut

Figure 9-17: Sheave Side Cover Removal

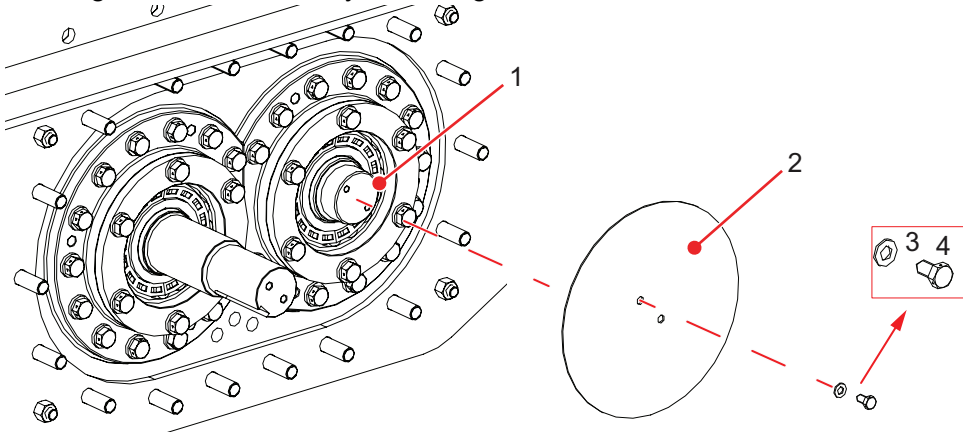
- 16. Remove the lifting device from the sheave side cover.
- 17. Remove the spacer from the shaft and store it.



1	Shaft	2	Spacer
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Figure 9-18: Spacer Removal

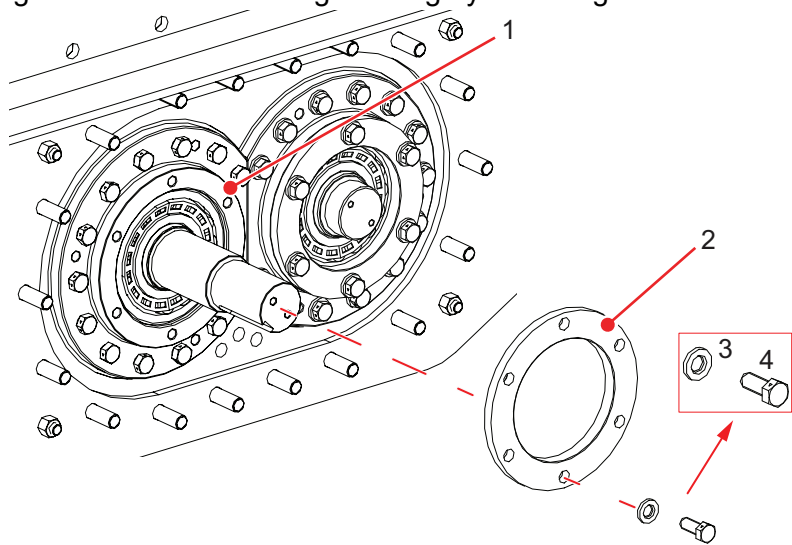
- 18. Remove the oil flinger from the shaft by removing the bolts and washers.



1	Shaft	3	Washer
2	Oil Flinger	4	Bolt

Figure 9-19: Oil Flinger Removal

- 19. Cut and remove the locking wires from the bearing cover bolts.
- 20. Remove the bearing cover from the bearing housing by removing the bolt and washer.



1	Bearing Housing	3	Washer
2	Bearing Cover	4	Bolt

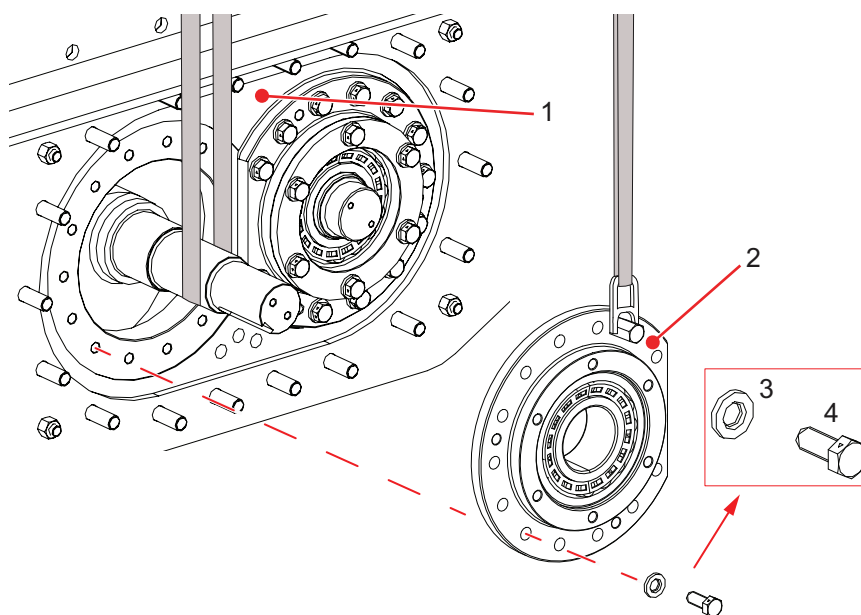
Figure 9-20: Bearing Cover Removal

- 21. Cut and remove the locking wires from the bearing housing bolts and remove the bolts and washers from the bearing housing.
- 22. Attach an appropriate lifting device to the bearing housing.
- 23. Install the three jacking screws to the bearing housing, release the bearing housing from the vibrator tube by tightening the jacking screws.
- 24. Remove the bearing housing from the vibrator tube, as shown in [Figure 9-21 on page 9-13](#).

CAUTION

DAMAGE TO SHAFT

Using a lifting sling or provide suitable support to maintain the shaft in the center position, to avoid bending and damage, as shown in [Figure 9-21 on page 9-13](#).



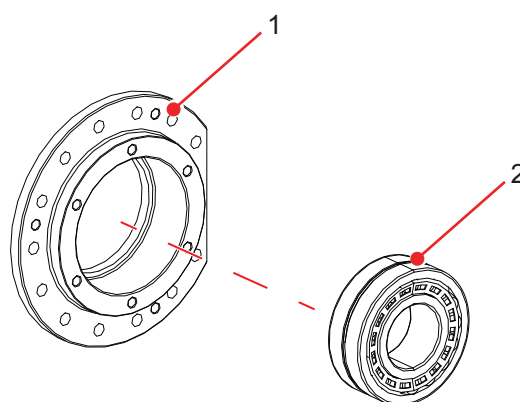
1	Vibrator Tube	3	Washer
2	Bearing Housing	4	Bolt

Figure 9-21: Bearing Housing Removal

**NOTICE****POSITION OF BEARING HOUSING**

If the disc is placed wrong side up, you cannot remove the bearing out from the bearing housing. The flange on the inside face of the bearing housing has a lip that holds the bearing in place, the flange on the outside face does not have a lip.

25. Place the bearing housing on the bench and remove the lifting device from the bearing housing.
26. Remove and store the bearing from the bearing housing, by placing brass drift against outer race and striking drift with a hammer. Use a hammer and drift in an alternating pattern around an outer race to ensure that it doesn't jam in the housing.



1	Bearing Housing	2	Bearing
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Figure 9-22: Bearing Removal

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10 Maintenance

Only authorized persons must perform maintenance activities.

As the equipment is used, worn parts must be adjusted and/or replaced to keep the machinery operating efficiently and properly. Do the periodic maintenance as per the manufacturer's recommendations.

During a periodic inspection, if any items need attention for repair, replacement or adjustment, report it immediately. A minor defect could result in more severe equipment damage if the machine is continued to be operated.

10.1 Maintenance Safety

	 DANGER
	RISK OF PERSONNEL INJURY OR DEATH DISCONNECT ALL ENERGY SOURCES AND APPLY LOCKOUT AND TAGOUT. ENSURE THAT THE MACHINE IS COMPLETELY STOPPED BEFORE DOING ANY MAINTENANCE ACTIVITY.

	 WARNING
	RISK OF PERSONNEL INJURY During feeder operation, do not do the following: • Perform maintenance work such as lubricating the parts of the feeder. • Stand beside the feeder while it is operating.

	 WARNING
	RISK OF PERSONNEL INJURY DURING CRANE OPERATION • When the crane is in operation, do not stand in the area. • Do not stand, crawl, walk, or lay under the load, hanging from a crane. • There is a risk of fall of the suspended loads.

10.2 Inspection

Scheduling a periodic inspection is the only way to keep the feeder performing safely and efficiently.

10.2.1 Start-up

All new equipment needs to be checked according to the maintenance schedule for start-up, as shown in [Table 10-1](#). This also needs to be done after a major shut-down or after the equipment was standing for a long period without use.

Inspection Parts	Interval	Work	Remarks
Fastening Bolts	First 24 hr	Check if fasten (M)	Torque again if loose. Refer to “Torque Specifications” on page 15-1 for torque values.
Oil	First 40 hrs	Initial oil change (A)	
	300 - 500 hrs/ 1 months	Second oil change (A)	Sample oil every 500 hrs
	1000 hrs/ 6 months	Third oil change (A)	Change oil every 1000 hrs

Table 10-1: Maintenance Schedule for Start-up

M - Measurements

A - Action

10.3 Maintenance Check Lists

FEEDER NAME: _____

SERIAL NO.: _____

DATE: _____

CHECKED BY: _____

Maintenance Check List For Feeder		Daily	Weekly	Fortnightly	Monthly	3 Monthly	Checked	Action Required		Action Taken
								No	Yes	
Stop the machine and lock out. Remove all guards if required.										
1.	Check the gear case side or non drive side and drive side to make sure that the bolts are not loose.				x					
2.	Check that all vibrator bolts are tight.				x					
3.	Check the oil level and top up if necessary. Follow the instruction in the O&M manual. Check for any oil leaks.	x								
4.	Clean air breather.		x							
5.	Check the feeder body for material build-up and spillage.	x								
6.	Check that all bolts are tight and locked. Re-torque regularly.			x						
7.	Check sheave and bushing. Tighten if necessary.	x								
8.	Inspect air breather.	x								
9.	Check vibrator housing temperature.	x								
10.	Inspect the feeder frame deformation, broken welds, cracks, etc. Repair as required.				x					
11.	Inspect all liners and grizzly bars for wear.				x					
12.	Grease the seal as prescribed in Lubrication section.		x							
13.	Check electrical connections and wiring.		x							
14.	Check suspension springs for any broken coils and deformation.	x								

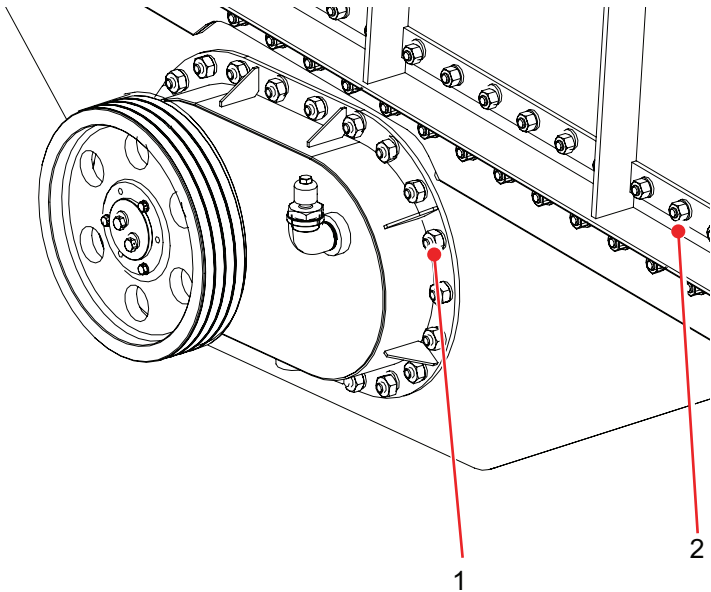
Table 10-2: Maintenance Check Lists

Maintenance Check List For Feeder		Daily	Weekly	Fortnightly	Monthly	3 Monthly	Checked	No Action Required	Yes	Action Taken
15.	Clean the material build-up from the feeder. Clean the material away from the support springs, drive guard, motor base and all moving parts. Clean out all of the hoppers and chutes of material build-up near the feeder.		x							
16.	Make a visual inspection of the feeder				x					
17.	Check the pivot type motor base for freedom of movement.		x							
18.	Check V belt condition and tension. Check the alignment of the sheaves.		x							
Start the machine (Make sure the guards are fitted)										
19.	Check that the feeder is not bumping or touching any static items whilst starting, running, or shutting down.	x								
20.	Listen for any noises on the feeder, indicating loose parts.	x								
21.	Inspect and listen for any loose bolts on the guards.	x								
22.	Check Wear on snubber assemblies				x					

Table 10-2: Maintenance Check Lists (Continued)

10.4 Gear Case Bolt Torque

After the first week run, recheck the tightness of all bolted connections, refer to [Table 10-3 on page 10-4](#). Re-torquing the feeder involves tightening the bolts, nuts, and ensure that the bolts are not too tight or too loose. Too tight, and the nuts can damage the bolt threads or break fasteners.



1	Gear Case Bolt	2	Feeder Body Bolts
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Figure 10-1: Gear Case Bolt

Bolt Location	Size	Torque Value		Frequency
		Dry	Lube	
Gear Case Bolts	M20 Grade 10.9	563 Nm (415 ft-lbs)	433 Nm (319 ft-lbs)	* Every 500 Hours
	or M24 Grade 10.9	956 Nm (705 ft-lbs)	736 Nm (543 ft-lbs)	
* Re-torque after first 40 hours of operation and then after 200 hours of operation. Finally follow by checking torque every 500 hours.				

Table 10-3: Torque Values for Bolts



 **WARNING**

RISK OF COMPONENT DAMAGE

Loosen gear case bolts can lead to leakage of lubricant and can damage the vibrator assembly components.

10.4.1 Bearing Temperature

The shafts are subject to continual stress, it is important to check bearing temperatures periodically.

The heat from bearings transfers to the oil, check an average bearing operating temperature by monitoring oil temperature.

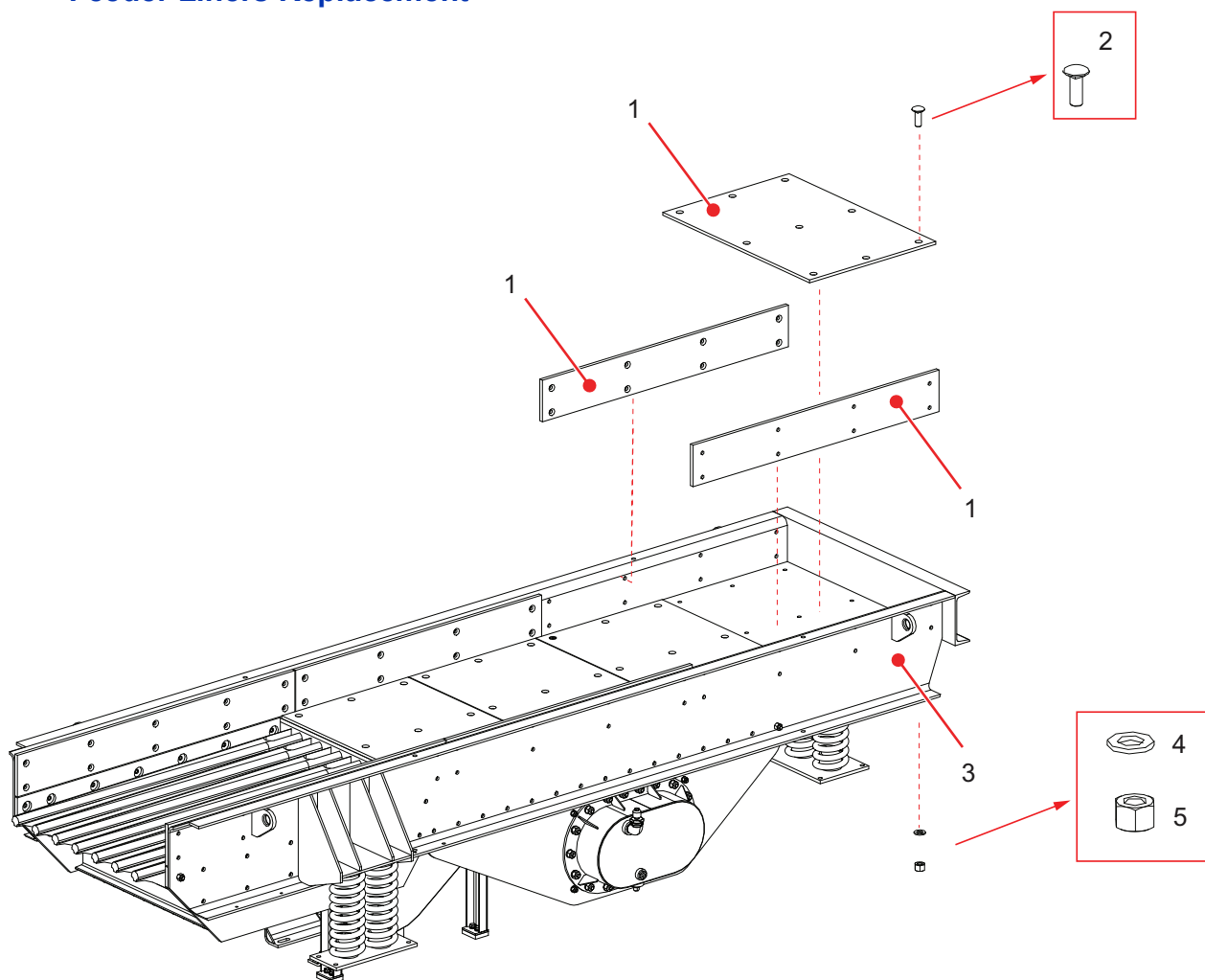
Place a magnetic-type thermometer on the gear case cover below the oil level plug. During normal operation, temperature should fall between 50° - 80° C (130° - 180° F). If temperature reaches 93° C (200° F) or higher, stop operation, determine the cause, and take corrective steps to lower the temperature to the proper range.

The most common causes of elevated temperatures are:

- Oil level is too high.
- Wrong type or grade of oil.
- Feeder being operated at a faster speed than recommended.
- Feeder basket not level.

Overheating could also be an indicator that bearings may be beginning to fail.

10.5 Feeder Liners Replacement



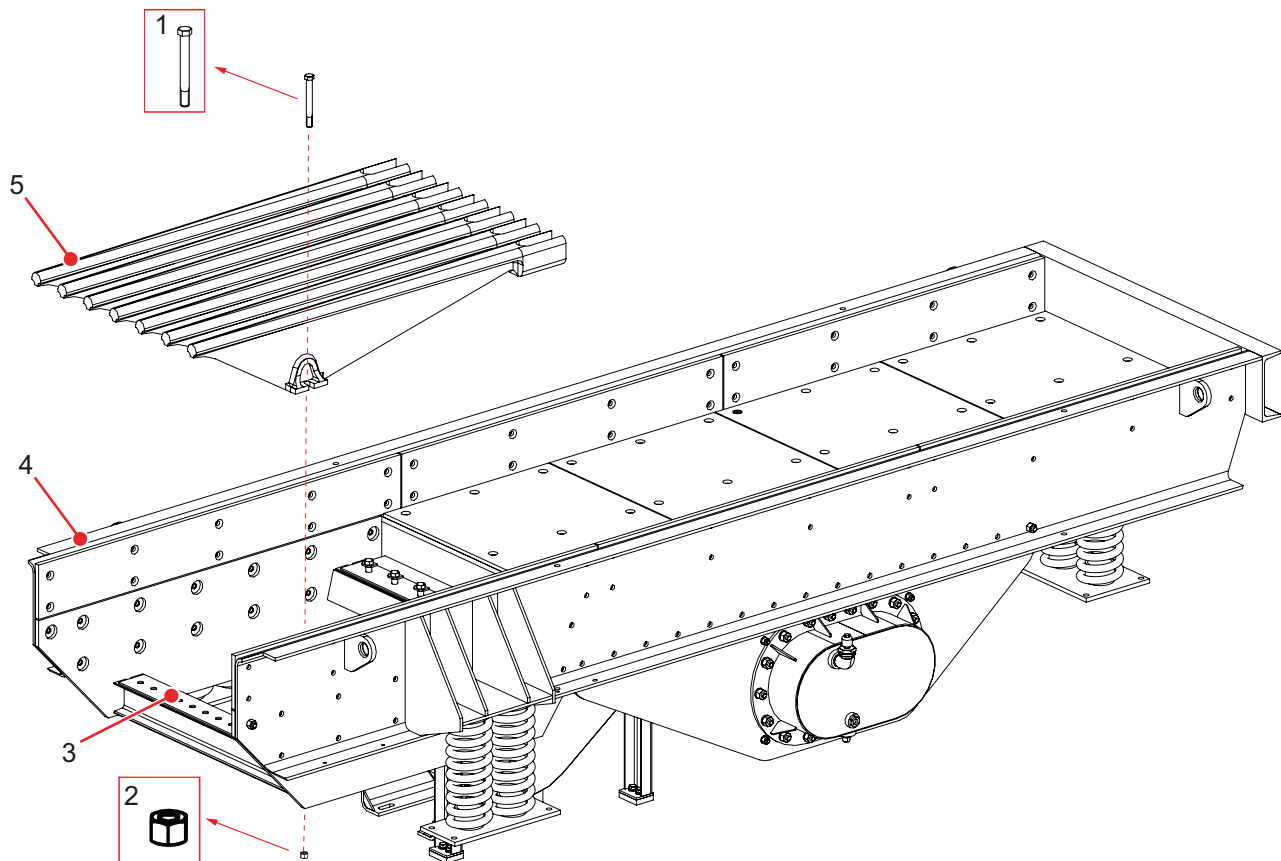
1	Liner	4	Washer
2	Carriage Bolt	5	Cleveloc Nut
3	Feeding Box		

Figure 10-2: Liner Replacement

Follow the steps to replace the liner in the main frame pan:

1. Follow the plant shut-down sequence and apply LOTO.
2. Remove the cleveloc nuts, washers, and carriage bolts from the liner.
3. Use an appropriate lifting device to lift the liner.
4. Remove and place the liner on a stable surface and remove the lifting device from the liner.
5. Attach lifting equipment (if required) to the new liner. Align the holes of the liner with main frame pan holes and install it by means of carriage bolt, washer and cleveloc nut, as shown in [Figure 10-2](#).
6. Remove the lifting equipment from the liner, if it is attached.
7. Follow the same procedure to replace the remaining liners.
8. After liners replacement, follow the plant start-up sequence.

10.6 Grizzly Bar Replacement



1	Hexagonal Bolt	4	Feeder Frame
2	Nut	5	Grizzly Bar
3	Frame and Pad		

Figure 10-3: Grizzly Bar Replacement

Follow the steps to replace the grizzly bar:

1. Follow the plant shut-down sequence and apply LOTO.
2. Remove the cleveloc nuts, hexagon bolts from grizzly bar frame [Figure 10-3](#).
3. Attach an appropriate lifting device to the grizzly bar.
4. Remove and place the grizzly bar on a flat surface and remove the lifting device from the grizzly bar.
5. Attach a lifting device to the new grizzly bar and align the grizzly bar holes with grizzly bar frame holes.
6. Install the grizzly bar on the grizzly bar frame by tightening the bolts, washers, and cleveloc nuts, refer to [“Torque Specifications” on page A-I](#).
7. Remove the lifting device from the grizzly bar.

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11 Lubrication

11.1 General Information

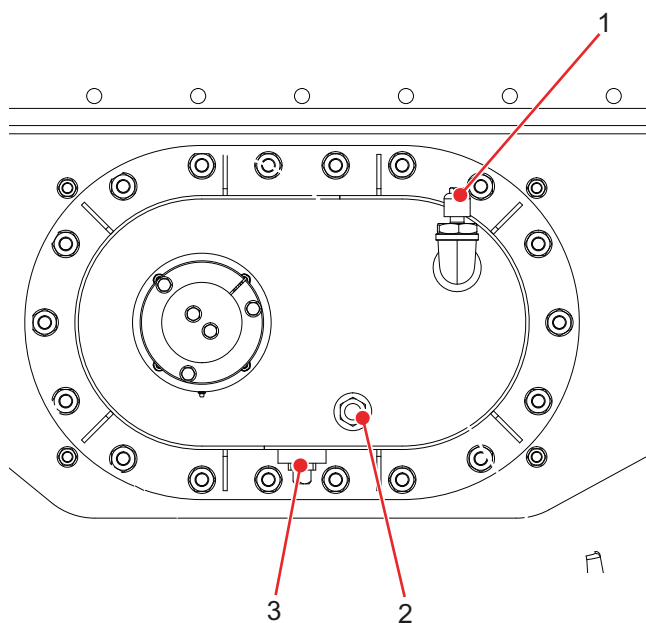
Lubrication procedures are performed primarily to deliver grease or oil to rotating or sliding surfaces to reduce friction and carry away heat. Pressure lubrication is also used to expel contaminants from seals and other openings. Certain lubricants are used to protect parts from corrosion.

11.2 Oil Lubrication

Maintaining adequate lubrication in the machine is the most important maintenance function. Many premature failures can be traced to inadequate or improper lubrication. Therefore, lubrication of the machine must be scheduled into the daily maintenance schedule.

The oil level sight gauges should have oil visible in the center of the site gauge on both sides to make sure oil is at the proper level. See [Figure 11-1](#). If level is low, add the proper lubricant as described in [Table 11-1 on page 11-2](#)

- During initial start-up of the new machine, change the oil after first forty (40) hours of operation. This procedure should also be followed if the machine has been idle for an extended period of time.
- Oil change refer to maintenance schedule on [Table 10-1 on page 10-1](#).
- Drain the oil immediately upon shut-down so any contaminants that are suspended are easily flushed out. Refer to [Figure 11-1](#) for drain plug details.
- If, during draining of the oil, contaminants are noticed, it is highly recommended that the lubrication mechanism is flushed with light EP oil (EP-2 or EP-3). The flushing procedure is as follows:
- Fill the gear cases to the proper level with light EP oil.
- Run the machine for approximately fifteen (15) minutes.
- Immediately drain the oil out of the machine upon shut-down. This picks up additional contaminants that otherwise would have remained in the system.



1	Oil Fill and Breather Cap	3	Drain Plug
2	Oil Level Sight Gauge		

Figure 11-1: Gear Case Oil Level (Drive Side Shown)

	 CAUTION
	EQUIPMENT DAMAGE
	Weir Trio feeders are shipped without oil in the gear cases, make sure to fill the recommended oil before operating the feeder. Do not overfill gear cases. Excessive oil can cause high oil temperatures in gear cases.

Table 11-1 provides the recommended lubrication quantities for different models and sizes of the feeder.

Feeder Model	Quantity (gallons)	Quantity (litres)
TF4010	7.5 - 10	28.4 - 37.9
TF4012	7.5 - 10	28.4 - 37.9
TF4016	7.5 - 10	28.4 - 37.9
TF4616	9 - 11	34.1 - 41.6
TF4620	9 - 11	34.1 - 41.6
TF5217	10 - 12	37.9 - 45.4
TF5220	10 - 12	37.9 - 45.4
TF5224	10 - 12	37.9 - 45.4
TF5820	18 - 20	68.1 - 75.7
TF5824	18 - 20	68.1 - 75.7
TF6420	20 - 22	75.7 - 83.2
TF6424	20 - 22	75.7 - 83.2
TF7220	17	65
TF7224	25	95

Table 11-1: Feeder Gear case Quantities

Lubricant service life can be extended if feeder has an optional filter pack installed. In this case the oil must be sent for testing.

Testing should be done for contamination of:

- (FE) Iron start at 55-60
- Chrome
- Aluminum
- (CU) Copper
- (SI) Silicon
- (H2O) Water

	 NOTICE
	OIL TESTING
	• Needs (90ml) of oil for testing. • Do not take the first start of the oil. • Do not mix oil brands in one machine.

11.2.1 Oil Lubricant Specifications

The lubricating oil should be extreme pressure lubricant with a non-foaming paraffinic base.

Lubricant Type	Ambient Temperature ° F (° C)	Specification	Recommended Standard Lubricant	Recommended Synthetic Lubricant
Grease	All Temperatures	NLGI 2, with EP Additives	Mobil Delvac Xtreme	Mobilith SHC
Gear case Oil	Below 40 (Below 4)	ISO 100, AGMA 3 EP	Mobilgear 600, XP 100	-
	40 to 80 (4 to 27)	ISO 150, AGMA 4 EP	Mobilgear 600, XP 150	Mobilgear SHC 150
	80 to 95 (27 to 35)	ISO 220, AGMA 5 EP	Mobilgear 600, XP 220	Mobilgear SHC 220
	Above 95 (Above 35)	ISO 320, AGMA 6	Mobilgear 600, XP 320	Mobilgear SHC 320

Table 11-2: VGF Feeder Lubricant Specifications

i

NOTICE

MOTOR START UP

Do not start motor until the gear case has been filled with oil.

The recommended lubrication characteristic for the feeder are given in [Table 11-3](#):

Specifications	Ambient Temperature		
	Below 40° F (4.5°C)	40-80°F (4.5-27°C)	Above 80°F (27°C)
ISO Grade	ISO 100 EP Gear Oil	ISO 150 EP Gear Oil	ISO 220 EP Gear Oil
AGMA Grade #250	3 EP	4 EP	5 EP
Viscosity SUS@100°F	469-575	709-871	1047-1283
Viscosity SUS@210°F	61-67	74-82	90-101
Viscosity Index (V.I.)	95+	95+	95+
Timeken OK Loads (LBS.)	60+	60+	60+
Extreme Pressure (EP)	Yes	Yes	Yes
Rust & Oxidation Agent	Yes	Yes	Yes
Anti Foaming Agent	Yes	Yes	Yes
Copper Strip Corrosion Test	Pass	Pass	Pass
- Oil must not be corrosive to metal generally used in bearings, shaft, pumps and housings. - Oil must have no effect on rubber seals, synthetic seals, packing or hoses.			

Table 11-3: Grades Lubricant Specifications

11.2.2 Oil Change Specifications

Change the oil every 400-500 hours or if oil samples show high contaminant levels.

Oil samples should be collected after every 250 hours of operation and sent to a professional lab for analysis.

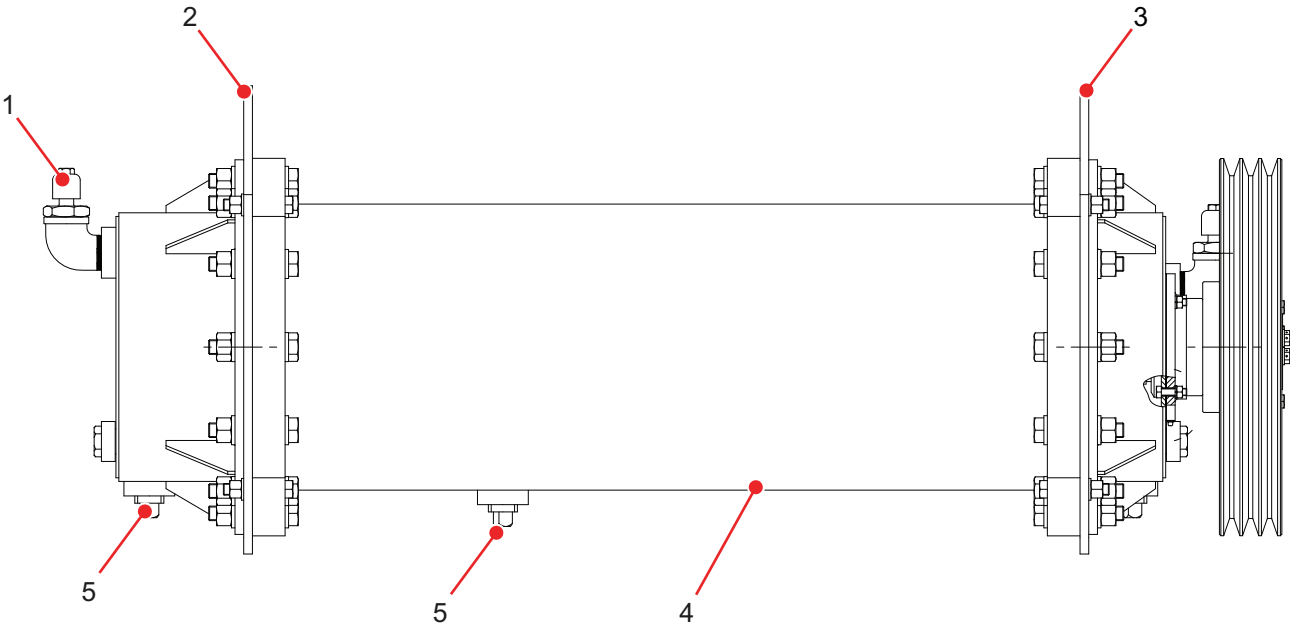
The acceptable ranges in PPM (parts per million) are listed in [Table 11-4](#).

Contaminant	Acceptable Range
Iron	125-150 (start at 55-60)
Chrome	25-30
Aluminum	45-50
Copper	100-125 (start at 5-7)
Silicone	25-30
Water	0

Table 11-4: Viscosity Specifications

11.2.3 Lubrication Oil Change Procedure

To drain oil from and refill the gear cases:



1	Breather Cap	4	Vibrator Tube
2	Gear Case Drive Side	5	Drain Plug
3	Gear Case Non Drive Side		

Figure 11-2: Lubricating of Vibrator Assembly

1. Follow the plant shut-down procedure and apply LOTO.
2. Remove drain plugs from the gear cases and vibrator tube, refer [Figure 11-2](#).
3. Allow oil to drain into the containers for recycling or disposal.
4. Clean the drain plugs and re-install to the gear cases and vibrator tube.

i NOTICE

Breather Cap

- Do not allow breather cap on either gear case to become plugged. A plugged breather cap could result in damage to seals and bearings.
- Do not replace the cap with an air-tight cap. Keep dirt and material build-up away from the caps.

- 5. Remove the oil level sight gauge from the both gear cases, refer [Figure 11-1 on page 11-1](#).
- 6. Remove the breather caps from the both gear cases, refer [Figure 11-2](#).
- 7. Clean the openings of all loose material and build-up.



 **NOTICE**

Oil Level

- The oil level should be at check plug not free flowing.

- 8. Fill the gear cases to recommended level as shown in [Figure 11-1 on page 11-1](#), using the recommended lubricant as shown in [Table 11-3 on page 11-3](#).
- 9. Clean the breather caps and oil level sight gauges and re-install on to the gear cases.

11.2.4 Oil Seal Grease

Grease the oil seal located on the drive shaft with 28 grams (1 oz.) from a hand operated grease gun every 40 hours of operation to prevent material from entering into the vibrator gear case. See ([Figure 11-3](#)).

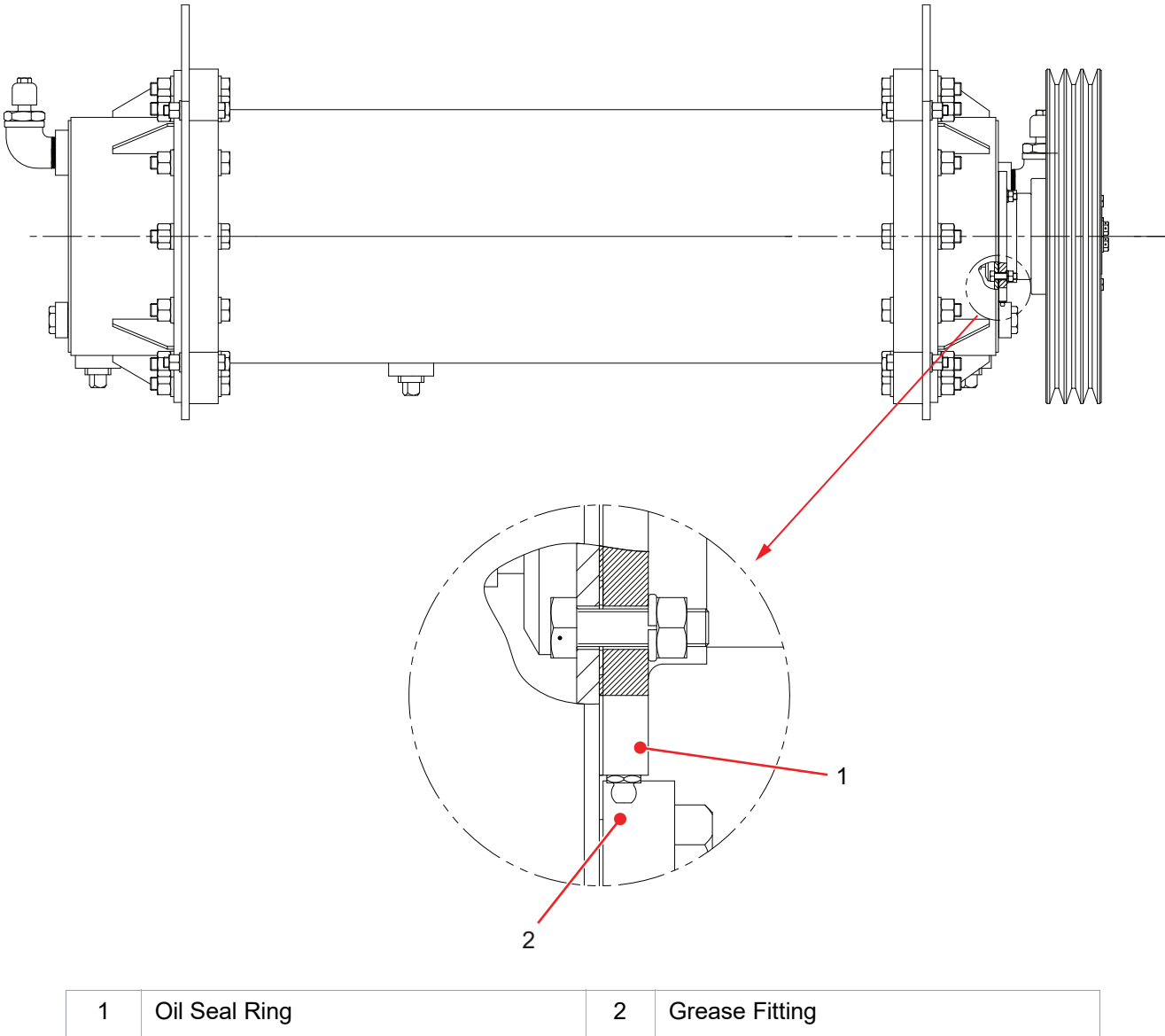


Figure 11-3: Grease Fittings

11.2.5 Bearing Temperature

The operating temperature range for bearing is between 50 °C - 80 °C (130 °F - 180 °F).

If the oil temperature reaches 93 °C (200 °F) or higher, determine that cause and take corrective steps to bring the temperature back to the operating range.

Some common causes of elevated oil temperatures are:

- Excessive oil in the tube
- Wrong type or grade of gear lubricant
- Feeder is operating faster than recommended

Overheating could be a sign that the bearings are beginning to fail.

- Take an oil sample to check for higher than recommended contaminant levels.
- Inspect and replace bearings if all indicators point to worn bearings.

11.3 Lubrication Interval

Item	Daily	Weekly or 40 Operating Hours	400-500 Operating Hours	1000 Operating Hours
Gear Case Oil Levels (Both Sides)	Check	(Change after initial 40 hours) Then Check Weekly	Sample Oil	Change
Gear Case Oil Seal	Check	28 grams (1 oz)		

Table 11-5: Lubrication Intervals

11.4 Cold Weather Operation

In extreme cold weather, it is recommended to lubricate the machine during mid-shift or at noon time. In case the machine is lubricated at night or in morning, before the unit is started, then it would require additional horse power to run the mechanism.

If pre-heating is necessary, then it is recommended to use a series of heat lamps (or a salamander type heater) which are directed towards the bearing and/or shaft assembly. Do not use an acetylene torch.

12 Decommissioning and Disposal

This section describes the decommissioning and disposal of the feeder units.

12.1 Decommissioning Safety

	 DANGER LIFTING EQUIPMENT SAFETY • THE EQUIPMENT MAY ONLY BE OPERATED OR MAINTAINED BY TRAINED PERSONNEL, WHO HAVE DEMONSTRATED THEIR ABILITY TO DO SO SAFELY. • USE LIFTING EQUIPMENT TO LIFT HEAVY OR AWKWARD COMPONENTS. • THE LIFTING EQUIPMENT MUST BE IN GOOD CONDITION, CERTIFIED AND TAGGED. • THE LIFTING EQUIPMENT MUST BE OF ADEQUATE CAPACITY AND MUST BE USED WHENEVER THEY ARE REQUIRED. • PERSONNEL MUST NEVER WORK UNDER SUSPENDED LOADS.
	 DANGER LIFTING POINT SAFETY REFER TO THE LIFTING INSTRUCTIONS AT ALL TIMES. • TAPPED HOLES (FOR EYE BOLTS) AND CAST-ON LUGS (FOR SHACKLES) ON WEIR PARTS ARE FOR LIFTING INDIVIDUAL PARTS ONLY. • SOME HEAVY PARTS OF THE UNIT HAVE THREADED HOLES FOR LIFTING. DURING ASSEMBLY, EYE BOLTS ARE SCREWED INTO HOLES TO ENABLE THE PARTS TO BE LIFTED WITH A CRANE.
	 WARNING TRIPPING OR FALLING During decommissioning, ensure that the feeder components are secured properly, so that they do not fall or tip over, or cause injury.
	 WARNING SHARP EDGES • Before starting any work, identify and examine any parts that could be hazardous due to sharp edges. • Worn out components can have sharp or jagged edges. Handle worn parts carefully, to prevent damage to slings or personnel injury. • Assess the components before moving or lifting and use mechanical devices wherever possible.
	 WARNING HAZARDOUS CHEMICALS During disassembly, personnel may come in contact with hazardous chemicals. These chemicals must be identified before disassembly, and the correct SDS must be made available and appropriate safety management precautions put in place.

	CAUTION WORKING AREA SAFETY • Barricade and clearly mark the working area. • While working on heights, place the following safety sign near the work area. Working above the ground level. Keep away from the work area.
	CAUTION FOLLOW SAFE WORKING PRACTICES Follow safe working practices during all assembly and maintenance work.
	NOTICE TOOLING Use only specifically designed impact sockets when using air-driven or electrically-driven impact tools.
	NOTICE PPE Correct PPE must be worn when disassembling feeder.

12.2 Decommissioning

To decommission the feeder, proceed as follows:

1. Flush the unit with clean water.
2. Confirm the hazards associated with the feeder and adopt necessary safety precautions.
3. Isolate the feeder electrically, hydraulically, and mechanically.
4. Disconnect the feeder electrically, hydraulically, and mechanically.
5. Decontaminate by a static flush and drain.

	WARNING HAZARDOUS CHEMICALS Personnel may come in contact with hazardous chemicals. Adopt necessary safety precautions such as wearing appropriate PPE.
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12.3 Disposal

To sell or dispose of a complete feeder unit proceed as follows:

1. Confirm the hazards associated with the feeder and adopt necessary safety precautions.
2. Ensure that all lifting points are suitable for use.

	 DANGER
	LIFTING POINT FAILURE LIFTING POINTS, INCLUDING LIFTING LUGS, MAY NOT BE FIT FOR PURPOSE DUE TO CORROSION BY TIME OF DISPOSAL. IF YOU ARE IN ANY DOUBT ABOUT THE STRENGTH OR INTEGRITY OF ANY LIFTING POINT, DISASSEMBLE THE FEEDER INTO SMALLER SECTIONS TO SAFELY REMOVE IT FROM SITE.

3. Remove complete feeder unit to a decontamination area.
4. Decontaminate by a static flush and drain.

	 WARNING
	HAZARDOUS CHEMICALS Personnel may come in contact with hazardous chemicals. Adopt necessary safety precautions such as wearing appropriate PPE.

5. Grease, oil, rubber, and urethane must be disposed under applicable waste and environmental regulations.
6. All metal and alloys can be recycled if they are decontaminated and are not classified as hazardous material under applicable environmental regulations.
7. If you are unsure of the method of disposal, contact the local supplier.

	 CAUTION
	RISK OF PERSONAL INJURY OR DAMAGE TO ENVIRONMENT The feeder should be disposed of in accordance with local regulations and authorities.

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13 Troubleshooting

Problem	Cause	Solution
Feeder does not start or start slow	Electrical fault	Check power supply and correct
	Belts slipping or breaking	- Belt breakage, turning over in grooves, or jumping off the sheave may result in loss of speed and bogging down of machine. If necessary to replace belts, replace complete set to maintain uniform belt stretch. - Check drive and driven sheave for wear and proper alignment. - Check belt tension. Adjust to correct tension if required
	Oil issue	- Check oil level - Check oil grade for ambient temperature.
Feeder running incorrectly	Feeder contacts something during operation (base, chute, material, ice etc.)	Arrange space for feeder to run correctly.
	Belt pulleys and/or belts are worn	Change new ones.
	Drive equipment is wrongly positioned in relation to vibrating shaft	Move and adjust according to instructions.
	Feeder base not solid enough	Support.
	Springs broken	Change springs.
	Springs tired	- Change springs. - Feeder is galloping-overloaded - add springs.
Drive equipment deflects and vibrates during operation	Fastening has loosened	Tighten the fastening bolts.
Feeder supporting frame deflects and vibrates more	Construction is too light or insufficient bracing	Add support and brace.
Unusual noise during operation	There are loose or damaged parts in the feeder	Tighten, repair or change the part. At the same time check all screws for tightness. Check the reason for damage.
	Feeder contacts something during operation	Arrange space for feeder to run.
	Feed Arrangement	Check feed arrangement to unit.
	Bearing damage	See Bearing damage.
	Broken spring	Change spring after checking the cause.
	There are stones between springs	Clean the spring systems regularly and prevent material from building up between the springs.

Table 13-1: Troubleshooting

Problem	Cause	Solution
Feeder bearings overheat	Incorrect amount of oil in vibrating shaft	Check oil level.
	Wrong type or old oil in shaft	Change oil.
	Bearing damage	Check amount of operating hours and always replace damaged bearing with a new one. Check also possible reasons for the damage
	High temperatures in the environment or material	Arrange ventilation and change the oil suitable for high temperatures.
	Too high speed	Lower speed.
Oil leaks in the feeder	Breather is clogged	Clean or change breather regularly.
	- Seals are worn out or damaged (for instance during assembly). Do not reuse old seals - Damaged gaskets	- Change the seals being careful that the new ones are not damaged during installation. - Check all gaskets for leaking or damage.
	Loose drain plugs	Check all plugs in the assembly for looseness.
	Oil level sight gauge loose or leakage	Tighten it.
Material flows mostly on one side of feeder	Incorrect feed to the feeder	Re-position the feed so that it is evenly spread across the whole width of the feeder.
	Feeder is not running in the right position	Check the horizontal position of the spring pedestals.
Material not moving in the pan	- Moisture level too high - Clay stuck in the pan - Rocks stuck in the grizzly bars	- Feed dry material - Clean pan section area - Remove rocks from grizzly bars - possible timing angle change depending on conditions
	- Worn out springs - Discharge end blockage	- Change springs - Do not enclose the front of feeder
Too much material collecting at feed end of feeder	Too high feed rate or uneven feed	Adjust feed rate
	Material fed against the conveying direction of the feeder	Change feed direction
	Too heavy material load from the chute above	Adjust feed chute to remove head load
Material spills from chutes to the maintenance platforms etc	Feed or discharge chutes unsuitable or incorrectly installed	Re-install and seal. It is not permitted to connect extra parts to the feeder body as this can impair the operation or damage the feeder.
	Feeder worn so badly that material is leaking from feed or discharge areas or side plates. This can happen if wear plates are not changed in good time or if they are loose	Check the condition and fixing of wear plates and change in good time according to instructions. Check also tightening of the bolts regularly.

Table 13-1: Troubleshooting

Problem	Cause	Solution
Springs tilted	Fastening screws in the spring system loose.	- Tighten and change if necessary. - Material built up between the feeder pan liners and the hopper flashing. Insure correct clearances, clean out and feed dry material
Loss of amplitude	Material build-up	Material build-up on vibrating members of the unit will increase the vibrating body weight and decrease the amplitude (stroke). Inspect the unit and clean off all material build-up.
	Belt slippage	Loss of speed will reduce material travel across the unit. Increasing bed depth and vibrating body weight with the resultant loss in amplitude.
	Power failure	Low plant voltage will result in reduced motor speed with results similar to "Belt Slippage" above. It can also result in internal damage to motor windings.

Table 13-1: Troubleshooting

13.1 Description of Damage

Damage that occurs on the equipment must be documented in the damage form with necessary notes and drawings included. The completed damage form should be send to Weir Trio. This feedback helps us to maintain a high quality of service level to our customers.

With warranty claims, the damage form must be filled immediately when the damage has occurred. The damage form must be sent to Weir Trio without delay with a copy of the maintenance log.

The person reporting the damage must be property trained and authorized. He or she must be able to describe the damage as much in detail as possible and photographs must be attached.

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14 Special Tools

14.1 Eye Bolts



i NOTICE

SAFE WORKING LOAD

It is a reference chart given for convenience. Do not consider this as a safe working load face value. Always confirm the safe working load from the supplier of the lifting equipment. For safe instructions, refer to [“Safety” on page 2-1](#).

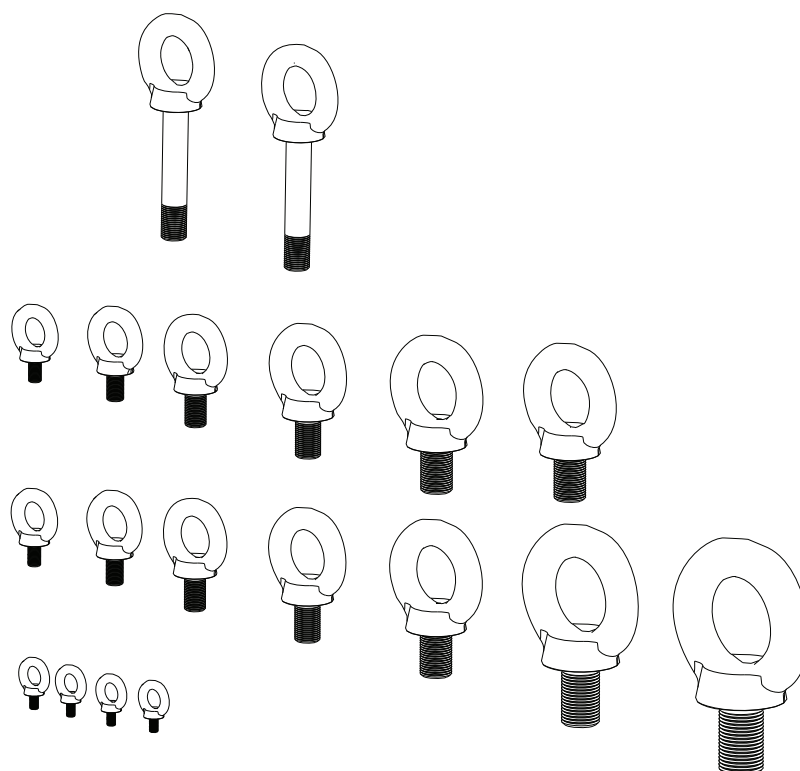


Figure 14-1: Eye Bolts

14.1.1 Shoulder Eye Bolt Chart

Shank Diameter	Working Load Limit In Linepull (lbs.)	Working Load Limit 60° Sling Angle (lbs.)	Working Load Limit 45° Sling Angle (lbs.)	Working Load Limit Angle Less Than 45° (lbs.)
1/4"	650	420	195	160
5/16"	1200	780	360	300
3/8"	1550	1000	465	380
1/2"	2600	1690	780	650
5/8"	5200	3380	1560	1300
3/4"	7200	4680	2160	1800
7/8"	10600	6890	3180	2650
1	13300	8645	3990	3325
1-1/4"	21000	13600	6300	5250
1-1/2"	24000	15600	7200	6000

Table 14-1: Shoulder Eye Bolt Chart

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Appendix 1 Torque Specifications

Appendix 1.1 Torque Charts

The following pages contain charts for use as guides when tightening bolts and nuts. These specifications must be followed unless specific torques are given elsewhere in this manual.

Appendix 1.1.1 SAE Bolt Torque

Table 15-1 on page 15-2 is a list of recommended torque values for SAE bolts. Use this torque chart to avoid over stressing SAE nuts and bolts used on equipment. These specifications must be followed unless specific torques are given. Weir Trio uses various bolts Grades 2, 5, and 8. The maximum torque values on 75% of the specified minimum proof strength of the bolt steel in order to provide a safety factor. The term “lube” applies to cadmium plating and/or application of thread lubricants. Hardened washers must always be used, regardless of whether standard nuts or lock nuts are employed.

Appendix 1.1.2 Metric Bolt Torque

Table 15-2 on page 15-4 is a list of torque values recommended for metric bolts. The maximum torque values are based on 75% of the specified minimum proof strength. The term “lube” includes the application of thread lubricants, cadmium plating and the use of hardened washers regardless of whether standard or lock nuts are used.

TF-Series Vibrating Grizzly Feeders

SAE Bolt Torque Chart

Diameter (inch)	Pitch (thd/ inch)	S.A.E Grade 2				S.A.E Grade 5				S.A.E Grade 8			
		Dry		Lube		Dry		Lube		Dry		Lube	
		(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm
1/4	20	5	7	4	5	8	11	6	8	12	16	9	12
1/4	28	6	8	5	7	10	14	7	9	14	19	11	15
5/16	18	11	15	9	12	17	23	13	18	25	34	19	26
5/16	24	12	16	10	14	19	26	15	20	27	37	21	28
3/8	16	20	27	15	20	31	42	24	33	44	60	34	46
3/8	24	23	31	17	23	35	47	27	37	49	66	38	52
7/16	14	32	43	25	34	49	66	38	52	70	95	54	73
7/16	20	36	49	27	37	55	75	42	57	78	106	60	81
1/2	13	49	66	38	52	75	102	58	79	106	144	82	111
1/2	20	55	75	42	57	85	115	65	88	120	163	92	125
9/16	12	70	95	54	73	109	148	84	114	154	209	118	160
9/16	18	79	107	60	81	121	164	93	126	171	232	132	179
5/8	11	97	132	75	102	150	203	115	156	212	287	163	221
5/8	18	110	149	85	115	170	231	131	178	240	325	185	251
3/4	10	157	213	121	164	266	361	205	278	376	510	289	392
3/4	16	180	244	133	180	297	403	229	311	420	570	323	438
7/8	9	210	285	160	217	430	583	330	447	606	822	466	632
7/8	14	230	312	177	240	473	641	364	494	668	906	514	697
1	8	320	434	240	325	644	873	495	671	909	1,233	699	948

Table 15-1: Torque Values for Standard Hardware

TF-Series Vibrating Grizzly Feeders

Diameter (inch)	Pitch (thd/ inch)	S.A.E Grade 2				S.A.E Grade 5				S.A.E Grade 8			
		Dry		Lube		Dry		Lube		Dry		Lube	
		(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm
1	14	340	461	255	346	704	955	542	735	995	1,349	765	1,037
1-1/8	7	350	475	270	366	794	1,077	611	829	1,288	1,747	990	1,342
1-1/8	12	395	536	305	414	891	1,208	685	929	1,445	1,959	1,111	1,507
1-1/4	7	495	671	380	515	1,120	1,519	862	1,169	1,817	2,464	1,398	1,896
1-1/4	12	550	746	425	576	1,241	1,683	954	1,294	2,012	2,728	1,548	2,099
1-3/8	6	655	888	500	678	1,469	1,992	1,130	1,532	2,382	3,230	1,832	2,484
1-3/8	12	745	1,010	570	773	1,673	2,269	1,287	1,745	2,712	3,677	2,086	2,829
1-1/2	6	865	1,173	665	902	1,949	2,643	1,500	20,34	3,161	4,286	2,432	3,298
1-1/2	12	975	1,322	750	1,017	2,194	2,975	1,687	2,288	3,557	4,823	2,736	3,710
1-3/4	5	1,370	1,857	1,055	1,430	2,286	3,100	1,758	2,384	4,988	6,764	3,837	5,203
2	4.5	2,060	2,793	1,585	2,149	3,438	4,662	2,644	3,585	7,500	10,170	5,769	7,823
2-1/4	4.5	3,015	40,88	2,320	3,146	50,27	6,817	3,867	5,244	10,969	14,874	8,438	11,442
2-1/2	4	4,125	5,593	3,170	4,298	6,875	9,323	5,288	7,171	15,000	20,340	11,538	15,646
2-3/4	4	5,590	7,579	4,300	5,830	9,321	12,639	7,170	97,23	17,794	24,129	13,688	18,561
3	4	7,385	10,013	5,680	7,701	12,313	16,696	9,472	12,844	23,507	31,875	18,082	24,519

Table 15-1: Torque Values for Standard Hardware (Continued)

TF-Series Vibrating Grizzly Feeders

Metric Bolt Torque Chart

Diameter (mm)	Pitch (mm)	Property Grade = 8.8				Property Grade = 10.9				Property Grade = 12.9			
		Dry		Lube		Dry		Lube		Dry		Lube	
		(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm
6		6.9	9.3	5.2	7	9.7	13.2	7.3	9.9	11.6	15.7	8.7	11.8
8		17.7	23.9	13.3	18	24.9	33.8	18.7	25.4	29.8	40.4	22.4	30.4
10	1.5	30	41	24	32	44	60	34	46	52	71	40	54
10	1.25	32	43	24	33	46	63	36	49	55	74	42	57
12	1.75	53	72	41	55	77	105	60	81	91	123	70	95
12	1.25	58	78	44	60	85	115	65	88	99	134	76	103
14	2	85	115	65	88	124	168	95	129	145	196	111	151
14	1.5	91	124	70	95	134	181	103	139	156	212	120	163
16	2	131	178	101	137	192	260	148	200	225	305	173	234
16	1.5	140	190	108	146	204	277	157	213	240	325	184	250
18	2.50	181	246	139	189	265	359	204	276	311	421	239	324
18	1.5	204	276	156	212	297	403	229	310	349	473	268	364
20	2.5	256	347	197	267	374	507	288	390	439	595	338	458
20	1.5	284	385	218	296	415	563	319	433	487	660	375	508
22	2.5	349	473	268	364	510	692	392	532	599	812	460	624
22	1.5	383	519	294	399	560	759	431	584	657	891	505	685
24	3	443	600	340	461	647	877	497	674	759	1,029	583	791
24	2	482	654	371	503	705	956	543	736	828	1122	637	863
27	3	648	879	499	676	949	1,286	729	989	1,112	1,508	856	1,160

Table 15-2: Torque Values for Metric Hardware

TF-Series Vibrating Grizzly Feeders

Diameter (mm)	Pitch (mm)	Property Grade = 8.8				Property Grade = 10.9				Property Grade = 12.9			
		Dry		Lube		Dry		Lube		Dry		Lube	
		(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm	(ft-lbs)	Nm
27	2	700	949	538	730	1,023	1,387	787	1,067	1,200	1,627	923	1,252
30		985	1,335	741	1,004	1,389	1,883	1,044	1,416	1,665	2,257	1,252	1,697
36		1,716	2,326	1,290	1,749	2,419	3,280	1,819	2,466	2,900	3,931	2,180	29,56
42		2,753	3,732	2,070	2,806	3,882	5,263	2,919	3,957	4,652	6,307	3,498	4,742
48		4,156	5,634	3,124	4,236	5,860	7,944	4,406	5,973	7,023	9,521	5,280	7,159
56		6,662	9,032	5,009	6,791	9,393	12,735	7,063	9,575	11,259	15,264	8,465	11,477
64		9,954	13,496	7,484	10,147	14,035	19,028	10,553	14,307	16,822	22,807	12,648	17,148
72		14,410	19,536	10,835	14,689	20,319	27,547	15,277	20,712	24,353	33,016	18,310	24,824

Table 15-2: Torque Values for Metric Hardware (Continued)

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Appendix 2 Oil Viscosity Classification

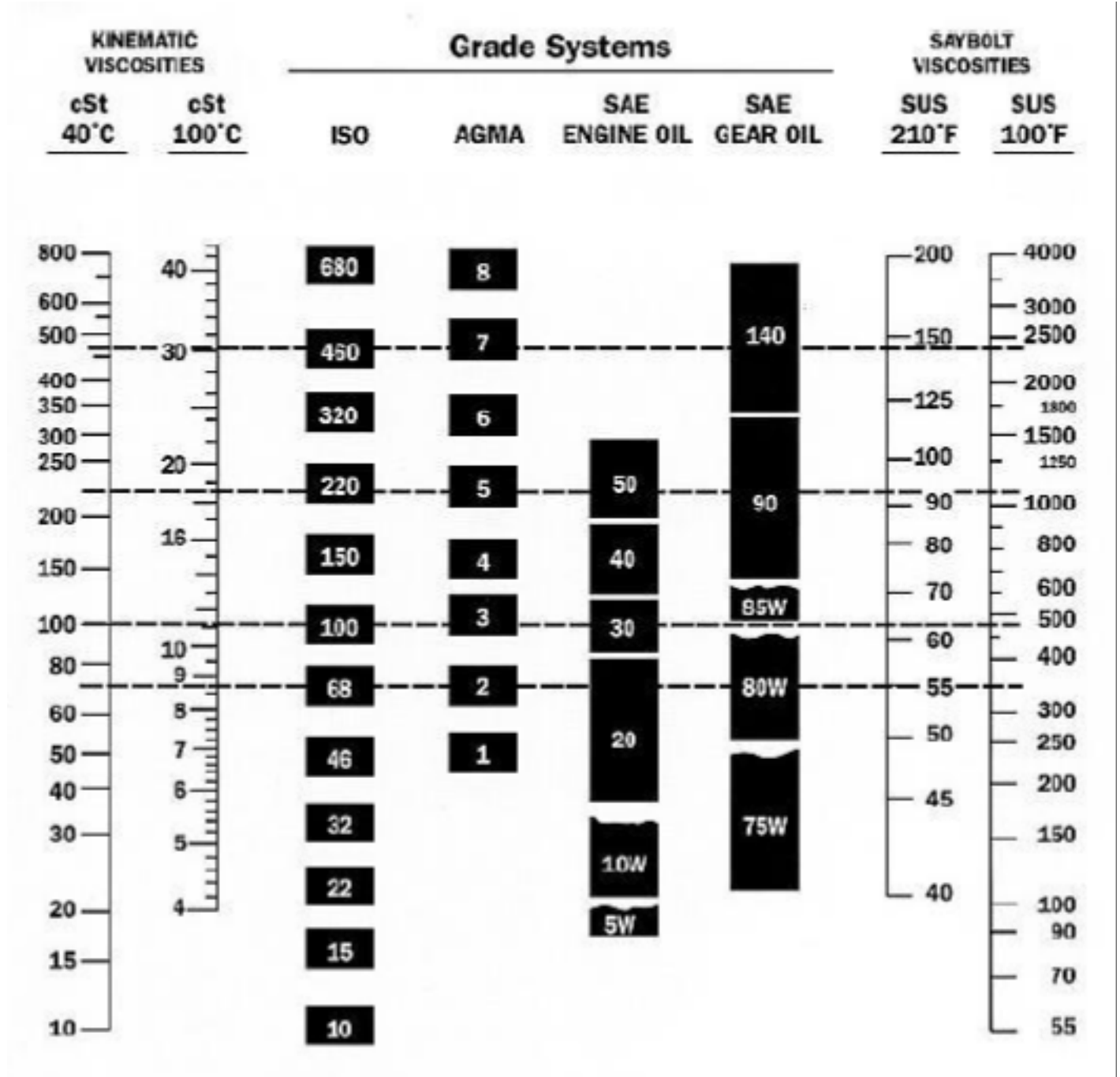


Figure 16-1: Oil Viscosity Classification

Viscosities can be related horizontally only. For example, the following oils have similar viscosities:

- ISO 460
- AGMA 7
- SAE Gear Oil 140

The viscosity/temperature relationships are based on 95 VI oils and are usable only for mono grade engine oils, gear oils and other 95 VI oils.

Crankcase oils and gear oils are based on 100 °C viscosity. The “W” grades are classified on low temperature properties. ISO oils and AGMA grades are based on 40 °C viscosity.

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Appendix 3 V-Belt Drive Information

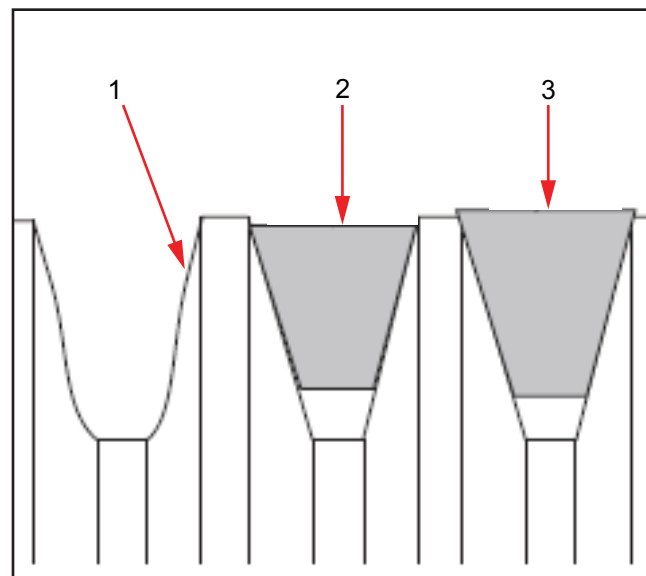
Appendix 3.1 General Belt Care

V-belts achieve more service life if kept clean and dry. Rock dust and rust from metals are abrasive. Oils and acids deteriorate belt material. Dust and moisture on belts can cause slippage. Dust buildup in sheaves can cause harmful vibration.

Belt dressing compounds are available and are sometimes used for various reasons. While they may be used, belt dressing may actually shorten belt life. Clean, well cared for belts are the best way to care for belt problems. Daily cleaning of belts and sheaves by blowing them off with compressed air can increase their useful life. Store unused belts in a cool and dry environment with no direct sunlight. Store on shelves or in original containers. Do not allow belts to be bent to diameters smaller than what is recommended for the belt cross section.

Appendix 3.2 General Belt Recommendations

1. Keep belts and sheaves clean.
2. Avoid belt or sheave rubbing. Don't let belts or sheaves rub against a guard (or anything else) while they are running.
3. Drive belts with motors mounted on adjustable bases. This allows convenient and proper installation of belts without prying on belts.
4. Keep belts properly tightened. Most belt stretch occurs in the first 30-40 hours of operation. Check belt tension frequently during this period.
5. Do not over-tighten belts. Over-tightening shortens the useful life of belts and sheaves. This can also cause premature bearing failures. Ideal belt tension is the lowest at which belts won't slip under peak loads.
6. Periodically inspect v-belt drives for wear on belts or sheaves. Refer to [Figure 17-1](#) and [Figure 17-2 on page 17-2](#). Replace worn sheaves.



1	Worn Sheave	2	Worn Belt or Sheave
3	Normal		

Figure 17-1: Typical Belt and Sheave Wear

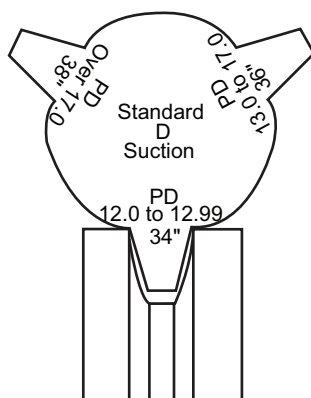


Figure 17-2: Sheave Gauge Showing Wear

7. Keep belts straight. Don't allow v-belts to run out of alignment. To run properly, sheaves must be parallel and grooves must be in line. Refer to [Figure 17-3](#).

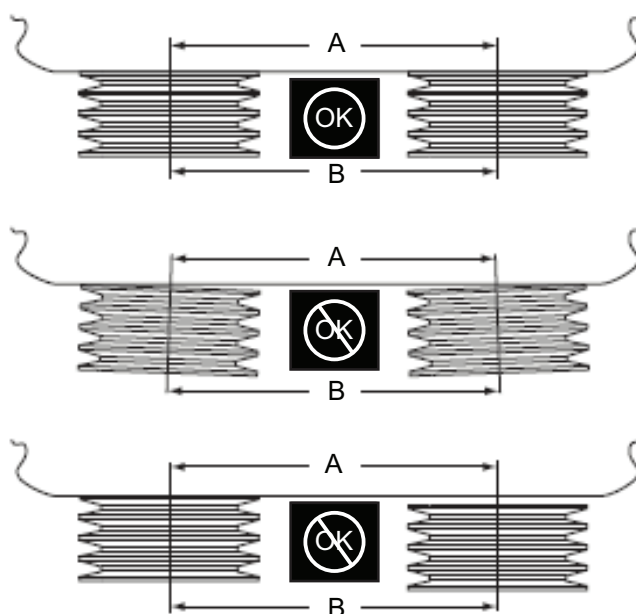


Figure 17-3: Sheave Alignment

8. If a drive uses multiple belts, use only matched belts. A matched set of belts last longer and loading is evenly divided among all the belts. Belts from different manufacturers must not be mixed in a set because material differences could reduce belt life. Always replace belts in full sets.
9. Don't overload belts. If the drive has 10 sheave grooves, use 10 belts. Using less than the designed number puts stress on the remaining ones. This can cause premature failure.

Appendix 3.3 Belt Drive Troubleshooting

Good maintenance practices must help you experience minimal trouble with belts and sheaves, however there may be times when troubles are experienced. The following [Table 17-1](#) can be used as a troubleshooting tool.

Trouble	Cause	Correction
Belt Slip (sidewalls glazed)	Insufficient tension	Replace belts; increase tension
Drive squeals	- Overloaded drive. - Arc of contact too short. - Heavy starting load.	- Find cause and correct. - Increase distance between hubs. - Increase tension.
Belts roll over	- Broken cord caused by prying. - Overloaded drive. - Impulse loads. - Misalignment. - Worn sheave grooves. - Flat idler sheave. - Excessive vibration.	- Replace belt correctly. - Redesign drive. - Apply proper tension. - Realign drive. - Replace sheave. - Align and reposition. - Check drive design, mounting. - Consider banded belts.
Mismatched belts	- New belts installed with old. - Sheave grooves worn unevenly. - Improper groove angle. - Shafts not parallel.	- Replace belts with matched set. - Replace sheaves. - Align drive.
Belt breaks	- Shock loads. - Heavy starting load. - Belt pried over sheaves. - Foreign object in drive.	- Apply proper tension. - Use compensator starting. - Replace belt correctly. - Determine source of object, design guard to correct problem.
Belt wears prematurely	- Sheave grooves worn. - Sheave diameter too small. - Mismatched belts. - Drive overloaded. - Belt slips. - Sheaves misaligned. - Oil present. - Excessive heat present.	- Replace sheaves. - Redesign drive. - Replace with matched set. - Redesign drive. - Increase tension. - Align sheaves. - Eliminate oil. - Ventilate drive.

Table 17-1: Belt Drive Troubleshooting

Appendix 3.4 Maintaining V-Belt Tensions

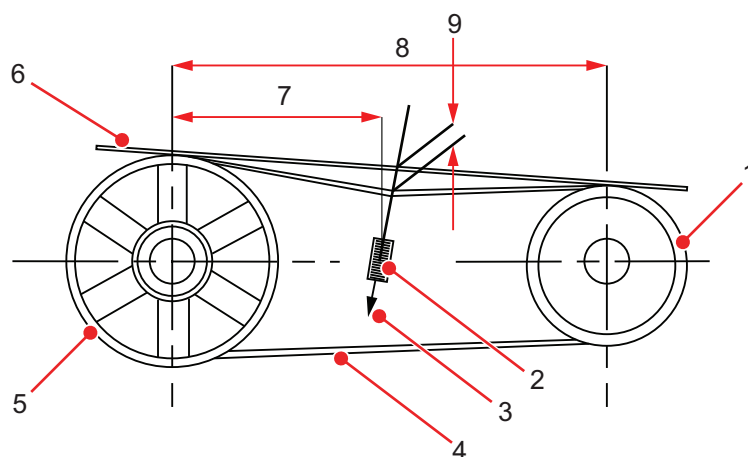
V-belt tension must be maintained to provide proper drive effectiveness. The following method of tensioning v-belts has been established as an acceptable and preferred method.

Appendix 3.4.1 Tension-Deflection Method

1. Place a straight edge across the top of both sheaves. Refer to [Figure 17-4 on page 17-4](#) measure the span length.
2. Using a spring scale at right angle to belt at 1/2 of span length, apply a force to the scale on one of the belts enough to deflect the belt 1/64 inch per each inch of span length.

Example: For a 100 inch span, the deflection would be 100/64, or approximately 1-1/2 inches.

3. Compare the force measurement to the recommended ranges shown in [Table 17-2 on page 17-5](#) below for the appropriate belt cross section type shown in.
4. Tighten or loosen belt to bring it into the recommended range.
5. When installing new belts, tighten them to “Initial Installation” tension. This tension will drop during the run-in period.



1	Motor Driven Sheave	6	Straightedge
2	Spring Scale	7	1/2 Span length
3	Force Measurement	8	Span Length
4	V-belt	9	Deflection Measurement - 1/64 Inch for each Inch of Span Length
5	Driven Sheave		

Figure 17-4: Tension-Deflection Diagram

TF-Series Vibrating Grizzly Feeders

V-Belt Cross Section	Smallest Sheave Diameter Range (Inches)	Recommended Belt Deflection Force (Pounds)		
		Initial Installation	Re-tensioning	
			Maximum	Minimum
A	- 3.0	3.6	3.1	2.4
	3.1 - 4.0	4.2	3.6	2.8
	4.1 - 5.0	5.2	4.6	3.5
	5.1 -	6.1	5.3	4.1
B	- 4.6	7.3	6.4	4.9
	4.7 - 5.6	8.7	7.5	5.8
	5.7 - 7.0	9.3	8.1	6.2
	7.1 -	10.0	8.8	6.8
C	- 7.0	12.5	10.7	8.2
	7.1 - 9.0	15.0	13.0	10.0
	9.1 - 12.0	18.0	16.3	12.5
	12.1 -	19.5	16.9	13.0
D	12.0 - 13.0	25.5*	22.1	17.0
	13.1 - 15.5	30.0*	26.0*	20.0
	15.6 - 22.0	32.0*	28.0*	21.5
E	18.0 - 22.0	45.0*	39.0*	30.0*
AX	22.1 -	52.5*	45.5*	35.0*
	- 3.0	5.1	4.4	3.4
	3.1 - 4.0	5.5	4.8	3.7
	4.1 - 5.0	6.0	5.2	4.0
	5.1 -	6.7	5.9	4.5
BX	- 4.6	10.0	8.7	6.7
	4.7 - 5.6	11.0	9.5	7.3
	5.7 - 7.0	11.5	9.9	7.6
	7.1 - 12.0	10.1	7.8	
CX	- 7.0	18.0	15.6	12.0
	7.1 - 9.0	19.5	16.9	13.0
	9.1 - 12.0	20.0	17.6	13.5
	12.1 -	21.0	18.2	14.0

Table 17-2: Belt Deflection Force

V-Belt Cross Section	Smallest Sheave Diameter Range (Inches)	Recommended Belt Deflection Force (Pounds)		
		Initial Installation	Re-tensioning	
			Maximum	Minimum
3V	2.65 - 3.35	4.6	4.0	3.1
	3.65 - 4.50	5.5	4.8	3.7
	4.75 - 6.0	6.4	5.6	4.3
	6.5 - 10.6	7.3	6.4	4.9
5V	7.1 - 10.3	16.5	14.3	11.0
	10.9 - 11.8	19.5	16.9	13.0
	12.5 - 16.0	21.0	18.2	14.0
8V	12.5 - 16.0	39.0*	33.8*	26.0*
	17.0 - 20.0	45.0*	39.0*	30.0*
	21.2 - 24.4	51.0*	44.2*	34.0*
3VX	2.2 - 2.5	4.8	4.2	3.2
	2.65 - 4.75	5.7	4.9	3.8
	5.0 - 6.5	7.2	6.2	4.8
	6.9 -	8.7	7.5	5.8
5VX	- 5.5	15.0	13.0	10.0
	5.9 - 8.0	19.0	16.9	13.0
	8.5 - 10.9	21.0	18.2	14.0
	11.8 -	22.0	19.5	15.0

Table 17-2: Belt Deflection Force (Continued)

*. For banded belts, multiply the force in table by the number of belts in the band.

Appendix 3.5 Belt Type Cross-Sections

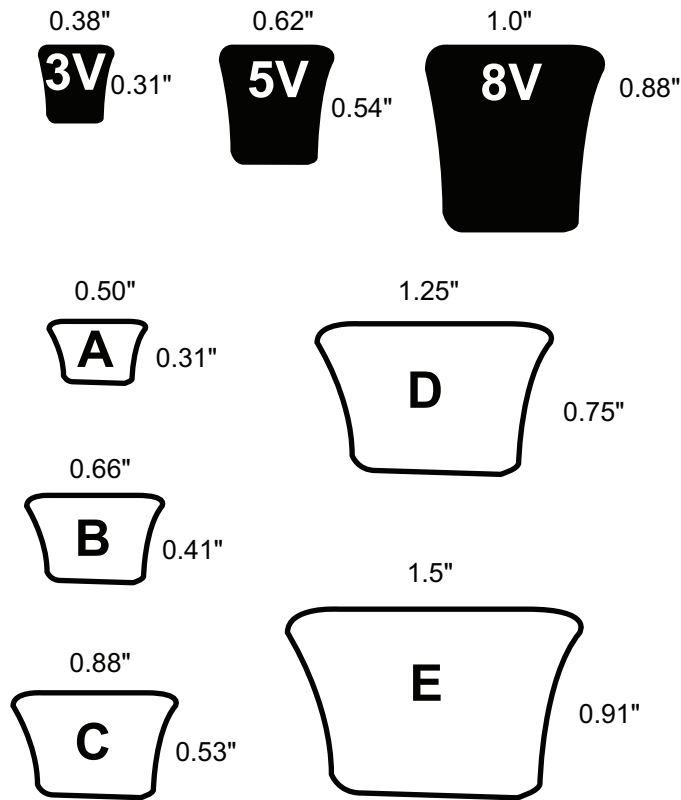


Figure 17-5: Belt Type Cross-Sections

Appendix 3.6 Belt Idlers

A properly designed v-belt drive does not require an idler to deliver full horsepower if proper belt tension can be maintained. Idlers must be avoided if possible because it puts additional bending stress points on the belt. Due to drive design parameters, idlers are sometimes used. The following are general descriptions of the types of idlers that may be used.

Appendix 3.6.1 Inside Idler

A v-grooved idler located on the inside of the belts on the slack side of drive would be recommended over a backside idler. The idler must be located near the large sheave to minimize reduction of the arc of contact on the small sheave. The diameter must be equal to or larger than that of the small sheave.

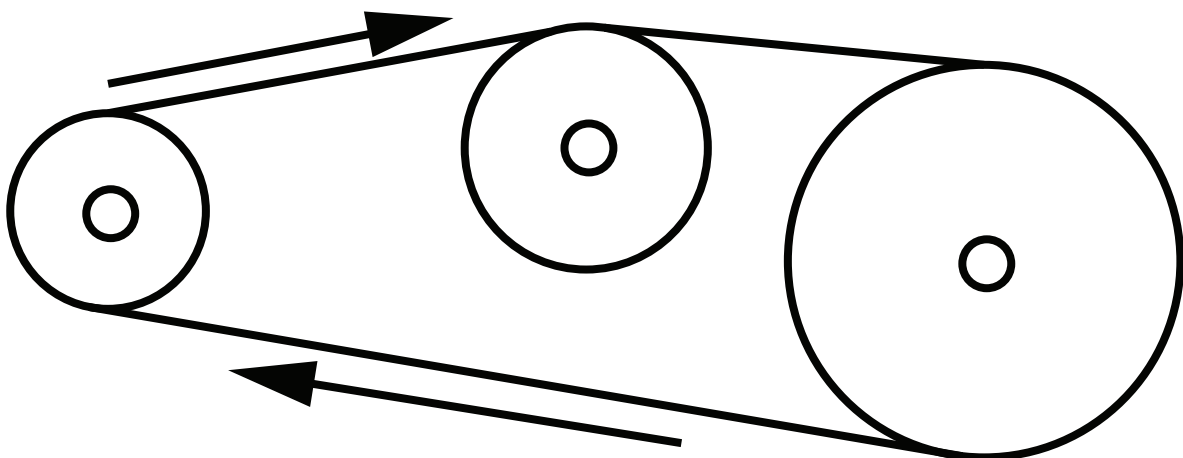


Figure 17-6: Inside Idler

Appendix 3.6.2 Backside Idler

A backside idler increases the arc of contact on both sheaves. However, it forces a backward bend in the v-belt that can contribute to premature failure. The idler puts additional stress on the bottom portion of the belt which will result in bottom cracking. If a backside idler must be used, the diameter must be at least 1 - 1/2 times the diameter of the small sheave. It must also be located as close as possible to the small sheave.

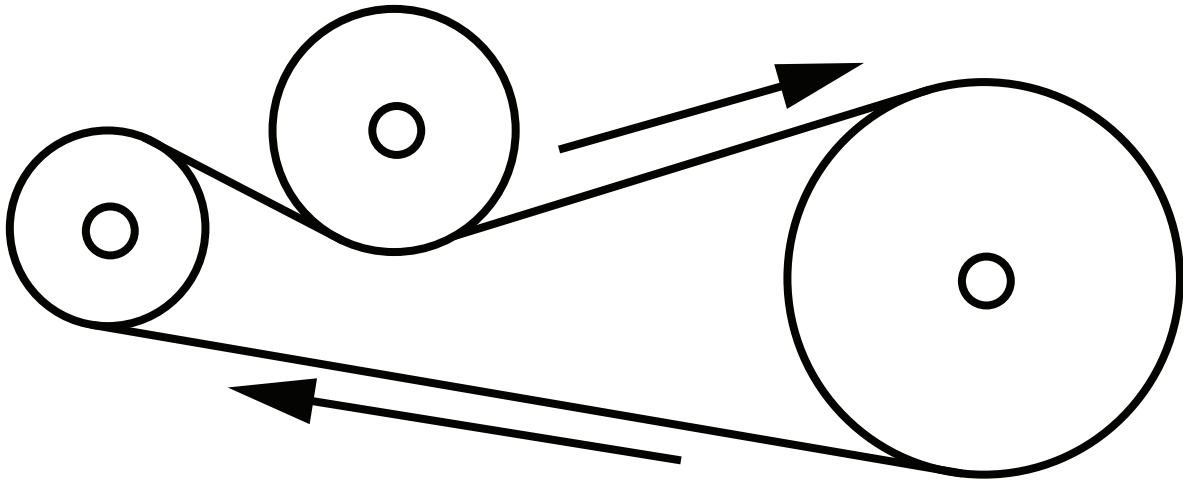


Figure 17-7: Backside Idler

Appendix 3.6.3 Kiss Idler

A kiss idler differs from the backside idler because the kiss idler does not penetrate the belt span and create a backward bend on the belt. A kiss idler does not contribute to premature belt failure. Use of a kiss idler can help control belt vibration and whip on drives subject to shock and pulsating loads. If a kiss idler is used, the diameter must be at least 1 - 1/2 times the diameter of the small sheave.

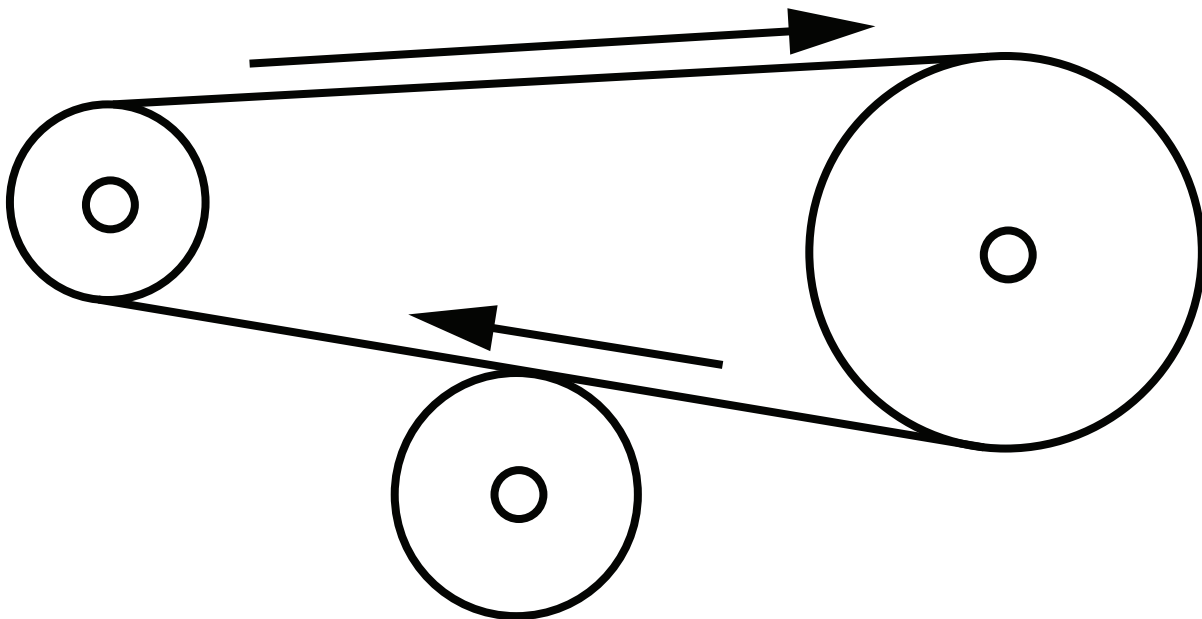


Figure 17-8: Kiss Idler

Appendix 4 Reference Documents for Customers

The following reference documents will be provided in additional documents:

1. General Arrangement Drawing
2. Parts Manual

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